

Photoshop Intermediate: From Practitioner to Pro

The full teaching manual for the eight-session course (one 3-hour session per module). Timing, live-demo scripts, the beginner mistakes to pre-empt, lab facilitation, and checks for understanding — for all eight modules.

8 modules • 23 lessons

8 sessions × 3 h • 24 h with labs & breaks

Instructor copy • not for students

CONTENTS

MOD 1 Advanced Selection & Masking

- L01 Luminosity Masks: Select by Tone
- L02 Channel Masking & Blend-If
- L03 Select & Mask Mastery: The Hard Cutout

MOD 2 Color & Tone, Deep

- L04 Color Management & Working Spaces
- L05 Grading with Curves, Gradient Maps & LUTs
- L06 Match, Split-Tone & Soft-Proof

MOD 3 High-End Retouching

- L07 Skin & Frequency Separation
- L08 Dodge & Burn Sculpting
- L09 Eyes, Teeth & Hair: Natural Finish

MOD 4 Commercial Compositing

- L10 Match Perspective, Scale & Lens
- L11 Light, Shadows & Reflections
- L12 Atmosphere & the Unifying Grade

MOD 5 Design Systems & Type

- L13 Type Systems, Styles & Grids
- L14 Smart-Object Templates & Linked Assets
- L15 Campaign from One Master

MOD 6 Generative + Automation, Advanced

- L16** Generative for Compositing
- L17** Advanced Actions & Conditionals
- L18** Data-Driven Graphics & Scripting

MOD 7 Multi-Shot Techniques

- L19** Panoramas, HDR & Focus Stacking
- L20** Timeline Animation & Export

MOD 8 Capstone

- L21** The Brief & the Plan
- L22** Studio Session: Build It
- L23** Present & Case Study

Module 1: Advanced Selection & Masking

How to open the intermediate course and teach the three lessons that rebuild selection and masking at a professional level: luminosity masks that let the image paint its own stencil, channel masks and Blend-If for subjects that tone or color makes distinct, and a four-step pipeline for the hardest common cutout, flowing hair. Everything later in the course (grading, retouching, compositing) leans on this control.

3 lessons

3 hrs session

Lessons L01–L03

Level Intermediate

Maps to Session 1

HOW TO USE THIS GUIDE

- Same rhythm as the intro kit: **Frame** → **Demo** → **Guided build** → **Solo variation** → **Check**. Every lesson ends with a recipe card, a checklist, and a journal prompt; use them as your closers.
- **This room already knows Photoshop**. Everyone finished the intro course (or its equivalent): layers, masks, selections, Camera Raw, Curves, Smart Objects, a basic composite. Don't re-teach "black hides, white reveals." Name it, then build on it. The lessons call this "the spiral, one turn deeper."
- **Demo live, step by step**, but faster than in the intro: demo a whole step, then let the room do it. Say both shortcut forms (Ctrl/⌘, Alt/⌥): the Channels-panel modifiers are the whole game in L01.
- **Teal** = hands-on, **amber** = watch-outs. Lesson codes are the file numbers (L01–L03); the site's own numbering (Lesson 1.1–1.3) is shown alongside.

THE #1 RISK OF MODULE 1: "WHICH PANEL AM I IN?"

This is the first time most learners live in the **Channels** panel. Nearly every stuck moment is a panel-state problem: Ctrl/⌘-clicking a *layer* thumbnail instead of the RGB *channel*; running Levels while the RGB composite (not the channel copy) is highlighted; returning to Layers with a single channel still active so the photo looks gray; painting with the image thumbnail selected instead of the mask. Teach the room a four-point check before any click: "**Which panel? Which channel or layer is highlighted? Which thumbnail has the border? Is a selection active?**" Write it on the board.

BEFORE SESSION 1

- ☐ Copy `labs/module-1/practice/` to each machine (or a shared drive). Keep `labs/module-1/instructor/l03-hair-cutout-reference.png` on your machine only: it's a catch-up file for L03. Never hand out the course site's licensed `images/` folder.

- ◆ ☐ Email learners a week ahead. Ask them to bring, on the laptop (not only the phone): a **high-contrast landscape** (bright sky, darker foreground); a **bare tree, power lines, or other intricate subject against sky**; a **portrait with loose or curly hair** against a fairly plain background; optionally a hair or pet shot against a busy background. Details and consent notes in [Labs/module-1/README.md](#).
- ◆ ☐ Every machine is on a **current release** of Photoshop (the course home page recommends it), signed in and launched once. Check with Help ► About Photoshop (Windows) or Photoshop ► About Photoshop (Mac).
- ◆ ☐ On the projector machine: **Window ► Workspace ► Reset Essentials**, make sure the **Channels** panel is one click away (**Window ► Channels**), and raise the interface size if the room is large.
- ◆ ☐ Open the course home page (for orientation and "Before You Start"), [recipes.html](#), and [lesson_01.html](#), [lesson_02.html](#), [lesson_03.html](#) in browser tabs for the before/after sliders and diagrams.
- ◆ ☐ A way to share results for the wrap-up: a shared folder, or learners turn their laptops toward the room.

SESSION 1 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	NOTES
0:00–0:20	Welcome, orientation & setup check	20	Introductions, the course path, then the "Before You Start" baseline: color settings sanity, the 16-bit habit, "diagnose before you edit."
0:20–1:05	L01 (Lesson 1.1) Luminosity Masks: Select by Tone	45	Build: rescue the landscape with Lights and Darks curves.
1:05–1:53	L02 (Lesson 1.2) Channel Masking & Blend-If	48	Build: extract a bare tree with the Blue channel, then a Blend-If texture.
1:53–2:06	Break	13	Leave the L02 recipe card on the projector for stragglers.
2:06–2:55	L03 (Lesson 1.3) Select & Mask Mastery: The Hard Cutout	49	Build: the curly-hair cutout (the Module 1 deliverable). Solo variation is homework.
2:55–3:00	Wrap-up & exit ticket	5	Two-minute pressure-test gallery, recap the arc, assign homework, preview Session 2.
Total		180	Stand-and-stretch for one minute midway through L01; it comes out of demo time.

The lesson pages estimate $55 + 60 + 60 = 175$ minutes for a solo learner, but after the orientation, break and wrap the room has 142. The room runs each lesson's frame, demo and full Guided Build; the "Now You" solo variations start in class and finish at home (L03's moves entirely to homework). If you run long, protect the **L03 guided build**: it's the

Module 1 deliverable. Trim L01's and L02's solo starts first, then the L02 Blend-If step (Step 4 can be demoed in two minutes and practiced at home).

0:00

Welcome, Orientation & Setup Check

20 min

◆ TIMING

6m	Welcome & introductions
6m	Course orientation: the path, Recipes II, journaling
8m	Setup check: "Before You Start"

◆ SEQUENCE & TALKING POINTS

- 1 Welcome (6m).** Introduce yourself briefly. Round of names, quick: name, what you shoot or make, and *one edit you gave up on* (a cutout that never looked right, a sky you couldn't save). Jot these down: they are ready-made examples for today and for the Capstone in Session 8. Housekeeping in one breath: break time, Wi-Fi, where files live, how to ask for help.
- 2 Orientation (6m).** Show the course home page. **Say:** "You can already drive Photoshop. This course is about working like a professional: fewer first steps, much deeper craft." Walk the path in three tiers: **Core Craft** (1 Selection & Masking, 2 Color & Tone, 3 Retouching), **Commercial Work** (4 Compositing, 5 Design Systems, 6 Generative + Automation), **Going Further** (7 Multi-Shot, 8 Capstone). One module per session; every module ships a portfolio-worthy deliverable. Open **recipes.html** and filter for "luminosity" to show the **Build a luminosity mask** card: "Recipes II are the harder how-tos you'll look up out of order." Remind them the journal continues: five things after every lesson, plus each lesson's own prompt.
- 3 Setup check (8m).** Scroll to **Before You Start** on the home page. **Say:** "Two minutes of setup keeps the whole course honest." Run three quick checks with the room:
 - Color settings sanity.** Everyone opens **Edit ► Color Settings** and reads out their RGB working space. **Say:** "Just know what it says today. Don't change it: Module 2 goes deep on this." Cancel out of the dialog.
 - The 16-bit habit.** Open `101-high-contrast-landscape.jpg` and show **Image ► Mode ► 16 Bits/Channel**. **Say:** "Converting doesn't add detail to an 8-bit original, but it stops new banding as you build masks and push curves. Every build today starts here."
 - Diagnose before you edit.** **Window ► Histogram**. Ask: "Where do the tones sit?" (A pile on the right, the bright sky; a pile on the left, the dark valley.) **Say:** "Before you touch a slider, look. That's a habit for every module." Also confirm **Window ► Channels** opens the Channels panel: it's today's main stage.

Leave the file open: it's the L01 guided-build file.

◆ WATCH-OUTS

- ◆ **Someone changes their Color Settings "to be safe."** Ask them to Cancel. Changing policies mid-course causes confusing profile warnings; Module 2 sets this up properly.
- ◆ **Photos only on the phone.** They use the practice files today and sync their own photos during the break, ready for homework.
- ◆ **Missing prerequisite.** If someone has never made a layer mask or used Select and Mask, pair them with a confident neighbor and point them at the intro course's Modules 1 and 4 for homework. Don't slow the room.

MODULE AT A GLANCE

LESSON	TOPIC	ROOM	LEARNERS LEAVE ABLE TO...
L01 (1.1)	Luminosity Masks: Select by Tone	45m	Load Lights from RGB, invert for Darks, intersect to a band, and move highlights and shadows in opposite directions without a visible seam
L02 (1.2)	Channel Masking & Blend-If	48m	Shop the channels, forge a mask from a duplicate with Levels, cut out an intricate subject, and drop a texture's lights with Blend-If
L03 (1.3)	Select & Mask Mastery: The Hard Cutout	49m	Run rough in → refine → rescue → decontaminate on flowing hair and pressure-test the edge on a saturated background

The module is one story: **let the image's brightness draw the stencil → let one color's brightness draw it → combine every tool into a pipeline for the hardest edge.** The through-line is the lesson mantra "let the image paint the mask": the less you paint by hand, the better the edge.

L01

Luminosity Masks: Select by Tone (Lesson 1.1)

45 min

◆ OBJECTIVES

- ◆ Explain a luminosity mask: a selection built from the image's own luminance
- ◆ Load a Lights or Darks selection from the channels
- ◆ Adjust highlights and shadows independently, edges self-feathered and invisible
- ◆ Narrow a mask to a tonal band (just the brightest highlights) by intersecting

◆ TIMING

3m	The goal: two exposures, one frame (lesson slider)
5m	Mental model: the image paints the mask
4m	Lights, Midtones & Darks; narrowing to a band
3m	Loading luminosity from Channels: the four modifiers
22m	Guided build: rescue the landscape (Steps 1–4: 4 + 7 + 6 + 5)
4m	Now You: start the highlight glow (finish at home)
4m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (3m).** Show the L01 slider. **Say:** "Expose for the sky and the land goes black; expose for the land and the sky blows out. On the left, both rescued, and no brush ever touched the mask."
- 2 The image paints the mask (5m).** **Say:** "You know the rule: white reveals, black hides. A luminosity mask says: let the picture's own brightness be that stencil. Bright pixels become white, dark pixels black. The mask is literally a grayscale copy of the photo's luminance." Two payoffs: **the edges are perfect** (as soft as the light in the scene, no halos, no hand-feathering) and it's **self-targeting** (the sky is the bright part, so a Lights mask already contains it). Show Figure 1 on the lesson page: invert Lights and you have Darks. **Say:** "Nothing new to believe. It's two intro ideas fused: selections become masks, and masks are grayscale."
- 3 A family of masks (4m).** Show Figure 2: the tonal bar with overlapping Darks, Midtones, Lights, and narrower variants. Then the key move: **intersecting**. **Say:** "Load Lights, then intersect it with Lights again. Only what both share survives, so the selection slides further into the extreme tones. That's how

you target a blown sky without touching a bright shirt below it." In today's file, the pale cabin plays the bright shirt.

- 4 **The four modifiers (3m).** Show Figure 3 and write on the board: **Ctrl/⌘-click RGB = load** · **+ Shift = add** · **+ Alt/⌥ = subtract** · **+ Shift + Alt/⌥ = intersect**. **Say:** "Click the *channel* thumbnail in the Channels panel, not the layer thumbnail in Layers. Ctrl/⌘-clicking a layer loads its transparency, which on a photo is everything."
- 5 **Step 1 (4m).** The landscape is already open from the setup check. **Image ► Mode ► 16 Bits/Channel**, then **File ► Save As** `luminosity-working.psd`. Read the histogram out loud: highlights piling near the right, shadows near the left. In **Channels**, **Ctrl/⌘-click the RGB thumbnail**. **Say:** "Marching ants appear around the brighter half. The ants only outline pixels more than half selected; the real selection fades smoothly into the darks."
- 6 **Step 2 (7m).** With the selection active, add a **Curves** adjustment layer. Point at the mask: "The selection wrote the mask for us." Pull the curve down to bring the sky back. **Say:** "Watch the land: the darkening fades out on its own." Then the spill check: if the pale cabin or the far hills darken too, undo the layer, reload Lights, and intersect once or twice (**Ctrl/⌘ + Shift + Alt/⌥-click RGB**) before adding the Curves layer again. Demo one intersect so the room sees the ants shrink toward the sky.
- 7 **Step 3 (6m).** Reload the RGB luminosity selection, then invert it (**Ctrl/⌘ + Shift + I**): the dark valley is selected. Add another **Curves** layer and lift the shadows. The path, rocks and grass appear out of the black. **Say:** "One S-curve would brighten the sky while it lifts the land and undo our recovery. Two masked curves move highlights and shadows in opposite directions at once. That's a range only masks can reach."
- 8 **Step 4 (5m).** Zoom to 100% (**Ctrl/⌘ + 1**) along the horizon and the pine edges. **Say:** "Try painting a mask around those pine needles by hand. Now look: no seam, no halo." If anyone sees a glow, their masks overlap: lower one layer's opacity or re-intersect. Optional: a gentle global Curves on top for overall contrast. **File ► Save**. Toggle both Curves eyeballs for the before/after.

◆ LAB: RESCUE THE LANDSCAPE, THEN NOW YOU

Guided build on `l01-high-contrast-landscape.jpg` (or a learner's high-contrast landscape). Solo variation on `l01-still-life-window-light.jpg` or a learner's portrait or still life: load Lights and add a subtle glow by lifting only the brightest highlights; build a Midtones mask from saved Lights and Darks channels and add contrast to only the mids; compare a luminosity-masked sky darken with a hand-painted mask. Start the glow in class; the Midtones mask and the comparison are homework. Details in `labs/module-1/README.md`.

Circulate for: 16-bit in the title bar; two Curves layers whose masks look like a grayscale photo (Lights) and its negative (Darks), not solid white; no glow along the horizon or around the pines at 100%.

Definition of done: the lesson's five-item checklist: highlights recovered through a Lights (RGB) mask, shadows lifted through an inverted Darks mask, no seam or halo at the horizon at 100%, every edit a non-destructive adjustment layer with a luminosity mask, 16-bit where possible. `luminosity-working.psd` saved.

◆ MISTAKES TO PRE-EMPT

- ◆ **Ctrl/⌘-clicking the layer thumbnail in Layers.** That loads the layer's transparency: the whole image, so the mask is solid white. Go to the Channels panel and click the RGB thumbnail.
- ◆ **Curves added with no active selection.** The mask comes out all white. Undo, reload Lights, then add the layer.
- ◆ **Wrong modifier combination.** Ctrl/⌘ + Shift alone *adds* (the selection grows); Ctrl/⌘ + Alt/⌘ alone subtracts. Intersect needs all three keys.
- ◆ **"No pixels are more than 50% selected."** After heavy intersecting, Photoshop warns that the selection edges won't be visible. The selection is still there, just faint. Click OK and add the adjustment; Alt/⌘-click the mask thumbnail to see it.
- ◆ **"The ants look hard-edged, so the mask must be."** Ants only outline the 50% boundary. Alt/⌘-click the mask thumbnail to show the real, soft mask (click again to return).
- ◆ **Darks reloaded from an already-darkened image.** The Channels panel reads what's visible, so after the sky Curves the RGB luminosity has changed a little. For the cleanest Darks, hide the first Curves layer before you reload, then turn it back on. (Saving selections with Select ► Save Selection before adjusting avoids this entirely.)
- ◆ **Banding in the sky.** They skipped the 16-bit step. Convert and rebuild the masks.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** A selection built from the image's own brightness, saved as a mask; **Q2 D)** Ctrl/⌘-click the RGB thumbnail in the Channels panel; **Q3 A)** Undo the layer, reload Lights, and intersect once or twice (Ctrl/⌘ + Shift + Alt/⌘-click RGB) before adding the adjustment; **Q4 C)** Reload the RGB luminosity selection, then invert it (Ctrl/⌘ + Shift + I); **Q5 B)** A single curve would brighten the sky while it lifts the land; two masked curves move highlights and shadows in opposite directions at once; **Q6 D)** You're working in 8-bit; convert to 16-bit (Image ► Mode ► 16 Bits/Channel) before building the masks. Full set in the Answer Key.

Journal (3 min, finish at home if needed): *Where have you fought a hand-painted mask that a luminosity mask would have solved instantly? Name one past edit you would redo with this technique.* Many will name their intro sky rescue: that's a perfect answer.

◆ DIFFERENTIATION

Fast finishers: build the Midtones mask now: save Lights and Darks as channels with **Select ► Save Selection**, then **Select ► All** and Ctrl/⌘ + Alt/⌘-click each saved channel to subtract it. Ask why Lights minus Darks is *not* midtones (it narrows to the brightest tones, like a second intersect). Then the lesson's "Going further": record an Action that loads Lights, Darks and Midtones as saved channels in one click.

Struggling learners: Steps 1–3 only, no intersecting. One Lights curve down, one Darks curve up is the whole idea. Put the four-modifier line from the board on a sticky note next to their keyboard.

◆ OBJECTIVES

- ◆ Read the Red, Green & Blue channels as ready-made grayscale masks
- ◆ Turn the highest-contrast channel into a selection (duplicate, push with Levels)
- ◆ Use Blend-If to drop out a layer's lights or darks instantly
- ◆ Pick the right technique for the subject: luminosity, channel, Blend-If, or a blend of all three

◆ TIMING

3m	The goal: mask the unmaskable (lesson slider)
4m	Mental model: three masks you already have
4m	Reading the channels (Figure 2)
5m	Blend-If: masking without a mask; when to use which
22m	Guided build: extract a tree (Steps 1–4: 4 + 8 + 5 + 5)
4m	Now You: start the Blend-If darks drop (finish at home)
6m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (3m).** Show the L02 slider. **Say:** "Thousands of twigs. Nobody paints that, and Select Subject chokes on it. But against a bright sky, one channel already separates them almost perfectly. Every twig came along because the mask *was* the Blue channel."
- 2 Three masks you already have (4m).** **Say:** "Every RGB image is secretly three grayscale images stacked together. Each one is a potential mask. You're not making a mask from scratch; you're shopping among three for the one with the most separation." Walk the lesson's pipeline (Figure 1): find the highest-contrast channel → duplicate it → Levels to near black and white → Ctrl/⌘-click to load → apply as a layer mask. **Say:** "The whole skill is choosing the channel. The rest is two clicks and a Levels push."
- 3 Reading the channels (4m).** Open `l02-bare-tree-sky.jpg` and click Red, Green, Blue in the Channels panel while the room watches. **Say:** "Skies are blue, so in the Blue channel the sky is bright and the tree is dark. In Red, the top of the sky goes nearly as dark as the branches." Mention the lesson's aside: for skin, Blue is where blemishes stand out most. Same habit, different job (Module 3 uses it).

- 4 **Blend-If (5m).** Double-click a layer's thumbnail or the empty area beside its name (not the name itself, which renames it) to open **Layer Style ► Blending Options**. Point at **Blend If: This Layer and Underlying Layer**. **Say:** "Drag This Layer's white slider inward and the layer's brightest pixels vanish; drag the black slider and its darkest pixels drop out. **Alt/⌘**-drag splits a slider for a soft, feathered transition." Then the rule of thumb from "When to use which": **Blend-If is instant and non-destructive but tone-global; a channel mask is editable and spatial.**
- 5 **Step 1 (4m).** **Image ► Mode ► 16 Bits/Channel, Save As** `channel-working.psd`. Everyone clicks Red, Green, Blue and picks the channel where the tree is darkest and the sky brightest. Hands up: which one? (Blue.)
- 6 **Step 2 (8m).** Drag **Blue** onto the New Channel icon: **Blue copy** appears. With only the copy highlighted, **Image ► Adjustments ► Levels** (**Ctrl/⌘ + L**). **Say:** "Adjustment layers can't target a channel, so this is the menu command, on a copy we're allowed to wreck." Pull the black and white points inward until the sky is white and the tree black. Zoom into the fine twigs and back off if they start to vanish. **Say:** "Push until the big areas are solid. The mask only has to be perfect at the edges." Paint any stray gaps with a hard black or white brush directly on the channel (on this file, the dark hedge and field are already black).
- 7 **Step 3 (5m).** **Ctrl/⌘**-click **Blue copy**. **Say:** "White loads as the selection, so right now the sky is selected." Invert (**Ctrl/⌘ + Shift + I**). Click the **RGB** channel to bring back full color, go to Layers with the image layer active (click the lock first if it's still the Background), and click **Add layer mask**. Place `102-new-sky.jpg` (**File ► Place Embedded**) and drag it below the tree layer. Zoom to 100%: every twig is there.
- 8 **Step 4 (5m).** **File ► Place Embedded** `102-grunge-texture.jpg` on top. Open Blending Options and pull the **This Layer** white slider left until only the texture's darker parts remain. **Alt/⌘**-drag to split it for a soft blend. **Say:** "No mask at all, purely by tone." **File ► Save.**

◆ LAB: EXTRACT A TREE, THEN NOW YOU

Guided build on `102-bare-tree-sky.jpg` (or a learner's tree, power lines, or other intricate subject against plain bright sky), with `102-new-sky.jpg` behind and `102-grunge-texture.jpg` blended in with Blend-If. Solo variation: Blend-If the other direction, dropping a layer's shadows (`102-fireworks-on-black.jpg` over `102-new-sky.jpg` or a learner's night scene); extracting an intricate subject against a busy background; inspecting a portrait's Blue channel (`103-curly-hair-portrait.jpg` works). Start the fireworks in class; the rest is homework. Details in `labs/module-1/README.md`.

Circulate for: Levels run on **Blue copy**, not on Blue or RGB (the photo's color should be untouched); the RGB channel clicked before the mask is added; fine twigs intact over the new sky at 100%.

Definition of done: the lesson checklist: chose the highest-contrast channel by eye, built a black-and-white mask on a duplicate channel with Levels, fine detail survived against a new background, blended a texture with Blend-If and no mask, worked non-destructively in 16-bit. `channel-working.psd` saved.

◆ MISTAKES TO PRE-EMPT

- ◆ **Levels on the real channel or the composite.** If the photo suddenly turns yellow or the whole image goes contrasty, the original Blue (or RGB) was highlighted. Undo; click only **Blue copy**.
- ◆ **Looking for an adjustment layer.** They can't target a channel. Image ► Adjustments ► Levels on the copy.
- ◆ **The photo looks gray back in Layers.** A single channel is still active. Click **RGB** in the Channels panel.
- ◆ **The sky got cut out, not the tree.** They forgot to invert after loading. Undo the mask, or select the mask and press **Ct r l / ⌘ + I** to invert it.
- ◆ **Over-pushed Levels.** The finest twigs snapped off (went white). Undo and push less; fix gaps by brush instead.
- ◆ **Double-clicking the layer name.** That renames the layer. Double-click the thumbnail or the empty area beside the name for Blending Options.
- ◆ **This Layer vs Underlying Layer.** Dragging the Underlying Layer slider forces the *scene's* tones through instead. For today's texture, it's This Layer, white slider.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 A)** Each of Red, Green and Blue is a full-resolution grayscale image, and a mask is just a grayscale image; **Q2 C)** Blue; **Q3 D)** The original channel is part of the photo's color; pushing it would damage the image, while the copy is a safe scratch mask; **Q4 B)** You over-pushed: the delicate twigs went white; back off until the big areas are solid and the twigs survive; **Q5 A)** White loads as the selection; invert (**Ct r l / ⌘ + Shift + I**), click RGB, then add the layer mask on the image layer; **Q6 D)** Drag the This Layer white slider left (**Alt / ⌥**-drag to split it for a soft blend). Full set in the Answer Key.

Journal (3 min): *Channel masking feels like a cheat code the first time it works. What subject have you given up on cutting out before that you would now attempt?* Link it to the "one edit you gave up on" from the introductions.

◆ DIFFERENTIATION

Fast finishers: the lesson's "Going further": Blend-If lives inside Layer Style, so it travels with copied layer styles and can be combined with a real mask on the same layer. Try a fast sky swap with no channel mask at all: duplicate the original tree layer above the new sky, open Blending Options, set **Blend If to Blue**, and drag This Layer's white slider left. Compare the twigs with the channel-mask version. Then read the FAQ on why Underlying Layer results change when layers move, and how stamping (**Ct r l / ⌘ + A l t / ⌥ + Shift + E**) locks the look.

Struggling learners: Steps 1–3 on the practice file only. The practice tree separates cleanly in Blue, so Levels needs only a modest push. Skip Step 4 in class; the recipe card covers it for homework.

◆ OBJECTIVES

- ◆ Run the full Select & Mask workflow, tuned for hair
- ◆ Rebuild wispy strands with the Refine Edge Brush
- ◆ Rescue what refinement misses with a channel mask (Lesson 1.2)
- ◆ Kill the fringe with Decontaminate Colors and pressure-test the edge

◆ TIMING

3m	The goal: every strand survives (lesson slider)
4m	Mental model: a pipeline, not a tool
3m	The Select & Mask workspace, tuned for hair (Figure 2)
32m	Guided build: the hard cutout (Steps 1–4: 5 + 10 + 8 + 9)
2m	Now You: brief the solo variation (homework)
5m	Self-check, recipe card, journal, Module 1 recap

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (3m).** Show the L03 slider and zoom into the curls. **Say:** "This is the Module 1 deliverable and the true test of everything so far. Flowing, curly hair is the hardest common cutout there is."
- 2 A pipeline, not a tool (4m).** **Say:** "Beginners look for the one button that cuts out hair. There isn't one." Write the four steps on the board and leave them up: **1 Rough in** (Select Subject: the solid mass, ignore the wisps) → **2 Refine** (Refine Edge Brush on the hairline) → **3 Rescue** (a channel mask painted into the layer mask) → **4 Decontaminate** (the old background's color lets go). **Say:** "No single step is hard. The skill is the order, and not expecting step one to do step three's job."
- 3 The workspace (3m).** Show Figure 2. For hair, three things matter: the **Refine Edge Brush** on the hairline; **Smooth** and **Feather** kept low with **Shift Edge** nudged slightly negative; **Output To: New Layer with Layer Mask**. Decontaminate Colors is ticked only on the final pass.
- 4 Step 1 (5m).** Open `l03-curly-hair-portrait.jpg`, **Image ▶ Mode ▶ 16 Bits/Channel, Save As cutout-working.psd**. Duplicate the layer (**Ct r L/⌘ + J**) so the original stays intact beneath, and hide the original now. **Select ▶ Subject**. Tidy obvious misses with Quick Selection (paint to add, **Alt t/⌘** to subtract). **Say:** "It nails the face and the solid hair and gives up on the wisps. Expected. Don't chase

strands yet." (On this illustrated file Select Subject may wobble; the Object Selection tool with a box around the head and shoulders is the fallback.)

- 5 **Step 2 (10m). Select ► Select and Mask** ($\text{Ctrl} + \text{Alt} + \text{R}$). View: **On Layers** or the red **Overlay**. Under **Edge Detection**, tick **Smart Radius** and raise **Radius** a few pixels. **Say:** "That makes Photoshop search a wider band for strands." Try **Refine Hair** in the options bar for a first pass. Then the Refine Edge Brush *along* the curly outline, never across the solid hair. Deliberately drag it across the middle of the hair once so the room sees it go translucent and mushy, then undo. Keep Smooth and Feather low, Shift Edge slightly negative.
- 6 **Step 3 (8m). Output To: New Layer with Layer Mask**, Decontaminate **off**, OK. Inspect at 100%: some fine flyaways are still missing. Now Lesson 1.2 returns: in Channels, find the channel with the most hair-vs-background contrast (on this teal backdrop, Blue or Green; Red is the weakest), duplicate it, push it with **Image ► Adjustments ► Levels**. $\text{Ctrl} + \text{Click}$ it and invert so the hair is selected (the backdrop loads first). Click RGB, then click the new layer's **mask thumbnail** and paint **white** with a soft, low-opacity brush along the hairline. **Say:** "The selection limits the paint to the strands. We're painting lost detail back." $\text{Ctrl} + \text{D}$ to deselect when done.
- 7 **Step 4 (9m).** With the rescued layer mask selected, reopen **Select and Mask**, tick **Decontaminate Colors**, raise **Amount** until the teal leaves the edge. **Say:** "Decontamination rewrites edge pixels, so it can only output to a new layer or document." Choose **New Layer with Layer Mask**: a fresh, decontaminated layer appears and the previous one is hidden. Pressure test: add a saturated **Solid Color** fill layer (hot magenta or lime) *below* the cutout. **Say:** "Any surviving fringe screams against this." Fix, then place `103-new-background.jpg` above the test color (or hide the test layer), and **File ► Save**. **Say:** "Module 1 deliverable: done."

◆ LAB: THE HARD CUTOUT (MODULE 1 DELIVERABLE)

Guided build on `103-curly-hair-portrait.jpg` (or a learner's portrait with loose or curly hair against a reasonably plain background), placed on `103-new-background.jpg` or a learner background. Solo variation (homework): a subject against a busy background (`103-hair-busy-background.jpg`: same sitter, auburn hair, out-of-focus foliage); something non-human and tricky (a glass, a potted fern, a pet); one cutout on a dark and one on a light background. Details in `labs/module-1/README.md`.

Circulate for: the original layer hidden (a visible original underneath makes every cutout look perfect); the channel copy loaded and inverted before painting; white painted on the *mask* thumbnail; the saturated test layer *below* the cutout.

Definition of done: the lesson checklist: solid mass selected then the hairline refined with the Refine Edge Brush; missing strands rescued via a channel mask painted into the layer mask; edge decontaminated, no fringe against a saturated test background; output as a layer + mask with the original kept as a hidden backup; subject placed convincingly on a new background. `cutout-working.psd` saved.

Catch-up: anyone whose cutout collapses with 10 minutes left can place `labs/module-1/instructor/103-hair-cutout-reference.png` to see the target edge and still practice Step 4's pressure test. Hand it out only then.

◆ MISTAKES TO PRE-EMPT

- ◆ **Original still visible underneath.** The cutout looks flawless because the old background shows through the transparent edges. Hide the bottom layer before judging anything.
- ◆ **Refine Edge Brush across solid hair.** Translucent, mushy patches. Undo or paint with the Brush tool in Select and Mask to restore, and stay on the boundary.
- ◆ **Smooth or Feather cranked up.** Smooth erases curls into a helmet; Feather blurs the whole edge. Keep both low.
- ◆ **Decontaminate ticked on the first pass.** It forces a new layer and hides the previous one, and the rescue then works on the wrong layer. Leave it for the final pass.
- ◆ **Painting with no selection, or on the image thumbnail.** White everywhere on the mask brings the teal back; white on the image paints the photo. Check the thumbnail border and the ants.
- ◆ **Rescue channel not inverted.** The backdrop is selected, so painting white brings back background, not strands. Invert before painting.
- ◆ **Test color layer on top.** It covers the subject. The test layer goes *below* the cutout.
- ◆ **"Where did my layer go?"** Decontaminate's output hid the previous layer. Nothing is lost: show its eyeball.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** C) Rough in, refine, rescue, decontaminate; **Q2** A) Along the boundary between hair and background, never across the solid hair; **Q3** B) They make Photoshop search a wider band around the edge for strands; **Q4** D) It comes last: it rewrites edge pixels and forces a new layer, so you run it on the final, rescued mask; **Q5** B) Click the layer mask thumbnail and paint white with a soft, low-opacity brush along the hairline; **Q6** A) Raise the Decontaminate Colors Amount, consider a tiny inward Shift Edge, and test on the darkest background it will sit on. Full set in the Answer Key.

Journal (3 min, finish at home): *Which pipeline step did the heavy lifting on your image, and which did you almost skip but shouldn't have? Noticing that tells you where your cutouts usually go wrong.*

◆ DIFFERENTIATION

Fast finishers: start the busy-background version on `103-hair-busy-background.jpg`: no channel is clean against mixed foliage, so the rescue step has to be combined with hand painting, exactly as the lesson's solo variation predicts. Or the "Going further": Select and Mask has no Smart Object output, but you can output to a Layer Mask and later select that mask and reopen Select and Mask on it to adjust the edge without starting over.

Struggling learners: Steps 1, 2 and 4 on the practice file are enough to hit the three-part standard (fine strands kept, no colored halo on a garish background, a soft edge). Do the channel rescue at home with the recipe card. Pair them with a neighbor for the Select and Mask settings.

CLOSING THE SESSION

MODULE 1 WRAP-UP (5 MIN, 2:55–3:00)

Two-minute pressure-test gallery: three volunteers toggle their saturated test layer on and off while the room hunts for fringe. Recap the arc: **let the image's brightness draw the stencil → let one channel draw it → combine everything into rough in, refine, rescue, decontaminate.** Exit ticket (sticky note): "My subject is dark against a bright blue sky. I'd reach first for ____ because ____." Homework from the lessons' "Before the next lesson/module": rescue two more high-contrast scenes with Lights + Darks curves and build a saved Midtones mask; channel-mask two intricate subjects against sky and knock a white-background texture into a scene with Blend-If; do three hard cutouts of increasing difficulty (studio hair, busy-background hair, a glass); save a reusable Action for your Select and Mask output settings; finish all three journal entries. Preview Session 2 (Module 2, Color & Tone, Deep): "You can now select almost anything. Next we make sure the colors you see are the colors everyone else sees: color management, grading with Curves, gradient maps and LUTs, and soft proofing for print." Remind learners the Color Settings dialog they peeked at today is where Session 2 begins.

Module 2: Color & Tone, Deep

How to teach the three lessons that take color from "looks fine on my screen" to a professional pipeline: set up the container (working space and bit depth) and diagnose before editing, author a cinematic grade and bottle it as a LUT, then split-tone, match a companion image, soft-proof for print and deliver one look to two destinations. One file (`grade-working.psd`), then (`finish-working.psd`) carries the grade from L05 into L06, so protect it.

3 lessons

3 hrs session

Lessons L04–L06 (2.1–2.3)

Level Intermediate

Maps to Session 2

HOW TO USE THIS GUIDE

- Same rhythm as Module 1: **Frame** → **Demo** → **Guided build** → **Solo variation** → **Check**. Every lesson ends with a recipe card, a completion checklist and a journal prompt; use them as your closers.
- **Demo the clicks on the projector; judge color on your own screen.** This is the one module where the projector actively lies. Show *where* things are and *what order* they go in; send learners to their own monitors to judge *how much*.
- These learners already know Curves, masks and Camera Raw from the intro. The new material is mostly **invisible**: profiles, bit depth, conversions, gamut. Make it visible every time: the banding test, the lagoon file with and without its profile, the gamut warning graying out neon. Seeing beats explaining.
- **Teal** = hands-on, **amber** = watch-outs. In Module 2 the watch-outs are **Assign instead of Convert**, **editing the master instead of a duplicate**, and **overcooked grades** (Gradient Map at 100%, split-tone in the mids).

THE #1 RISK OF MODULE 2: "IT CHANGED COLOR WHEN I..."

Every profile dialog is a chance to relabel a file by accident. The two costly clicks are **Assign Profile** where **Convert to Profile** was meant, and answering an open-time mismatch dialog with "**Convert document's colors to the working space**" or "**Discard the embedded profile**" without reading it. Give the room one rule and write it on the board: "**When Photoshop asks about a profile, read the dialog out loud before clicking. On open: use the embedded profile. At the end: Convert, on a duplicate.**" Also ask learners to switch off Night Light (Windows) or True Tone / Night Shift (Mac) and set brightness to about the middle; a warm screen makes every grade too cool.

BEFORE SESSION 2

- ◆ ☐ Copy `labs/module-2/practice/` to each machine (or a shared drive): the 16-bit sky TIFF, five histogram drills, the lagoon pair (tagged and untagged), the studio portrait, the pier, the grade test chart, the companion portrait, the three-frame match set and the neon soft-proof file. Keep `labs/module-2/instructor/` (histogram answers, fallback LUT) on your machine only. As always, never hand out the course site's licensed `images/` folder.
- ◆ ☐ Remind learners (in the Session 1 wrap-up or by email) to bring: **a favorite photo they want to grade** (a portrait is ideal, because L06 split-tones the same file), **a second photo that should sit in the same set**, and **one deliberately vivid image** (neon, sunset, saturated product). Largest original files, not messaging-app copies. Details in `labs/module-2/README.md`.
- ◆ ☐ On every machine, open **Edit ► Color Settings** once and note what is there. If your lab machines are shared, decide in advance whether learners may change them (they should, for this session) and how you will reset them afterward.
- ◆ ☐ Do a dry run of **File ► Export ► Color Lookup Tables** on a Background layer + one Curves layer, so you have seen the dialog before learners do. Load `labs/module-2/instructor/l05-teal-gold-fallback.cube` once in a Color Lookup layer so you know it works on your version.
- ◆ ☐ Find out which press or printer profile your learners are likely to use (a local lab, an online printer). If none, **U.S. Web Coated (SWOP) v2** is the lesson's stand-in and ships with Photoshop.
- ◆ ☐ Open `lesson_04.html`, `lesson_05.html` and `lesson_06.html` in browser tabs for the before/after sliders and figures.

SESSION 2 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	NOTES
0:00–0:10	Warm-up: Module 1 recap + screen check	10	Two volunteers show their hard cutout. Quick recall: "which mask would you use for a bright sky?" (a luminosity mask). Screen settings reminder; state today's destination: one look, two deliveries.
0:10–0:55	L04 (Lesson 2.1) Color Management & Working Spaces	45	Build: Color Settings, histogram diagnosis, the banding proof, a correct sRGB export. Solo variation starts here, finishes at home.
0:55–1:50	L05 (Lesson 2.2) Grading with Curves, Gradient Maps & LUTs	55	Build: a teal-and-gold grade on the portrait, saved as <code>grade-working.psd</code> , bottled as a <code>.cube</code> and poured onto the pier.
1:50–2:02	Break	12	Everyone saves <code>grade-working.psd</code> before leaving the seat.

CLOCK	SEGMENT	MIN	NOTES
2:02–2:52	L06 (Lesson 2.3) Match, Split-Tone & Soft-Proof	50	Build: split-tone, match the companion, soft-proof, deliver web + print (the Module 2 deliverable). Solo variation is homework.
2:52–3:00	Wrap-up & exit ticket	8	One volunteer shows their web JPG next to the soft-proofed master. Recap, homework, preview Session 3.
Total		180	Add a one-minute stand-and-stretch inside L06, between Step 2 and Step 3.

The lesson pages estimate $50 + 60 + 55 = 165$ minutes for a solo learner; the room has 150 once the warm-up, break and wrap are in. The difference comes out of the "Now You" solo variations: each one **starts in class and finishes at home** (L04's five-histogram drill and two-way export, L05's three LUTs, L06's three-photo match). If you run long, protect the L06 Guided Build, especially **Step 4, deliver both files**: that is the Module 2 deliverable. Shorten the L04 Color Settings tour first (the figure on the lesson page carries it), then the L05 local-refinement step.

MODULE AT A GLANCE

LESSON	TOPIC	TIME	LEARNERS LEAVE ABLE TO...
L04 (2.1)	Color Management & Working Spaces	45m	Set Color Settings deliberately, work in 16-bit, diagnose from the histogram in one sentence, and Convert (not Assign) to sRGB on a duplicate for the web
L05 (2.2)	Grading with Curves, Gradient Maps & LUTs	55m	Build a grade in order (neutral base, contrast, per-channel color, a blended Gradient Map, a masked refinement) and export it as a reusable LUT
L06 (2.3)	Match, Split-Tone & Soft-Proof	50m	Split-tone by zone, match a companion image with Color Sampler + Curves, soft-proof and fix gamut warnings, and deliver web + print files

The module is one story: **build a container you can trust → author a look and bottle it → finish it and ship it correctly to screen and paper**. The mantras stack up: *capture wide, edit deep, convert down last · correct, then grade, then bottle · what you see is only trustworthy once you have proofed it*.

L04

Color Management & Working Spaces (Lesson 2.1)

45 min

◆ OBJECTIVES

- ◆ Explain working spaces (sRGB, Adobe RGB, ProPhoto) and pick the right one for the job
- ◆ Say why you edit in 16-bit and what banding actually is
- ◆ Read the histogram and diagnose an image before touching a slider
- ◆ Set up Photoshop's Color Settings once, correctly, for the rest of the course

◆ TIMING

4m	The goal (banding slider) + mental model: a color container
6m	Which working space? Convert vs. Assign (lagoon demo)
3m	Color Settings, as a diagram
24m	Guided build: Steps 1–4 (6 + 8 + 6 + 4)
3m	Now You: start the histogram drill (finish at home with the two-way export)
5m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (2m).** Drag the L04 slider slowly across the banded sky. **Say:** "Same adjustment on both sides. The only difference is the container. This is the least glamorous lesson in the course and quietly the most important: get the pipeline right and every grade after it is trustworthy."
- 2 A color container (2m).** Draw a box on the board. Its **width** is the working space (gamut); the **lines inside it** are the bit depth. **Say:** "8-bit gives you 256 steps per channel; 16-bit gives you over 32,000. And a number like R 220 means nothing until you know the container: 220 in sRGB is a different real-world color from 220 in Adobe RGB. Color management is just keeping track of the container." Show Figure 1: **capture wide, edit wide and deep, convert down to the destination at the very end.**
- 3 Which working space? (3m).** Walk the three-row table: **sRGB** for web, social, email ("assume the internet is sRGB"); **Adobe RGB** for print (it holds cyans and greens sRGB clips); **ProPhoto** for heavy RAW grading and archival masters, **always with 16-bit**. **Say:** "Bigger isn't automatically better. ProPhoto in 8-bit bands badly, and much of it is imaginary color no device shows."
- 4 Convert vs. Assign (3m).** Open `104-lagoon-adobergb.jpg` (answer the mismatch dialog with **Use the embedded profile**). Then open `104-lagoon-untagged.jpg`: same numbers, no label, and the

water and parrot look duller. **Say:** "Identical pixel values. The only thing missing is the container label, so Photoshop reads them as the wrong space." On the untagged copy, **Edit ► Assign Profile** and pick Adobe RGB (1998): it snaps to match. **Say:** "That's the one honest use of Assign: labeling a file whose real space you know. At the end of a job you never relabel; you **Convert**, which remaps the numbers so the color keeps its appearance."

- 5 **Color Settings as a diagram (3m).** Show Figure 3, then open **Edit ► Color Settings** live. Point at the four decisions: RGB working space, CMYK (your press profile or the default), **Preserve Embedded Profiles**, and the mismatch check boxes (**Ask When Opening**). **Say:** "Preserve means a file's own tag is respected. The warnings mean you always know which container you're in."
- 6 **Step 1: configure the pipeline (6m).** Everyone sets RGB to **Adobe RGB (1998)** for print-capable work (or sRGB if strictly web), CMYK to their provider's press profile (for example **Coated GRACoL 2006** or **Coated FOGRA39**) or the default, policies to **Preserve Embedded Profiles**, mismatch warnings on. Then open `104-smooth-sky-16bit.tif` or their own image and check the title bar: **RGB/16**. **Say:** "Anything you'll grade hard: **Image ► Mode ► 16 Bits/Channel**."
- 7 **Step 2: diagnose from the histogram (8m).** **Window ► Histogram**. Open `104-histogram-a.jpg` and think aloud: "Crammed left? Right? A spike against a wall? Narrow?" **Say:** "Now one sentence: what does this image need? That sentence is your edit plan." Learners write their sentence for their own image (or `104-histogram-c.jpg`) in the workbook. **Say:** "Amateurs grab a slider and push; pros read, name, then reach. Thirty seconds of diagnosis saves ten minutes of flailing."
- 8 **Step 3: prove the container matters (6m).** On the sky TIFF: **Image ► Duplicate**, then on the copy **Image ► Mode ► 8 Bits/Channel**. On both, add the same aggressive Curves (drag the shadow and highlight ends inward hard), then a strong brighten. Zoom to 100% on the upper sky. **Say:** "Stripes on the 8-bit copy. Silk on the 16-bit. That's why the pipeline is not optional." Tip: arrange the two windows side by side (**Window ► Arrange ► 2-up Vertical**).
- 9 **Step 4: export correctly (4m).** Save the layered master as `color-master.psd` (wide space, 16-bit). Then **Image ► Duplicate**, tick **Duplicate Merged Layers Only**; on the copy **Edit ► Convert to Profile ► sRGB**, **Image ► Mode ► 8 Bits/Channel**, and export. **Say:** "File ► Export ► Export As can also convert to sRGB for you. Either way the master is untouched, and the color you graded is the color the world sees."

◆ LAB: SET UP & DIAGNOSE, THEN NOW YOU

Guided build on the learner's own favorite image (or `104-smooth-sky-16bit.tif` for the banding proof and `104-histogram-a...e.jpg` for diagnosis). Solo variation: read the five histogram drills cold (guess, then check; answers in `labs/module-2/instructor/104-histogram-answers.txt`), then at home export `104-lagoon-adobergb.jpg` (or their own Adobe RGB/ProPhoto version) two ways, Converted to sRGB and with the profile stripped, view both in a browser, and set up a wide-gamut 16-bit document template. Details in `labs/module-2/README.md`.

Circulate for: Color Settings actually changed (not just opened and cancelled); "RGB/16" in the title bar; a written one-sentence diagnosis before any edit; the 8-bit copy showing bands at 100%.

Definition of done: the lesson's five-item checklist: Color Settings configured with a deliberate working space; working file 16-bit; a one-sentence histogram diagnosis; banding seen with their own eyes; web export Converted to sRGB, not assigned.

◆ MISTAKES TO PRE-EMPT

- ◆ **Assign where Convert was meant.** Symptom: the colors jump the moment they pick sRGB. Undo, and use **Edit ► Convert to Profile**. Assign relabels; Convert remaps.
- ◆ **Clicking through the open-time dialog.** With Preserve Embedded Profiles and the mismatch warnings on, nearly every file will ask. The answer for this course is **Use the embedded profile (instead of the working space)**.
- ◆ **"I don't see banding."** The Curves move wasn't aggressive enough, they're zoomed out, or the file was never set to 8-bit. Push harder (stack a second Curves), zoom to 100% on the upper sky.
- ◆ **Converting the master.** They Convert to sRGB and 8-bit on `color-master.psd` itself and save. The wide, deep master is gone. Always **Image ► Duplicate** first.
- ◆ **ProPhoto in 8-bit.** Someone "plays safe" with ProPhoto and leaves the file 8-bit. That combination bands the worst.
- ◆ **Misreading a high-key image as overexposed.** A bright scene piles to the right but shouldn't slam into the wall. Histogram drill `b` is built to catch this.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** Working space (how wide the gamut is) and bit depth (how finely each channel is divided); **Q2 D)** The edit demands more tonal steps than 8-bit's 256 per channel; 16-bit has about 128 times more; **Q3 A)** The file was left in Adobe RGB or ProPhoto and never Converted to sRGB; **Q4 C)** On a merged duplicate: **Edit ► Convert to Profile ► sRGB, 8 Bits/Channel**, then export; **Q5 B)** Highlights are clipped: detail at the bright end is lost; **Q6 D)** ProPhoto RGB for heavy RAW grading and archival masters, always paired with 16-bit. Full set in the Answer Key. Journal (3 min, finish at home if needed): *Have you ever had colors "change" on export or a print come back wrong? Now that you understand containers, what was probably going on?* Invite one story from the room; almost everyone has one.

◆ DIFFERENTIATION

Fast finishers: the lesson's "Going further": if their monitor is calibrated, turn calibration off for a minute and watch a careful grade fall apart. Or open the lagoon in ProPhoto (**Convert to Profile ► ProPhoto RGB**) in 8-bit and then 16-bit, run the Step 3 Curves on each, and report which container bands. Have them save the 16-bit wide-gamut template now (**File ► New, 16 bit, Advanced Options ► Color Profile**).

Struggling learners: reduce it to two habits: **16-bit while editing** and **Convert to sRGB on a duplicate at the end**. Let them leave the working space on sRGB (the lesson allows it for strictly web work). Hand them the recipe card and the board rule about reading dialogs aloud.

◆ OBJECTIVES

- ◆ Grade with per-channel Curves for precise color in shadows, mids and highlights
- ◆ Use a Gradient Map to remap tones to a color ramp
- ◆ Apply and save a LUT (Color Lookup) to reuse a grade consistently
- ◆ Keep the whole grade non-destructive and dialable

◆ TIMING

4m	The goal (pier slider) + mental model: remapping color
5m	Gradient Maps and LUTs: what each is (test-chart demo)
5m	Set-up: open in the 2.1 pipeline, correct to a neutral base, Save As <code>grade-working.psd</code>
30m	Guided build: Steps 1–4 (10 + 8 + 6 + 6)
5m	Now You: start the first of three LUTs (finish at home)
6m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (2m).** Drag the L05 pier slider. **Say:** "Exposure barely moved. What changed is every color relationship: deeper teal water and sky, richer contrast, one mood. That's a look, not a fix."
- 2 Remapping color (2m).** **Say:** "Correcting asks 'what's wrong?' Grading asks 'what should it feel like?' Technically every grade is a remap: you decide what color each tone becomes. Shadows don't have to stay neutral." Name the three tools from surgical to sweeping: **per-channel Curves, Gradient Map, LUT**. Show Figure 1: **correct → contrast → color → refine locally → bottle it**.
- 3 Tones in, colors out (5m).** Open `L05-grade-test-chart.png`. Add a **Gradient Map** adjustment layer with a teal-to-gold ramp: the gray ramp at the top becomes the ramp exactly. **Say:** "Darkest pixels get the left color, brightest get the right." Now it looks garish. Set the layer to **Color** or **Soft Light** and drop opacity to about 20–30%. **Say:** "Blend it, don't bake it." Then hide it, add a **Color Lookup** layer, open the 3DLUT File menu and try two of the shipped looks on the chart. **Say:** "A LUT is a saved recipe: this input color becomes that output color, for every color. Photoshop ships dozens; today you'll save your own."
- 4 Neutral base (5m).** Everyone opens `L05-portrait-studio.jpg` (or their own portrait), sets **Image ► Mode ► 16 Bits/Channel**, and corrects the yellow cast and underexposure with a Curves layer (gray

eyedropper on the `GRAY CARD` middle patch) or **Filter ▶ Camera Raw Filter. Save As** `grade-working.psd`. **Say:** "A portrait is ideal because in Lesson 2.3 we split-tone this same file. And a neutral base is what makes a look portable."

- 5 **Step 1: contrast, then color (10m).** Curves layer with a gentle **RGB S-curve**. Then a **new** Curves layer, switch to **Blue**: lift the shadow end, drop the highlight end. **Say:** "Blue alone gives blue and yellow. For a truer teal, lift Green slightly or drop Red slightly in the shadows." Name the layers ("contrast", "teal-gold"). Say it again: **contrast before color**.
- 6 **Step 2: a Gradient Map accent (8m).** Gradient Map, teal-to-gold ramp, blend **Color** or **Soft Light**, opacity ~20–30%. **Say:** "It unifies the palette without flattening. If you can see the Gradient Map, it's too strong."
- 7 **Step 3: refine locally (6m).** Mask the teal-gold Curves or the Gradient Map so skin stays natural while the background goes moody (paint black on the face with a soft brush, low opacity). **Say:** "Module 1 pays off here. A luminosity mask from Lesson 1.1 is perfect for keeping the grade out of the brightest highlights."
- 8 **Step 4: bottle it (6m).** **File ▶ Save** first (Lesson 2.3 reopens this file). **Say:** "The export bakes every visible adjustment layer, not just the selected ones, and a LUT can't store a mask." **Shift**-click each mask thumbnail to disable it (a red X appears), hide anything that isn't part of the grade, keep the Background + adjustment-layer structure, then **File ▶ Export ▶ Color Lookup Tables** and save a `.cube`. Re-enable the masks. Open `l05-pier-dusk.jpg`, add a Color Lookup layer, **Load 3D LUT...**, pick the `.cube`. **Say:** "That's your grade, reusable forever. Building the grade is craft; saving it is professionalism."

◆ LAB: GRADE & BOTTLE IT, THEN NOW YOU

Guided build on `l05-portrait-studio.jpg` or the learner's own portrait (landscape is allowed, but they'll need a portrait to receive the grade in L06), saved as `grade-working.psd`; LUT transfer test on `l05-pier-dusk.jpg` or a second own photo. Solo variation (start in class, finish at home): three distinct LUTs (warm-summer, cold-thriller, faded-matte) applied to the same photo; reverse-engineer a film still with per-channel Curves; one LUT across five photos, noting per-image tweaks. If a learner's export fails, hand them `labs/module-2/instructor/l05-teal-gold-fallback.cube` so they can still do the transfer test, then troubleshoot the export at the break.

Circulate for: a neutral gray card *before* any grading; separate, named layers for contrast and color; a Gradient Map that is blended, not at 100% Normal; masks disabled (red X) at export time; a `.cube` that actually transfers.

Definition of done: the lesson's six-item checklist: contrast before color; per-channel Curves grade; Gradient Map at reduced opacity or a blend mode; at least one mask; LUT exported and re-applied to a second image; `grade-working.psd` saved for Lesson 2.3.

◆ MISTAKES TO PRE-EMPT

- ◆ **Grading on top of the cast.** They skip the neutral base; the "teal" fights a yellow cast and the LUT won't travel. Gray eyedropper on the card first.

- ◆ **Contrast and color on one Curves layer.** It works, but they can't dial either independently. Two named layers.
- ◆ **Gradient Map at 100% Normal.** Garish and detail-killing. Color or Soft Light, 20–30%.
- ◆ **Export Color Lookup Tables grayed out.** Adobe documents the export for a document with a Background layer plus adjustment layers. If a learner's photo is a Smart Object or an ordinary layer, or there are no adjustment layers yet, rebuild to that structure (a quick fix: **Layer ► New ► Background from Layer** on the photo).
- ◆ **The LUT includes the correction or a stray layer.** Everything visible is baked. If the transfer looks wrong on the pier, check what was visible during export. (Including the neutral-base correction is allowed, but then the LUT also "corrects" the next image; discuss which they want.)
- ◆ **Forgetting to re-enable the masks.** Their saved master now has an unmasked grade. **Shift-click** again, then Save.
- ◆ **Over-pushed channel curves.** A few pixels of drag is already a strong grade. Judge on your own screen, not the projector.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 C)** Correcting asks "what is wrong?"; grading asks "what should it feel like?" and deliberately remaps tones to chosen colors; **Q2 A)** Lift the shadow end and drop the highlight end; **Q3 D)** Set the layer to Color or Soft Light and/or drop its opacity to about 15–40%; **Q4 B)** The export bakes every visible adjustment layer, and a LUT cannot store a mask; **Q5 A)** The second photo wasn't corrected to a neutral base first, and a LUT assumes a similar starting point; **Q6 C)** Correct to neutral → contrast (S-curve) → color → refine locally with masks → bottle as a LUT. Full set in the Answer Key.

Journal (3–5 min): *Describe the "look" you are drawn to as an editor in terms of shadow color, highlight color, and contrast. Naming your taste is the first step to building it into a LUT.* Pair-share: each learner names a film or photographer whose look they'd reverse-engineer.

◆ DIFFERENTIATION

Fast finishers: the lesson's "Going further": stack a subtle Gradient Map under a Color Lookup and mask both. Or build the cold-thriller LUT now and pour both LUTs onto the test chart to compare how each remaps the hue field. Ask: "Where does your LUT break skin tones?" (Check the chart's skin patches.)

Struggling learners: Curves only: one S-curve, one Blue-channel split, then straight to Step 4. Skip the Gradient Map and the mask for now (the recipe card covers them). The must-haves are a saved `grade-working.psd` and one LUT that transfers.

◆ OBJECTIVES

- ◆ Match color across two images so a set feels like one shoot
- ◆ Refine a split-tone (warm highlights, cool shadows) with Color Balance / Curves
- ◆ Soft-proof for a printer and read the gamut warning
- ◆ Deliver a graded image as both a web file and a print-ready file

◆ TIMING

4m	The goal (portrait slider) + mental model: split-tone by zone
4m	Matching two images: Color Sampler, Info panel, tones before colors
4m	Soft-proofing: Customize Proof Condition, Gamut Warning (neon demo)
34m	Guided build: Steps 1–4 (10 + 10 + 8 + 6), the Module 2 deliverable
4m	Self-check, recipe card, journal (Now You is homework)

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (2m).** Drag the L06 slider. **Say:** "Cooler shadows, warmer highlights, a polished editorial feel. And this is the deliverable lesson: by the end you ship one look to two destinations."
- 2 Split-tone by zone (2m).** Sketch the ramp from Figure 1: cool on the dark end, neutral mids, warm on the bright end. **Say:** "It mimics warm sunlight and cool skylight, which is how we expect the world to look. Keep the mids roughly neutral so skin stays believable. Color Balance gives you explicit Shadows / Midtones / Highlights buttons; per-channel Curves do the same by which end you bend."
- 3 Matching (4m).** Put `l06-set-a-daylight.jpg` and `l06-set-b-tungsten.jpg` side by side. Pick the **Color Sampler Tool** (in the Eyedropper group, **Shift + I** cycles) and click the middle patch of the gray card in each. Open **Window ▶ Info**. **Say:** "The differences tell you which way each channel must move." Point at the rule: **match tones before colors**, then black point, white point, mid color. Mention **Image ▶ Adjustments ▶ Match Color** as the quick automatic pass. **Say:** "Hand-matching with Curves is more controllable, and it's what survives client scrutiny."
- 4 Soft-proofing (4m).** Open `l06-neon-proof.jpg`. **View ▶ Proof Setup ▶ Custom:** Device to Simulate **U.S. Web Coated (SWOP) v2**, Rendering Intent **Relative Colorimetric** (or Perceptual), **Black Point Compensation** on, **Simulate Paper Color** on. The image dulls. **Say:** "That's the paper." Toggle proof with **Ctrl/⌘ + Y**. Then **View ▶ Gamut Warning** (**Ctrl/⌘ + Shift + Y**): the neon grays out, the brick,

door and poster don't. **Say:** "Neon dies on paper. Pull its saturation down until the warning clears, and the print will match the proof."

- 5 **Set-up (in Step 1's time).** Open `grade-working.psd` from L05 and **Save As** `finish-working.psd`. If someone graded a landscape in L05, they drag their L05 grading layers onto the corrected portrait instead (select the layers, drag them onto the other document's tab).
- 6 **Step 1: split-tone it (10m).** Add a **Color Balance** layer (Preserve Luminosity ticked). **Shadows:** push toward cyan and blue a little. **Highlights:** toward red and yellow. **Midtones:** near neutral. **Say:** "Judge on skin. Warmly lit, not sunburnt. Ease off until it's believable."
- 7 **Step 2: match a companion (10m).** Open `l06-companion-portrait.jpg` (or their own second photo). Drop a Color Sampler on a comparable point in each (gray card middle patch, then the lit cheek). Add Curves to the companion; first the RGB curve for brightness and contrast, then each channel until its sampled values approach the reference's. **Say:** "Now they read as one shoot." (Stand-and-stretch minute here.)
- 8 **Step 3: soft-proof (8m).** Back on `finish-working.psd`: **View ▶ Proof Setup ▶ Custom** with their printer's profile (or SWOP v2), intent, BPC, Simulate Paper Color; proof on (Ctrl/⌘ + Y). Gamut Warning on. Anything gray gets a **masked Hue/Saturation** layer with Saturation pulled down until it clears. Learners whose portrait shows no warning repeat this on the neon file or their own vivid image.
- 9 **Step 4: deliver both files (6m).** **Save** the layered master first; it stays wide and 16-bit. **Web: Image ▶ Duplicate** (Duplicate Merged Layers Only), **Convert to Profile ▶ sRGB, 8 Bits/Channel**, export a JPG. **Print:** another merged duplicate, **Convert to Profile ▶** the printer/CMYK profile (or keep tagged RGB if the printer asks for it), save as PDF or TIFF. **Say:** "Module 2 deliverable: one look, delivered correctly for two destinations."

◆ LAB: SPLIT-TONE & PROOF (MODULE 2 DELIVERABLE)

Guided build on `finish-working.psd` (from `grade-working.psd`), companion match with `l06-companion-portrait.jpg` or an own second photo, soft-proof on the master (plus `l06-neon-proof.jpg` if the master shows no warnings). Solo variation is homework: match `l06-set-a/b/c` (or three own photos shot in different light) into one set; split-tone the same image cool/warm and green/magenta; soft-proof a deliberately vivid image and note how much saturation it surrenders; save the split-tone as a LUT. Details in `labs/module-2/README.md`.

Circulate for: Color Balance moves on the Shadows and Highlights radio buttons (not all on Midtones); Color Sampler readouts in the Info panel; proof actually on (the title bar shows the simulated profile); Hue/Saturation masked, not global; two merged duplicates, master untouched.

Definition of done: the lesson's five-item checklist: a believable split-tone; a second image matched to the set; soft-proofed with a deliberate intent and paper simulation, gamut warnings resolved; a web (sRGB) and a print (CMYK/printer-profile) file delivered; layered master kept wide + 16-bit.

◆ MISTAKES TO PRE-EMPT

- ◆ **All the Color Balance moves on Midtones.** That's the default radio button, and it tints the skin. Switch Tone to Shadows, then Highlights.
- ◆ **Sickly skin.** The split is reaching the mids. Ease the amount, keep it in the extremes, and mask the grade off the face if needed.
- ◆ **Matching color before tone.** The channels never settle because the exposures differ. RGB curve first.
- ◆ **Sampling different things.** A cheek in one and a backdrop in the other gives nonsense numbers. Comparable points only: the gray card patches are the safest.
- ◆ **Proof "does nothing".** Proof Colors is toggled off. When it's on, the document tab or title bar shows the simulated profile after the color mode; toggle with `Ctr L / ⌘ + Y`. A soft, low-saturation image may also change only slightly: try the neon file.
- ◆ **Converting the master to CMYK.** Then there is no wide RGB master left, and some adjustment layers behave differently or aren't available in CMYK. Always a merged duplicate.
- ◆ **Gamut warning left on while judging color.** The gray overlay hides the image. Toggle it off to judge, on to check.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** Cool shadows, warm highlights, with the mids kept roughly neutral; **Q2 D)** The color is reaching the midtones where skin lives: keep the split in the extremes, ease the amount, and mask the grade off the face if needed; **Q3 A)** Tones first (brightness and contrast), then black point, white point and mid color from the sampled values; **Q4 C)** The Info panel (Window ► Info); **Q5 B)** Lower the saturation there with a masked Hue/Saturation layer until the warning clears; **Q6 D)** Keep the layered master wide + 16-bit; from merged duplicates, Convert to sRGB (8-bit JPG) for web and to the printer's profile (PDF or TIFF, or tagged RGB if the printer asks) for print. Full set in the Answer Key.

Journal (5 min, in the wrap-up): *If you have ever printed a photo that disappointed you, what do you now suspect went out of gamut? How would soft-proofing have caught it?*

◆ DIFFERENTIATION

Fast finishers: the lesson's "Going further": save the split-tone as a LUT (Lesson 2.2 method) so it and the soft-proof workflow become a repeatable delivery pipeline. Or compare Relative Colorimetric and Perceptual on the neon file with proof on and note which one they'd hand a client, and why.

Struggling learners: split-tone with Color Balance (two radio buttons, small moves) and the web delivery first; that is a complete, valid result. Add the print duplicate with SWOP v2 as a stand-in. The companion match can move to homework with the recipe card.

CLOSING THE SESSION

MODULE 2 WRAP-UP (8 MIN, 2:52–3:00)

One volunteer puts their web JPG (in a browser) next to the soft-proofed master: the room should see the same look, with the print version slightly quieter. Recap the arc: **a trustworthy container → a grade you can bottle → a finish that survives screen and paper**. Exit ticket (on a sticky note): "At the end of a job I ____ to Profile, never ____." (Convert, Assign.) Homework from the lesson's "Before the Next Module": deliver one image as both a web and a print-ready file; match a small set of photos into one consistent look (the **106-set** frames count); finish the three solo variations started today (five histograms, three LUTs, the set of three); write the three journal entries. Preview Session 3 (Module 3, High-End Retouching): "Color mastered, we turn to people: frequency separation, dodge and burn, a natural finish. Bring a high-resolution portrait you have permission to retouch."

Module 3: High-End Retouching

How to teach the three lessons that make up a professional portrait pipeline: frequency separation at production depth (the correct Apply Image math for the bit depth, a radius chosen from pore size), dodge and burn on a 50% gray Soft Light layer that follows the bone structure, and the local, masked, restrained finish on eyes, teeth, and hair. One portrait runs through the whole session, from `skin-retouched.psd` to `portrait-sculpted.psd` to the finished Module 3 deliverable. The technique is the easy half; the judgment of where to stop is what you are really teaching.

3 lessons

3 hrs session

Lessons L07-L09

Level Intermediate

Maps to Session 3

HOW TO USE THIS GUIDE

- Same rhythm as every module: **Frame** → **Demo** → **Guided build** → **Now You** → **Check**. Every lesson ends with a recipe card, a checklist, and a journal prompt; use them as your closers.
- **Demo on the practice portrait, never on a learner's face.** The illustrated practice faces (`L07-` , `L08-` , `L09-` files) carry known flaws placed on purpose: blotchy tone, blemishes, pores, flat light, dull eyes, yellowed teeth, flyaways. Your demo always works, and nobody's appearance is discussed in front of the room.
- These learners have done a basic frequency separation and dodge and burn in the intro course. Skip the "what is a layer mask" material and spend the time on the **why** behind the settings: why the Apply Image numbers change with bit depth, why the radius follows the pore size, why the light map follows the bones.
- **Teal** = hands-on, **amber** = watch-outs. In Module 3 the watch-outs are **the wrong Apply Image recipe for the bit depth, working on the wrong layer** (Low vs High, gray layer vs photo), and **overdoing it** (poreless skin, chalky dodging, glowing eyes, toothpaste-white teeth).

CONSENT, RESPECT, AND HOW WE TALK ABOUT FACES

Portrait retouching is personal, and at this level it gets powerful enough to change how someone looks. Set three ground rules in the warm-up and hold them all session: **1)** Only edit portraits of yourself, of people who agreed to be edited *and* shown in class, or free-license portraits whose license allows it. **2)** Nobody's portrait goes on the projector without their say-so. **3)** In critique, **talk about the edit, never the person's looks**: "the dodge on the cheekbone is streaky," not "their skin is bad." The lesson's own position is the course position: technique is neutral, context sets the line, keep the person recognizably themselves, and be honest with clients about what was changed.

BEFORE SESSION 3

- ◆ ☐ Copy `labs/module-3/practice/` to each machine (or a shared drive): `l07-portrait-retouch.jpg`, `l07-beauty-closeup.jpg`, `l07-candid-distance.jpg`, `l08-portrait-clean-flat.jpg`, `l08-apple-flat.jpg`, `l09-portrait-sculpted.jpg`. Keep `labs/module-3/instructor/` for yourself: `l07-flaw-map-INSTRUCTOR.png` marks every blemish, tone patch, flyaway, and permanent feature on the practice face; `l08-light-map-INSTRUCTOR.png` shows where it wants dodging and burning. As always, never hand out the course site's licensed `images/` folder.
- ◆ ☐ Remind learners (in the Session 2 wrap-up or by email) to bring **one consented portrait** as a large original file, ideally with a flatly lit face, visible teeth, and some loose hair. For the L07 Now You, a second portrait at a different distance (a close beauty shot and a candid at arm's length) helps. Details in `labs/module-3/README.md`. Anyone who would rather not use a real face does the whole module on the practice files.
- ◆ ☐ Open a practice file and check **Image ► Mode**: it opens as **8 Bits/Channel**. The lesson recommends 16-bit and gives a different Apply Image recipe for it, so decide now which you demo. This guide demos 16-bit (convert with **Image ► Mode ► 16 Bits/Channel** right after opening) and shows the 8-bit numbers on the slide as the contrast case.
- ◆ ☐ Rehearse the L07 separation once on `l07-portrait-retouch.jpg` within the week. It is the one demo in the module where a single wrong field in Apply Image turns the room's screens into a gray mess.
- ◆ ☐ Open `lesson_07.html`, `lesson_08.html`, `lesson_09.html`, and `recipes.html` (the High-End Retouching cards) in browser tabs for the before/after sliders and the recipe cards.

SESSION 3 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	NOTES
0:00–0:12	Warm-up: Module 2 recap, ground rules	12	"What does 16-bit buy you?" (today it changes the Apply Image numbers). Consent and critique rules. Everyone opens a portrait (own or practice), runs global corrections if needed, and checks Image ► Mode .
0:12–1:05	L07 (Lesson 3.1) Skin & Frequency Separation	53	Build: Low + High, tone on Low, blemishes on High, group opacity. Save <code>skin-retouched.psd</code> .
1:05–1:52	L08 (Lesson 3.2) Dodge & Burn Sculpting	47	Build: 50% gray Soft Light layer, burn, dodge, B&W check. Save <code>portrait-sculpted.psd</code> .
1:52–2:05	Break	13	Everyone saves first. Leave the L08 checklist on the projector for stragglers.

CLOCK	SEGMENT	MIN	NOTES
2:05–2:52	L09 (Lesson 3.3) Eyes, Teeth & Hair: Natural Finish	47	Build: masked Curves + High Pass eyes, Hue/Sat teeth, hair layer, flip-and-toggle review in pairs. The Module 3 deliverable.
2:52–3:00	Wrap-up & exit ticket	8	Recap the pipeline order, assign homework, preview Session 4.
Total		180	The pair review inside L09 doubles as a short critique; volunteers only for the projector.

The lesson pages estimate $55 + 50 + 55 = 160$ minutes for a solo learner; the room has 147 once the warm-up, break, and wrap are in. The room runs each lesson's frame, demo, and Guided Build together, and each **Now You** starts in class and finishes at home. If you run long, protect the L09 guided build and final review (it is the Module 3 deliverable) and the L07 checkpoint where the stack adds back to the original. Cut in this order: the L09 optional iris sharpening, the L08 Now You start, then the L07 Now You start. All three Now You variations are homework anyway.

MODULE AT A GLANCE

LESSON	TOPIC	TIME	LEARNERS LEAVE ABLE TO...
L07 (3.1)	Skin & Frequency Separation	53m	Build a correct Low/High pair for the file's bit depth, even tone on Low, heal blemishes on High, and keep real pores at 100%
L08 (3.2)	Dodge & Burn Sculpting	47m	Sculpt a face on a 50% gray Soft Light layer at low flow, following the existing light, checked in black and white
L09 (3.3)	Eyes, Teeth & Hair: Natural Finish	47m	Finish eyes, teeth, and hair with local, masked, gentle edits and judge the natural line with a mirrored view

The module is one story on one face, and the order is the workflow: **global corrections → frequency separation (clean) → dodge & burn (sculpt) → eyes/teeth/hair (finish)**. Cleaning after sculpting undoes your light; finishing before cleaning wastes effort. Say the order out loud at the start and at the end of the session.

L07

Skin & Frequency Separation (Lesson 3.1)

53 min

◆ OBJECTIVES

- ◆ Explain what frequency separation splits an image into (low = tone and color, high = texture) and why it matters
- ◆ Build a correct Low (blurred) and High (Apply Image, Linear Light) pair, with the Apply Image settings matched to the bit depth
- ◆ Even out blotchy skin on the Low layer without smearing pores
- ◆ Remove a blemish on the High layer without leaving a soft patch
- ◆ Know when frequency separation is the wrong tool, and dial it back so the skin stays human

◆ TIMING

5m	The goal (lesson slider) + mental model: two frequencies
6m	Building the separation: the finished stack and the bit-depth warning
10m	Guided build Step 1: make the two layers (demo + room)
10m	Guided build Step 2: even the tone on Low
8m	Guided build Step 3: remove blemishes on High
5m	Guided build Step 4: dial it back, check pores, save <code>skin-retouched.psd</code>
5m	Now You (start): the radius follows the pore size
4m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 **The goal (2m).** Show the L07 slider. **Say:** "Look at what did *not* change: the texture. Tone is calm, blemishes are gone, and every pore is still there. That restraint is the whole point of frequency separation."
- 2 **Two frequencies (3m).** Draw two sheets. **Low:** the slow, gradual changes (cheek to shadow, a patch of redness, an uneven tan). **High:** the fast, tiny changes (pores, fine hairs, the grain). **Say:** "A blur or a soft brush destroys both at once, which is why it looks like melted plastic. Split them, and you calm the tone on one layer and touch texture only on the other." Remind them this is the spiral from the intro course: same idea, production depth.

- 3 **The stack and the math (6m).** Put the finished stack on the board: **High Frequency** (Linear Light, texture only) over **Low Frequency** (Normal, Gaussian Blur) over the untouched Background. **Say:** "Apply Image subtracts the blurred Low layer itself, so the radius matches automatically; there is no second radius to enter. What has to match is the bit depth." Write both recipes side by side: **8-bit:** Blending Subtract, Scale 2, Offset 128. **16-bit:** Blending Add, **Invert** ticked, Scale 2, Offset 0. **Say:** "Use the 8-bit numbers on a 16-bit file and the stack will not add back to the original. Check the title bar before you open Apply Image." Mention the High Pass shortcut only as the approximate fallback: same radius as the blur, and only approximate.
- 4 **Step 1: make the layers (10m).** Open `107-portrait-retouch.jpg`, **Image ▶ Mode ▶ 16 Bits/Channel**. **Say:** "Global corrections first, always. Frequency separation is a local cleanup, not a substitute for exposure and white balance." Duplicate the layer twice (**Ctrl/⌘ + J**, twice). Name the lower copy `Low` and the upper `High`. Hide `High` (the recipe card's tip: otherwise you can't see the blur). Select `Low`: **Filter ▶ Blur ▶ Gaussian Blur**, zoom to 100% on a cheek, raise the radius slowly. **Say:** "Stop the moment the pores vanish but the features stay. On this file that's around 5 or 6 pixels. Write your number down." Show `High`, select it, **Image ▶ Apply Image: Layer Low, Channel RGB, Blending Add, Invert** ticked, Scale 2, Offset 0. Read each field aloud. OK, then set the blend mode to **Linear Light**. **Say:** "Checkpoint. It looks exactly like the original again. If yours doesn't, hands up now, before anyone paints."
- 5 **Step 2: even the tone on Low (10m).** Select `Low`. Healing Brush, big and soft, **Sample: Current Layer**; **Alt/⌥**-click calm skin and gently push the redness beside the nose toward its neighbors. Then show the alternative: Mixer Brush at low opacity with Sample All Layers off. Do the cheek redness and the muddy patch under one eye. Toggle `High`'s eye off and on. **Say:** "Because you're only on the color layer, texture never moves. The pores are exactly as sharp as before."
- 6 **Step 3: blemishes on High (8m).** Select `High`. Healing Brush (or Clone Stamp), **Sample: Current Layer**, **Alt/⌥**-click clean texture near a blemish and heal over it. Do two on the cheek. **Say:** "Only the texture changes; the tone underneath stays put, so there's no soft patch or color smear where the spot was." If a faint pink ghost remains, go back to Low and even it: that is the two-layer logic in action.
- 7 **Step 4: dial it back (5m).** Select both layers, group them (**Ctrl/⌘ + G**), lower the group's opacity to about 80%. Zoom to 100% and pan across the face. **Say:** "Real pores everywhere? Good. If it looks poreless, you smoothed too hard. Letting a little of the original through is what keeps it human." **File ▶ Save As →** `skin-retouched.psd`, layers intact: **Say:** "Lesson 3.2 opens this exact file."
- 8 **Now You (5m, finish at home).** Hand out `107-beauty-closeup.jpg` and `107-candid-distance.jpg` (or learners' own two portraits). Each learner finds the radius where the pores just vanish on both and writes both numbers in the workbook. **Say:** "The close shot needs a bigger radius because its pores are bigger in pixels, and it needs far more restraint."

Guided build on the learner's own consented portrait (after global corrections) or `107-portrait-retouch.jpg`, following the lesson's four steps and saving `skin-retouched.psd`. Now You on `107-beauty-closeup.jpg` + `107-candid-distance.jpg` or two of their own portraits; the over-smooth-and-rescue, the plain-soft-brush comparison, and the Action recording are homework. Details in `labs/module-3/README.md`.

Circulate for: the checkpoint (the stack looks identical to the original before any painting); the title bar's bit depth matching the recipe used; `Low` below `High`; the right layer highlighted for each fix; **Sample: Current Layer** on the Healing Brush.

Definition of done: the lesson's six-item checklist: a correct Low + High (Linear Light) separation with Apply Image matched to the bit depth; tone evened on Low only; blemishes removed on High only; group opacity pulled back; real pores at 100%; `skin-retouched.psd` saved with layers.

◆ MISTAKES TO PRE-EMPT

- ◆ **Gray mess or a shifted image after Apply Image.** Almost always the wrong recipe for the bit depth (the lesson FAQ's number one), then Linear Light not set, then **Layer** in Apply Image not set to `Low`. Check the title bar: 8-bit = Subtract / Scale 2 / Offset 128; 16-bit = Add / Invert / Scale 2 / Offset 0. Fastest fix: delete `High`, duplicate the original again, redo Apply Image.
- ◆ **High shows eyes, lips, and face shapes.** The blur radius was too large, so real features moved into the texture layer. Rebuild with a smaller radius. A High layer that is almost blank means the opposite: the radius is too small and the pores stayed on Low.
- ◆ **Can't see the blur.** `High` is still visible on top. Hide it while blurring `Low`, then show it for Apply Image.
- ◆ **Healing on the wrong layer, or with the wrong Sample.** Heal on High with Sample: All Layers (or Current & Below) drags color onto the texture layer and leaves a smear. Mantra: **tone on Low, texture on High, each with Sample: Current Layer.**
- ◆ **Poreless, plastic skin.** Heavy strokes on Low that flatten the face's shading, or a group left at 100%. Lower the group opacity to 70–85% and look at 100% zoom, not fit-to-screen.
- ◆ **Skipping global corrections.** Trying to fix an overall color cast on Low, patch by patch. Fix exposure and white balance first (Module 2), then separate.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** Tone and color: the gradual shifts, patches of redness, and shading; **Q2 D)** Just until the pores vanish but the features stay, and note the radius; **Q3 A)** Blending Add, Invert ticked, Scale 2, Offset 0; **Q4 C)** On Low, with a big soft Healing Brush (Sample: Current Layer) or a low-opacity Mixer Brush; **Q5 A)** The blur radius on Low was too large, so real features moved into the texture layer; **Q6 D)** The close beauty shot, because its pores are bigger in pixels. Full explanations in the Answer Key.

Journal (homework): *Look at a magazine cover and a phone-beauty-filter selfie. Both smooth skin; describe, in frequency terms, why one reads as expensive and the other as fake.*

◆ DIFFERENTIATION

Fast finishers: the rest of the Now You: over-smooth one face on purpose, then rescue it with group opacity; do the same fix with a plain soft brush and no separation and compare side by side; then the lesson's "going further": record the whole layer setup as an Action (Module 6 automates this properly). Stretch: build the 8-bit recipe on an 8-bit copy and the 16-bit recipe on a 16-bit copy and confirm both add back perfectly.

Struggling learners: stay on the practice file (the radius is known: about 5–6 px) and 8-bit, which is one fewer checkbox. Pair them with a neighbor who is past the checkpoint. Success = the checkpoint plus one evened patch on Low; the High fix and the group opacity are the next rung.

◆ OBJECTIVES

- ◆ Explain why dodge and burn adds form that smoothing can't
- ◆ Build a non-destructive 50% gray Soft Light dodge and burn layer
- ◆ Work at low flow to build light gradually instead of streaking it on
- ◆ Sculpt the planes of a face: cheekbones, jaw, nose bridge, brow
- ◆ Check the work in black and white to judge form without color

◆ TIMING

5m	The goal (flat vs sculpted sphere slider) + painting with light
5m	The face map + the 50% gray layer and the B&W check
3m	Guided build Step 1: add the gray layer
10m	Guided build Step 2: burn the recesses
10m	Guided build Step 3: dodge the highlights
6m	Guided build Step 4: judge in B&W, temper, save <code>portrait-sculpted.psd</code>
4m	Now You (start): three lighting moods or a non-face object
4m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (2m).** Show the L08 slider: a flat sphere and a sculpted one. **Say:** "A little burn in the core shadow and a dab of dodge on the highlight, and you get obvious volume. Even, clean skin from L07 can still look flat. This is how we give the face its third dimension back."
- 2 Painting with light (3m).** **Say:** "A camera flattens a face; our brains read depth from where light falls. Dodging lightens and pulls a plane toward the viewer. Burning darkens and pushes it back. We exaggerate the light the lens already captured." Then the professional habit: **low and slow.** **Say:** "You don't lighten a cheekbone in one stroke. Many passes, each a whisper. Streaky dodge and burn is just impatient dodge and burn."
- 3 The map and the gray layer (5m).** Draw the face map: **dodge** forehead, cheekbones, nose (bridge and tip), brow bone; **burn** eye sockets, sides of the nose, under the cheekbones, under the jaw. **Say:** "Follow the bone structure the light already implies. Don't invent a light source." Then why the gray

layer: the Dodge and Burn tools edit pixels directly and are hard to undo; a 50% gray layer in Soft Light is invisible until you paint on it, and every stroke stays adjustable, reversible, and maskable. Finish with the check: **Say:** "Color hides uneven sculpting. A temporary Black & White layer on top shows every lump and streak."

- 4 **Step 1: the gray layer (3m).** Open `skin-retouched.psd` (or `l08-portrait-clean-flat.jpg`). Select the top of the stack; with the L07 group collapsed, click the group itself so the new layer lands above it, not inside it. **Layer ▶ New ▶ Layer, Mode: Soft Light**, tick **Fill with Soft-Light-neutral color (50% gray)**, name it `Dodge & Burn`. Brush (B): soft round, **Hardness 0, Flow 5–10%, Opacity 100%**. D for default colors. **Say:** "Flow, not opacity, is what we keep low today. Each pass adds a little, and passes build up."
- 5 **Step 2: burn the recesses (10m).** Black foreground (X swaps). Build shadow gently in the eye sockets, along the sides of the nose, under the cheekbones, under the jaw. Point out the practice file's window light comes from the viewer's left. **Say:** "Burn the right side a touch more than the left: we're amplifying the light that's there." Let the room work in parallel for most of these minutes.
- 6 **Step 3: dodge the highlights (10m).** X for white. Lift the tops of the cheekbones, the bridge and tip of the nose, the center of the forehead, the brow bone. **Say:** "Amplifying, not inventing. The left cheekbone gets more than the right." **Alt/⌘**-click the gray layer's eye to show your light map alone, then again to return.
- 7 **Step 4: judge and temper (6m).** Add a temporary **Black & White** adjustment layer on top. Toggle it: **Say:** "See that lump under the left cheekbone? Color hid it." Fix it with a few passes, turn the B&W layer off, and lower the gray layer's opacity until the sculpting is felt but not seen. Delete the temporary Black & White layer, **File ▶ Save As →** `portrait-sculpted.psd`. **Say:** "Lesson 3.3 finishes this file."
- 8 **Now You (4m, finish at home).** Two choices to start: sculpt the same face into three moods (soft beauty, dramatic side light, gentle top light), or sculpt `l08-apple-flat.jpg`. **Say:** "Product retouchers live on dodge and burn. The principle is identical."

◆ LAB: SCULPT THE FACE, THEN NOW YOU

Guided build on `skin-retouched.psd` from L07 (or `l08-portrait-clean-flat.jpg` for anyone whose L07 file isn't ready). Learners first write the light direction and their three dodge and three burn zones in the workbook, then follow the four steps and save `portrait-sculpted.psd`. Now You on the same face or `l08-apple-flat.jpg`; the over-dodge-and-rescue and the coarse/fine two-layer split are homework.

Circulate for: Soft Light mode *and* the 50% gray fill; the gray layer highlighted while painting (not the photo, not a layer inside the L07 group); Flow 5–10% in the options bar; strokes that follow the existing light; a B&W check actually done.

Definition of done: the lesson's six-item checklist: a 50% gray Soft Light layer; recesses burned and highlights dodged, following the real light; built at low flow with no visible streaks or edges; checked in black and white and corrected; layer opacity tuned so the effect is subtle; saved as `portrait-sculpted.psd`.

◆ MISTAKES TO PRE-EMPT

- ◆ **The gray layer lands inside the L07 group.** The top layer of the expanded group was selected, so the new layer inherits the group's opacity. Drag it above the group, or collapse the group and select it before adding the layer.
- ◆ **Solid gray over the photo.** The mode stayed on Normal. Set the layer to Soft Light. (A transparent layer instead means the fill checkbox was skipped; it only appears after choosing Soft Light in the dialog.)
- ◆ **Streaks and blotches.** Flow at 100% or a hard brush. Undo, set Hardness 0 and Flow 5–10%, and brush over the plane many times.
- ◆ **Painting on the photo.** White and black paint appears directly on the face. Undo, select the gray layer.
- ◆ **Inventing light.** Dodging the shadow side "because it looks dark," or a symmetric map on a face lit from one side. Follow the light that's already there.
- ◆ **Chalky, overdone sculpting.** If you can point to it, it's too strong. Lower the layer opacity; that is the finishing move, not an admission of failure.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** C) Darkens (burns) that area and pushes that plane back; **Q2** A) Soft round, Hardness 0, Flow 5–10%, Opacity 100%; **Q3** D) The eye sockets, the sides of the nose, under the cheekbones, and under the jaw; **Q4** B) Color hides uneven sculpting; without it, lumps and streaks in the light become obvious; **Q5** C) Lower the gray layer's opacity until the sculpting is felt but not seen; **Q6** A) It's non-destructive and maskable, and you can dial the whole effect up or down afterward. Full explanations in the Answer Key.

Journal (homework): *Study a portrait you admire and, without touching the file, describe where its retoucher likely dodged and where they burned. Training your eye to see the light is half the skill.*

◆ DIFFERENTIATION

Fast finishers: the lesson's "going further": split the work into two gray layers, one coarse (big global light shaping) and one fine (a catch-light, individual lashes). Then over-dodge on purpose until it goes chalky and rescue it with opacity and a mask on the gray layer.

Struggling learners: three zones done well: burn under the cheekbones and under the jaw, dodge the nose bridge. Put `108-Light-map-INSTRUCTOR.png` on your screen for them (not the projector) if they are stuck on where to paint. Skip the three-moods variation.

◆ OBJECTIVES

- ◆ Brighten and sharpen eyes with a masked local edit
- ◆ Whiten teeth by removing yellow first, then brightening, not the reverse
- ◆ Tame flyaway hair and fill gaps without a helmet-hair look
- ◆ Apply every finish through a mask so nothing bleeds where it shouldn't
- ◆ Judge the natural line: the point where more becomes worse

◆ TIMING

4m	The goal (the last five percent) + local, masked, restrained
11m	Guided build Step 1: eyes (Curves + optional High Pass)
9m	Guided build Step 2: teeth (Hue/Saturation)
9m	Guided build Step 3: hair (empty layer, Clone/Heal)
8m	Guided build Step 4: final review, flip and toggle, in pairs
4m	Self-check, recipe card, journal
2m	Now You briefing (homework): three portraits, timed

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The last five percent (2m).** Show the L09 slider. **Say:** "The skin is untouched in this comparison. Only the eyes and teeth changed, and it's enough to make the face read as more awake. Done with restraint it lifts a portrait; overdone, it screams 'edited.' This lesson is about knowing exactly where to stop."
- 2 Local, masked, restrained (2m).** **Say:** "Every finish has one shape: make an adjustment, hide it behind a black mask, paint it back only where it belongs." Then the warning box: real eye whites are faintly gray or pink, real teeth faintly warm. **Say:** "Aim for clean, never glowing. Back off until you doubt you did anything, then stop there."
- 3 Step 1: eyes (11m).** Open `portrait-sculpted.psd` (or `l09-portrait-sculpted.jpg`). Add a **Curves** adjustment layer, lift the midtones a little. Invert the mask to black (**Ctrl/⌘ + I**), then paint white with a small soft brush over the iris and the whites. Then the optional sharpen: select the top layer and make a merged copy of everything below (**Ctrl + Alt + Shift + E/⌘ + ⌘ + ⬆ + E**), **Filter ▶ Other ▶ High Pass** at about 1–3 px, set the layer to **Overlay** or **Soft Light** so the gray drops

out. **Alt/⌘**-click the Add Layer Mask button for a black mask and paint white over the iris only. **Say:** "Keep the catchlight sacred. Never paint it out; it's the life in the eye."

- 4 **Step 2: teeth (9m).** Add a **Hue/Saturation** layer. Choose **Yellows**, drop the saturation; then switch to **Master** and raise the lightness slightly. **Say:** "Yellow out first, brightness second. Brighten a yellow tooth and you get a bright yellow tooth." Invert the mask to black (**Ctrl/⌘ + I**) and paint the effect onto the teeth only, feathering the gum line so there's no hard edge. On the practice file, the gums sit right above the teeth: zoom in.
- 5 **Step 3: hair (9m).** Add an empty layer on top. Clone Stamp or Healing Brush, **Sample: All Layers**. Remove two flyaways against the background with short strokes that follow the hair direction. Then fill the thin gap in the hair on the viewer's right by cloning nearby strands along their natural flow. **Say:** "Leave a few stray hairs. Perfectly smooth hair looks like a wig." If anyone's hair layer ended up *below* the Curves or Hue/Sat layers, their clone samples the adjustments too; the fix is to keep it on top (or use the options bar's Ignore Adjustment Layers toggle).
- 6 **Step 4: final review (8m, pairs).** Zoom out to fit. **View ▶ Flip Horizontal**. **Say:** "A mirrored view exposes anything overdone. It only flips the view, not the pixels, so choose it again to come back. **Image ▶ Image Rotation ▶ Flip Canvas Horizontal** also works, but it flips the real pixels, so you must flip back before saving." Then toggle all Module 3 work off and on. Pairs swap seats for two minutes each and answer one question aloud: "**Does the after look like a good day, or like a retouch?**" Critique the edit, never the face. Save the finished layered file.
- 7 **Now You (2m briefing).** Homework: finish three portraits back to back and time it; make a deliberately over-finished version beside the restrained one; finish one shot for a magazine and for a LinkedIn headshot. "Going further": save the three finishing layers as a template group.

◆ LAB: FINISH THE PORTRAIT (THE MODULE 3 DELIVERABLE)

On **portrait-sculpted.psd** (or **109-portrait-sculpted.jpg**): masked Curves on the eyes, optional High Pass sharpen masked to the iris, Hue/Sat on the teeth (Yellows down, then Master lightness up), an empty hair layer for flyaways and the gap, then the flip-and-toggle review in pairs. The lesson doesn't name the final file; have learners **File ▶ Save As** a new layered PSD (for example **portrait-finished.psd**) so **portrait-sculpted.psd** stays as a checkpoint.

Circulate for: black masks on the Curves and Hue/Sat layers (mask thumbnail, not a white one); Yellows chosen before the saturation drop; the catchlight untouched; the hair layer on top with Sample: All Layers; a few strays left.

Definition of done: the Module 3 checklist: even skin from 3.1; form restored with dodge and burn from 3.2; eyes brightened and sharpened through a mask; teeth de-yellowed then gently lifted, masked; flyaways tamed with a few strays left, on their own layer; passes the flip-and-mirror test and reads natural, not retouched.

◆ MISTAKES TO PRE-EMPT

- ◆ **The whole face got brighter.** The Curves mask is still white. Select the mask thumbnail and press **Ctrl/⌘ + I**, then paint white only on the eyes.

- ◆ **Bright yellow teeth, or blue-white teeth.** Brightened before removing yellow, or the Yellows saturation and Master lightness pushed too far. Yellow out first, lift gently, stop at "faintly warm."
- ◆ **Gray halo or crunchy iris.** The High Pass copy is on Normal (set Overlay or Soft Light), the radius is too large, or the mask covers the lids and lashes. Small radius, iris only.
- ◆ **Catchlight painted out.** Healing or cloning over the eye, or a mask that dims it. Undo; the catchlight stays.
- ◆ **Helmet hair.** Every stray cloned away, or long strokes that ignore the hair's flow. Short strokes along the strands; leave some strays.
- ◆ **Forgot to flip back.** They used Flip Canvas Horizontal and saved the mirrored pixels. Flip the canvas back and resave; next time use **View ► Flip Horizontal**.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 A)** Make an adjustment, hide it behind a black mask, and paint it back only where it belongs; **Q2 D)** Drop Yellows saturation first, then raise Master lightness slightly; **Q3 B)** Merged copy, High Pass at about 1–3 px, blend Overlay or Soft Light, black mask painted white over the iris only; **Q4 C)** On an empty layer, Clone/Heal with Sample: All Layers, short strokes along the hair's direction, leaving a few strays; **Q5 D)** View ► Flip Horizontal flips only the view; Flip Canvas flips the pixels, so you must flip back before saving; **Q6 B)** Global corrections → frequency separation → dodge & burn → eyes/teeth/hair. Full explanations in the Answer Key.

Journal (in class if time, else homework): *Where is your natural line? Describe a finish you have seen (an ad, a friend's photo) that crossed it, and name exactly which move gave it away.*

◆ DIFFERENTIATION

Fast finishers: the template group from "going further": group the three finishing layers, name it **Finish**, and drag it into **109-portrait-sculpted.jpg** or a second portrait, then repaint only the masks. Then the magazine vs LinkedIn pair: same face, two amounts.

Struggling learners: eyes (Curves only, no High Pass) and teeth. Those two finishes plus the flip test are a complete deliverable; the hair layer can come at home. Keep them on the practice file, where the flyaways are obvious and against a plain background.

CLOSING THE SESSION

MODULE 3 WRAP-UP (8 MIN, 2:52–3:00)

Everyone should leave with three layered files: `skin-retouched.psd` , `portrait-sculpted.psd` , and the finished portrait. Recap the pipeline, in order: **global corrections** → **frequency separation (clean)** → **dodge & burn (sculpt)** → **eyes/teeth/hair (finish)**. Exit ticket (on a sticky note): "Tone on ____, texture on ____, and I stop when ____." Homework from the lesson's "Before the Next Module": take one portrait through the complete 3.1 → 3.2 → 3.3 pipeline and keep the layered file; save the finishing layers as a reusable group; finish the three Now You variations started today; write the journal entries. Preview Session 4 (Module 4, Commercial Compositing): "You can now perfect a single subject. Next we combine them: matching perspective, light, shadow, and atmosphere so two photos shot on different days look like one impossible scene."

GROWTH-MINDSET MOMENT

Restraint is the hardest retouching skill, and it doesn't feel like progress while you're practicing it: learners spend ten minutes on a cheekbone and can barely see the change. Name that out loud: "If you doubt you did anything, toggle the layer. That gap between 'I can barely see it' and 'wow, the before looks flat' is exactly what clients pay for." Celebrate the room's first clean checkpoint in L07 and the first flip test that passes.

Module 4 — Commercial Compositing

How to teach the three lessons that turn a pasted element into an advertising-style key visual: geometry first (horizon, scale, lens), then light (relighting, contact and cast shadows, reflections), then the whole frame (atmosphere and one unifying grade). One file carries all three lessons, and the finished key visual is the module deliverable.

3 lessons

3 h session incl. break

Lessons L10-L12 (4.1-4.3)

Level Intermediate

Maps to Session 4

HOW TO USE THIS GUIDE

- ◆ Same rhythm as every module: **Frame** → **Demo** → **Guided Build** → **Check**. Each lesson has a guided build, a recipe card (also in **Recipes II ► Compositing**), a completion checklist, and a learning-journal prompt. Self-check quizzes live in the participant workbook; answers are here and in the Answer Key.
- ◆ These learners already finished the intro course: they have cut subjects out, clipped adjustments, painted contact shadows, and run one unifying grade. Module 4 does not re-teach those moves. It adds the **geometry** they skipped (horizon, vanishing points, lens), the **built** cast shadow (silhouette, transform, graduated blur), reflections, and depth by contrast. Spend demo time on the new parts.
- ◆ **Demo on the projector, and step back often.** Geometry mistakes are invisible at 300% and obvious at "fit on screen." Zoom in for shadow contact points and reflection alignment; zoom out for horizon and scale.
- ◆ **Teal** = hands-on, **amber** = watch-outs. Shortcuts are given as **Windows / Mac**, for example **Ct r l + T / ⌘ + T**. Say both out loud.
- ◆ Lesson codes: the kit uses the file numbers **L10-L12**; the course site calls them **Lessons 4.1-4.3**. Both appear below.

THE #1 RISK OF SESSION 4: ONE FILE, THREE LESSONS, AND GEOMETRY THAT CAN'T BE FIXED LATER

L10, L11 and L12 all build the **same layered file** (`m4-key-visual.psd` , named in L10 Step 4). An element at the wrong scale or with the wrong horizon in L10 cannot be rescued by L11's shadows or L12's grade: shadows get built around the wrong size and the grade only makes the error prettier. Two safety nets: (1) the synthetic plaza `l10-plaza-background.jpg` was rendered with a real camera model, horizon at exactly $y = 700$ px and one vanishing point at (1260, 700), so you can check anyone's placement in seconds; (2) keep the instructor-only `labs/module-4/instructor/l10-plaza-answer-overlay.jpg` open on your machine. It shows the horizon, the vanishing point, the shadow direction, and correctly sized ghost outlines of the practice figure at three distances. A learner who is lost at 0:50 can copy one of those placements and join L11 on time.

BEFORE SESSION 4

- ◆ ☐ Copy `labs/module-4/practice/` to each machine (or a shared drive). Keep `labs/module-4/instructor/` on your machine only. As always, never hand out the course site's licensed `images/` folder.
- ◆ ☐ Remind learners (Session 3 wrap-up or by email) to bring: **a background with clear perspective and visible shadows** (a street, a room, a tiled floor, a table shot from standing height); **a cleanly cut-out element** from their Module 1 masking work, as a layered PSD or transparent PNG (full-length person, or a product with flat faces); optionally **a glossy-surface photo** (wet street, marble floor, polished table) and **a deep scene** with near, mid and far elements. Details in `labs/module-4/README.md` .
- ◆ ☐ On every machine, check **Preferences ► Performance ► Use Graphics Processor** is on (Photoshop ► Settings ► Performance on a current Mac). Perspective Warp is grayed out without it, and restarting Photoshop mid-lesson costs five minutes.
- ◆ ☐ Check everyone is on a current Photoshop version, where Free Transform scales proportionally by default (holding **Shift** switches it *off*). Older habits will fight this all morning.
- ◆ ☐ Open `lesson_10.html`, `lesson_11.html`, and `lesson_12.html` in browser tabs for the before/after sliders, plus `recipes.html` scrolled to the Compositing cards.
- ◆ ☐ Open `l12-key-visual-reference.jpg` (instructor folder) to show at the start of L12 as "where we are heading," and the horizon drill answers (`l10-horizon-practice-answers.jpg`) for the L10 frame.

SESSION 4 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	WHAT HAPPENS
0:00–0:10	Warm-up: recap Module 3 + "what gives a composite away?"	10	One line on Module 3 (the 50% gray Soft Light dodge & burn layer returns today in L11). Show a deliberately bad composite; the room names every tell. Sort them on the board into geometry · light · whole frame : today's three lessons.

CLOCK	SEGMENT	MIN	WHAT HAPPENS
0:10–0:57	L10 Match Perspective, Scale & Lens (4.1)	47	Build: find the horizon, place and scale one element by the ground-contact rule, warp its planes. Save the layered PSD. Now You starts here, finishes at home.
0:57–1:47	L11 Light, Shadows & Reflections (4.2)	50	Build: read the light, relight with clipped D&B + color, contact shadow, built cast shadow, reflection if glossy. Save.
1:47–2:00	Break	13	Leave the plaza answer overlay on the projector; stragglers can check their horizon against it.
2:00–2:48	L12 Atmosphere & the Unifying Grade (4.3)	48	Build: depth by contrast, painted haze, one grade on top, vignette + grain, export. The Module 4 deliverable.
2:48–2:55	Mini-gallery: the checklist in order	7	Three or four volunteers show their key visual; the room runs the lesson's checklist aloud: horizon, scale, lens, light, shadow, contact shadow, depth, grade.
2:55–3:00	Wrap-up & exit ticket	5	Recap the pipeline, assign homework, preview Session 5 (Design Systems & Type).
Total		180	Take a one-minute stand-and-stretch between L10 and L11; it comes out of L11's frame, not its build.

The site estimates $55 + 60 + 55 = 170$ minutes for a solo learner; the room has **145** after warm-up, break, gallery and wrap. The room runs each lesson's frame, demo and Guided Build together and trims the build times you see in each Timing table; the **"Now You" solo variations start in class and finish at home**. If you run long, protect the **L12 build**: the exported key visual is the Module 4 deliverable. Cut the mini-gallery first (show one volunteer), then L11 Step 4 (reflection) for anyone whose ground isn't glossy, which the lesson itself marks "if glossy." L10 Step 4 (lens sanity check) can shrink to one question asked aloud.

MODULE AT A GLANCE

LESSON	TOPIC	ROOM TIME	LEARNERS LEAVE ABLE TO...
L10 (4.1)	Match Perspective, Scale & Lens	47m	Find the horizon from converging lines, place and scale an element by its ground contact, warp strong planes to the vanishing point
L11 (4.2)	Light, Shadows & Reflections	50m	Read and match the scene's light, build a tight contact shadow and a graduated cast shadow, fake a reflection on a glossy surface

LESSON	TOPIC	ROOM TIME	LEARNERS LEAVE ABLE TO...
L12 (4.3)	Atmosphere & the Unifying Grade	48m	Build depth by lowering contrast and adding haze, lay one grade over the whole stack, and deliver a finished key visual

One story in one file: **make it belong there → make it lit and touching → make it one photograph.** The Module 4 summary on the site says it in the same order: perspective first, light and shadow next, atmosphere and one grade last.

L10

Match Perspective, Scale & Lens (Lesson 4.1)

47 min

◆ OBJECTIVES

- ◆ Find a photo's horizon line and read the camera's eye level from it
- ◆ Match an element's eye level and vanishing points to the background
- ◆ Scale an element correctly using the ground-contact rule, not guesswork
- ◆ Recognize and match a lens: wide vs long, and the distortion each brings
- ◆ Use Perspective Warp and Free Transform to correct a mismatch

◆ TIMING

3m	The goal: it belongs there (lesson slider)
5m	Mental model: everyone shares a horizon (+ drill sheet, panel 1)
4m	Scale by the ground, not the eye
4m	Matching the lens; Perspective Warp's two modes
25m	Guided build: Steps 1–4 (6 + 8 + 8 + 3)
2m	Now You: start "three distances" (finish at home)
4m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (3m).** Show the L10 slider: the same figure on the same ground, matched and ignored. **Say:** "A composite lives or dies in the first ten seconds, and it dies on geometry. Before light, before color, the element has to agree with the scene's horizon, its vanishing points, and its lens. Light can't save a giant."
- 2 Everyone shares a horizon (5m).** **Say:** "Every photograph has a horizon at the camera's eye level, whether or not you can see it. It's the height the lens was at." Put `l10-horizon-practice-sheet.jpg` on screen and extend two edges of panel 1 (the corridor) with a marker or the Line tool until they meet. **Say:** "Receding parallel lines meet at a vanishing point, and that point sits on the horizon. Every element you add must share that one line." Flash panel 3 or 4 (two-point): both vanishing points fall off the panel, yet on the same horizontal line.
- 3 Scale by the ground (4m).** Open `l10-plaza-background.jpg`. **Say:** "For people of similar height on the same flat ground, the horizon crosses each of them at the same point on their body. This camera

was at standing eye level, so the horizon cuts every person at the eyes, near or far." Point at the red-jacketed man in front and the green figure far back: both have their eyes on the horizon. Then the callout: **Say:** "Feet on the ground line first. The contact point fixes distance *and* size."

- 4 **Matching the lens (4m).** **Say:** "A wide lens (say 24mm) exaggerates depth and bends verticals near the edges; a long lens (say 200mm) compresses depth and flattens features. Paste a telephoto headshot onto a wide street and it looks subtly wrong even with perspective matched." Show Figure 3: **Edit ► Perspective Warp** has **Layout** (draw a quad over each plane) and **Warp** (drag its pins). Then the warning: **Say:** "Warp fixes geometry, not optics. You can't add depth a long lens removed. If the lenses are wildly different, find a better source photo."
- 5 **Step 1: Find the horizon (6m).** On a new empty layer, draw along two receding edges with the **Line tool** (or the Pen tool) and extend them until they meet. **Say:** "Guides can only be horizontal or vertical, so they can't follow an angled edge; that's why we draw." Show rulers (**Ct r l + R / ⌘ + R**) and drag a horizontal guide through the meeting point. On the plaza the guide lands at y = 700. Confirm it crosses the people at eye height. **File ► Save As ►** `m4-key-visual.psd`.
- 6 **Step 2: Place and scale (8m).** **File ► Place Embedded** `l10-figure-cutout.png` (or open the learner's cutout and drag the layer in). **Say:** "Decide where the feet touch the tiles. That point is now fixed." Free Transform (**Ct r l + T / ⌘ + T**); move the reference point to the bottom-center handle so scaling pivots on the feet, then drag a corner until the horizon guide crosses the eyes. **Say:** "Proportional by default in current versions. Don't hold **Shift**: Shift now switches proportional scaling off." The figure lands small (about a fifth of its original size if the feet sit near y = 950). **Say:** "Much smaller than your gut wanted. That's normal."
- 7 **Step 3: Match the perspective (8m).** Place `l10-product-box-cutout.png`, a carton shot from a high angle. Right-click ► **Convert to Smart Object** so the warp stays editable. **Edit ► Perspective Warp:** in **Layout** draw a quad on its front face and one on its side; switch to **Warp**; drag the pins until the box's bottom edges follow the tile grout toward the vanishing point. **Say:** "If it's grayed out, Use Graphics Processor is off, or you have more than one layer selected." For the figure: **Say:** "Organic subjects rarely need a warp; a touch of Free Transform ► Distort to align the lean is usually enough."
- 8 **Step 4: Sanity-check the lens (3m).** Step back from the screen. **Say:** "Does it feel flatter or rounder than its surroundings? If so, a small amount of Lens Correction or a subtle warp nudges it." **File ► Save.** **Say:** "This file is the geometry base you light in 4.2 and grade in 4.3. Don't flatten it."
- 9 **Now You (2m).** Start the first variation: the same cut-out person at three distances down the plaza. **Say:** "If your scaling is right, all three look like the same person walking away." Finish at home.

◆ **LAB: PLACE ONE ELEMENT, THEN NOW YOU**

Guided build on `labs/module-4/practice/l10-plaza-background.jpg` with `l10-figure-cutout.png` (and `l10-product-box-cutout.png` for the Perspective Warp step), or a learner's own background and cutout. Solo variations: three distances; the deliberate wrong horizon (the high-angle box is

exactly "a shot taken from a ladder"); a product on a phone-shot table; going further with Filter ► Vanishing Point. Details in [labs/module-4/README.md](#).

Definition of done (the lesson's checklist): horizon found from the scene's own converging lines; element's ground-contact point placed before scaling; scaled so the horizon crosses it at the correct body height; strong planes warped toward the scene's vanishing point; lens character sanity-checked, no obvious flat/round mismatch.

Circulate for: a horizon guide placed by eye instead of from a meeting point; feet floating above the tiles (look for a gap between shoe and grout line); a figure whose eyes sit well above or below the guide; Perspective Warp run on a raster layer, which leaves no way back.

◆ MISTAKES TO PRE-EMPT

- ◆ **Guessing the horizon at the visible skyline.** The skyline here is hills; the horizon is the eye level where the lines meet. On most street photos they differ.
- ◆ **Scaling first, placing second.** The contact point decides the size. Fix the feet, then scale from the bottom-center reference point.
- ◆ **Holding Shift out of intro-course habit.** In current versions that *distorts* the element. If a learner's figure looks stretched, this is why.
- ◆ **Warping a person.** Perspective Warp is for planes (boxes, signs, buildings). A person needs correct scale and eye level, plus at most a small Distort.
- ◆ **Expecting Warp to fix the lens.** It changes geometry, not optics. A big lens mismatch means a different source photo.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** the camera's eye level, the height the lens was at; **Q2 D)** extend two receding edges to their vanishing point and draw a horizontal guide through it; **Q3 A)** set the ground contact first, then scale until the horizon crosses it at eye height; **Q4 C)** it switches proportional scaling off; **Q5 D)** turn on Use Graphics Processor in Preferences ► Performance; **Q6 B)** warp fixes geometry, not optics, so pick a source on a similar focal length. Full set in the Answer Key.

Journal (write now if time, otherwise homework): *Find a composite ad or movie poster that feels "off." Can you locate the horizon and spot the element whose eye level disagrees with the rest?*

◆ DIFFERENTIATION

Fast finishers: the "deliberately break it" variation: place the high-angle box unwarped next to a warped copy and describe why the first refuses to sit. Then the lesson's "Going further": **Filter ► Vanishing Point**, define the plaza floor as a plane, and paste a flat graphic so it conforms automatically (floor graphics, wall posters). The horizon drill sheet's panel 6 is homework for everyone, but fast finishers can do it now with their own photo.

Struggling learners: stay on the plaza and the figure; skip the box and Perspective Warp. Give them the two numbers: horizon at $y = 700$, and "eyes on the guide, feet on a grout line." If they are still lost at the end of the build, let them copy the $Z = 8$ m placement from the answer overlay so they arrive at L11 with a correct base.

◆ OBJECTIVES

- ◆ Read a scene's light: direction, quality (hard/soft), and color
- ◆ Relight an element with dodge & burn and masked color so it matches
- ◆ Build a believable cast shadow (transform, blur, gradient opacity)
- ◆ Add the tight contact shadow that actually glues feet to the floor
- ◆ Fake a convincing reflection on a glossy surface

◆ TIMING

2m	The goal: anchored vs floating (lesson slider)
5m	Mental model: read the light first (hard-sun vs overcast references)
4m	Building a cast shadow; shadows are colored, not black
3m	Contact shadows & reflections; the two-shadow habit
30m	Guided build: Steps 1–4 (9 + 5 + 10 + 6)
2m	Now You: start hard sun vs overcast (finish at home)
4m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (2m).** Show the L11 slider. **Say:** "Same object, same spot. No shadow and the brain reads a sticker. A contact shadow plus a cast shadow running away from the light, and it's standing on the floor."
- 2 Read the light first (5m).** Open `l11-light-reference-hard-sun.jpg` and `l11-light-reference-overcast.jpg` side by side. Ask the room the lesson's three questions for each: **Where** is it coming from (shadows point away from the source)? **How hard** is it (crisp edges = hard, direct sun; soft edges = overcast or diffused)? **What color** is it? **Say:** "Whatever you read, the element must obey. A mismatch here is the second most common composite tell, after scale." Point out on the hard-sun file that the shadow is crisp at the base and softer at the tip.
- 3 Building a cast shadow (4m).** Show Figure 2's stack: **Contact shadow** and **Cast shadow**, both on **Multiply**, under the subject. **Say:** "Multiply keeps the ground texture showing through, so the shadow darkens the tiles instead of painting a flat gray blob on them." Then the warning callout: **Say:**

"Shadows are colored, not black. Outdoors they lean cool and blue because the sky lights them, not the sun. Indoors they pick up the room's bounced color."

- 4 **Contact shadows & reflections (3m).** Say: "The contact shadow, or ambient occlusion, is the small, dark, barely-blurred smudge exactly where the object meets the ground. It's the single most convincing element in the whole composite." Write the two-shadow habit on the board: **cast shadow = where the light is · contact shadow = touching the world**. Mention reflections: glossy floors, water, and polished tables owe one.
- 5 **Step 1: Read and match the light (9m).** Open `m4-key-visual.psd`. Read the plaza: sun high on the right, hard, warm; every shadow falls to the left. The practice figure was lit from the *left* in cool light, so it disagrees. Select the figure layer, **Layer ▶ New ▶ Layer**, Mode **Soft Light**, tick **Fill with Soft-Light-neutral color (50% gray)**, tick **Use Previous Layer to Create Clipping Mask** (or clip it afterwards with `Ctrl+Alt+G / ⌘+⌘+G`). Say: "Same 50% gray layer as Lesson 3.2, now clipped to the element." Paint white at low flow on the right side, black on the left. Add a clipped **Color Balance** or **Photo Filter** pushing warm toward the scene's sun.
- 6 **Step 2: Contact shadow (5m).** New layer *under* the element, **Multiply**, named `Contact Shadow`. Small soft black brush at low flow; darken exactly where the shoes meet the tiles. Say: "Tight, and only lightly blurred."
- 7 **Step 3: Cast shadow (10m).** `Ctrl/⌘`-click the element's layer mask thumbnail (or the layer thumbnail if it has no mask, as with the transparent practice PNG) to load the silhouette. New layer below the element, **Edit ▶ Fill** with a very dark color, deselect (`Ctrl+D / ⌘+D`), set **Multiply**. Say: "Filling a new layer works even when the cut-out uses a mask or is a Smart Object, where Lock Transparent Pixels and a direct fill would not." Free Transform ▶ **Distort/Skew** it onto the ground, running left, the same way as the lamp posts' shadows. For blur that grows with distance: **Filter ▶ Blur Gallery ▶ Tilt-Shift or Field Blur** (sharp pin near the feet, strong blur at the far end), or blur a duplicate heavily and blend the two with a gradient mask. Then a gradient mask so it fades, and tint it cool.
- 8 **Step 4: Reflection, if glossy (6m).** The plaza is matte, so switch to `l11-glossy-floor-scene.jpg` to demo. Duplicate the element, **Edit ▶ Transform ▶ Flip Vertical**, align bases, lower opacity to about **25–40%**, add Motion Blur or Gaussian Blur that increases downward (Blur Gallery, or a blurred copy blended with a gradient mask), and mask a fade. Point to the pedestal and sphere reflections already in the file as the target. Say: "Get the base alignment perfect; a reflection that starts a few pixels off is instantly wrong." **File ▶ Save** the layered PSD.
- 9 **Now You (2m).** Start the hard-sun vs overcast variation: the same object in each light reference. Say: "Feel how completely the shadow's edge and depth have to change." Finish at home.

◆ LAB: LIGHT & ANCHOR, THEN NOW YOU

Learners continue their L10 file (`m4-key-visual.psd`) through Steps 1–3. Step 4 applies only if their ground is glossy; everyone else practices it on `labs/module-4/practice/l11-glossy-floor-scene.jpg`

with the product box or figure, at home if time is short. Solo variations use `l11-light-reference-hard-sun.jpg` and `l11-light-reference-overcast.jpg`. Details in `Labs/module-4/README.md`.

Definition of done (the lesson's checklist): element re-lit to match the scene's direction, hardness, and color; tight contact shadow at the base (Multiply); cast shadow running away from the light, blurred and faded with distance; shadows tinted (not pure black) to match the scene; reflection added if the surface is glossy.

Circulate for: shadow layers sitting *above* the element; a cast shadow pointing right (toward the sun); one uniform Gaussian Blur on the whole cast shadow; the D&B layer not clipped (the ↵ is missing and the plaza gets painted too); pure black shadows at 100% opacity.

◆ MISTAKES TO PRE-EMPT

- ◆ **Shadow direction by guesswork.** Read the scene's existing shadows first. On the plaza every lamp post and person shadow falls to the left, so yours must too.
- ◆ **Using the Drop Shadow layer style as is.** It sits at a fixed offset and never lies on the ground in perspective. As a starting point only: Layer ► Layer Style ► Create Layer splits it onto its own layer, which can then be distorted, blurred and masked.
- ◆ **Too dark.** Match the scene's real shadows, then usually pull back 5–10%. Real shadows carry bounced light.
- ◆ **Blurring the contact shadow like the cast shadow.** The contact shadow stays tight; only the cast shadow gets progressive blur.
- ◆ **Relighting with the Dodge and Burn tools on the element's pixels.** Use the clipped 50% gray Soft Light layer from Lesson 3.2 so it stays editable.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** C) a hard, direct source such as the sun; **Q2** A) the ground texture shows through, so it darkens the floor instead of painting a gray blob; **Q3** D) Ctrl/⌘-click the mask thumbnail, fill the selection on a new layer below, deselect, Multiply; **Q4** B) crisp and dark near the object, softer and paler toward the end; **Q5** A) cool/blue, because the sky lights it; **Q6** C) duplicate, Flip Vertical, align bases, 25–40% opacity, blur increasing downward, mask a fade. Full set in the Answer Key.

Journal: *Look around the room and study one object's shadow. Where is the crisp contact shadow, where does the cast shadow soften, and what color is it? Retelling it in words trains your compositing eye.* Pair-share option: partners pick the same object and compare descriptions.

◆ DIFFERENTIATION

Fast finishers: the colored-source variation (let a sunset or neon color spill onto the element's lit edge; the glossy scene's magenta neon strip is a ready-made source) and the lesson's "Going further": a subtle **rim light**, a thin bright edge on a masked layer where the light wraps the element. Also: build the reflection on the glossy scene and nail the base alignment at 200%.

Struggling learners: contact shadow first, then the cast shadow without the progressive blur (one moderate Gaussian Blur plus a gradient mask). Skip the reflection. The lesson's own FAQ backs this up: a good contact shadow sells the grounding even when the cast shadow is subtle.

◆ OBJECTIVES

- ◆ Use atmospheric perspective (haze, fading contrast) to build depth
- ◆ Add fog, light rays, and dust that sit between elements, not just on top
- ◆ Lay a single unifying grade over the whole stack with one adjustment
- ◆ Understand why one grade over everything beats grading each piece
- ◆ Deliver a finished, cohesive advertising-style key visual

◆ TIMING

2m	The goal: one photograph (lesson slider + instructor reference)
4m	Mental model: distance has a look
4m	The unifying grade; where haze and grade sit in the stack
32m	Guided build: Steps 1–4 (7 + 9 + 8 + 8)
2m	Now You: start "one composite, three grades" (finish at home)
4m	Self-check, recipe card, journal, Module 4 recap

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (2m).** Show the L12 slider: a flat, neutral daytime scene vs the same scene under one grade (cooler shadows, warmer highlights, richer contrast). Then flash `l12-key-visual-reference.jpg`.
Say: "This is the deliverable. Perfectly placed and lit pieces can still look like separate photos sitting near each other. Atmosphere adds depth; one grade is the final handshake."
- 2 Distance has a look (4m).** **Say:** "The farther something is, the more air washes it out: contrast drops, colors desaturate and shift toward the sky's color, usually cooler and lighter, and edges soften. Mountains on the horizon are pale blue for this reason." Open `l12-far-clocktower-cutout.png` and drop it at the far end of the plaza at the right size. **Say:** "As crisp and saturated as the foreground. It feels pasted forward."
- 3 The unifying grade (4m).** Draw Figure 2's stack on the board, top to bottom: ★ **Unifying Grade** · **Subject + clipped light/color** · **Contact + cast shadows (Multiply)** · **Atmosphere/haze (over the far elements only)** · **Background**. **Say:** "The grade lives at the top so it rains down on everything. Unify last, grade once: get every piece neutral and correct first, then one dial." Name the options: a Color Lookup (LUT), a Gradient Map at low opacity, or a Curves + Color Balance group.

- 4 **Step 1: Set depth by contrast (7m).** Identify far, mid and near. On the far element (the clock tower layer), add a clipped Curves: lift the black point slightly and flatten the curve to lower contrast, and nudge it toward the sky color. **Say:** "Keep the nearest element the crispest and most saturated. The contrast falloff alone reads as distance." **Say:** "If your far elements are part of the background photo rather than their own layer, clipping to it would flatten the whole background. Use an unclipped Curves with a mask that covers only the distance."
- 5 **Step 2: Paint the atmosphere (9m).** New layer (Normal mode) above the far elements but below the subject and its shadows. Sample the pale sky near the horizon, soft brush, very low flow, glaze haze over the distance. **Say:** "Haze builds up in thin coats. If you can see a brushstroke, it's too strong." Optional: a Screen layer for light rays from the sun (upper right on the plaza), blurred and masked to shafts; a whisper of particles.
- 6 **Step 3: The unifying grade (8m).** At the very top: a **Color Lookup** LUT, or a Curves + Color Balance group (cool the shadows, warm the highlights, add gentle contrast). Toggle its eye. **Say:** "Watch every element lock together." Pull the opacity back until it's strong but not garish.
- 7 **Step 4: Finish and deliver (8m).** A subtle vignette to hold the eye, a touch of glow on the highlights (a merged copy, blurred, on Screen at low opacity), and grain to bind everything to one film stock. **File ► Save** the layered master. Then: flatten a copy (**Image ► Duplicate**, tick Duplicate Merged Layers Only), **Edit ► Convert to Profile ► sRGB**, and export (**File ► Export ► Export As ► JPG**). **Say:** "That's the Module 4 deliverable: one key visual that reads as one photograph."
- 8 **Now You (2m).** Start "one composite, three grades": warm blockbuster, cool thriller, faded indie. Finish at home, along with saving the grade as a LUT (Lesson 2.2).

◆ LAB: DEPTH & GRADE (MODULE 4 DELIVERABLE)

Learners finish `m4-key-visual.psd` with Steps 1–4 and export the key visual. On the plaza, the far element is `labs/module-4/practice/l12-far-clocktower-cutout.png` placed on the hills. The "prove the depth with contrast and haze alone" variation uses `l12-depth-bands-flat.jpg`, a five-band landscape with no atmosphere at all. Details in `labs/module-4/README.md`.

Definition of done (the lesson's checklist): element placed with correct perspective and scale (4.1); element lit, with contact and cast shadows (4.2); depth set by lowering contrast and adding haze on far elements; atmosphere (haze, optional rays/particles) between elements; one unifying grade over the whole stack; vignette + grain, exported as a finished key visual.

Circulate for: haze layers placed above the grade (they escape it and look pasted on); a grade clipped to the subject instead of sitting at the top; heavy fog over the foreground; learners who flattened the master instead of a copy.

◆ MISTAKES TO PRE-EMPT

- ◆ **Grading each piece separately.** They chase their tail forever. Neutral and correct first, then one grade on top.

- ◆ **Using the grade to hide a bad match.** The FAQ's order: horizon, scale, lens, light direction, shadow, contact shadow, depth/haze, grade. Ninety percent of "off" composites fail at scale or the missing contact shadow.
- ◆ **Haze on everything.** Atmosphere goes on the *far* elements. Haze over the subject flattens the thing the viewer should look at.
- ◆ **Forgetting the sRGB conversion.** A wide-gamut master exported without it looks dull on the web (Module 2 callback).

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** D) contrast drops, colors desaturate and shift toward the sky color, edges soften; **Q2** B) above the far elements, below the subject, its shadows and the grade; **Q3** C) one set of adjustments touches every element, so they share one color identity; **Q4** A) get every piece to neutral and correct; **Q5** B) a Color Lookup, a low-opacity Gradient Map, or a Curves + Color Balance group; **Q6** D) vignette, glow and grain, then flatten a copy, convert to sRGB, export. Full set in the Answer Key.

Journal: *Pick a film still you love. Describe its grade in words (where are the shadows pushed, where the highlights?) and where atmosphere creates its depth. Then try to rebuild that grade on one of your own images.*

◆ DIFFERENTIATION

Fast finishers: the deep-scene variation on [l12-depth-bands-flat.jpg](#) (prove depth with contrast and haze alone, before any grade), then save the grade as a LUT (File ► Export ► Color Lookup Tables, Lesson 2.2) and apply it to that landscape to feel a series identity. The lesson's "exam every senior compositor passes": composite into a scene shot at a completely different time of day, and use only atmosphere and grade to make them agree.

Struggling learners: Step 1 on the clock tower only, one haze layer, and one Color Lookup at reduced opacity; then save and export. Vignette, glow and grain are the last 10%. A grounded, graded key visual with those three moves is a legitimate deliverable.

MINI-GALLERY PROTOCOL (7 MIN)

- 1 Three or four volunteers put their exported key visual full-screen for about a minute each.
- 2 The room runs the lesson's diagnostic checklist aloud, in order: **horizon, scale, lens, light direction, shadow, contact shadow, depth/haze, grade**. Stop at the first item that fails; that is the fix.
- 3 The maker names one thing they are proud of and the next thing they would fix.
- 4 Keep feedback specific and kind: "the eyes sit on the horizon and the contact shadow sells it; the cast shadow could fade sooner" rather than "looks fake."

CLOSING THE SESSION

MODULE 4 WRAP-UP (5 MIN, 2:55–3:00)

Recap the pipeline: **share one horizon and place by ground contact → match the light and anchor with two shadows → push the distance back and lay one grade over everything.** Exit ticket (sticky note): "The contact shadow says ____; the cast shadow says ____." (Touching; where the light is.) Homework, from the lessons' "Before the Next Lesson/Module" lists: find the horizon in five photos (drill sheet panel 6 for one of them); study the contact and cast shadows of three real objects; finish the L10–L12 solo variations started in class; complete one composite through the full 4.1 → 4.2 → 4.3 pipeline and keep the layered file; save the unifying grade as a reusable LUT; write the three journal entries. Preview Session 5 (Module 5, Design Systems & Type): "You've made single images extraordinary. Next we produce at scale: type systems, Smart-Object templates, and a whole campaign from one master file. Keep your key visual handy; it makes a good test image for the templates."

Module 5: Design Systems & Type

How to teach the three lessons that turn one good design into a small production system: a type system built from a modular scale, named styles and a grid (L13), Smart-Object templates and linked assets that update themselves (L14), and a full campaign (poster, square, story, banner) from one master file (L15). This module opens the Commercial Work tier's production thinking: decide once, reuse everywhere.

3 lessons

3 hrs session incl. break

Lessons L13–L15

Level Intermediate

Maps to Session 5

HOW TO USE THIS GUIDE

- ◆ Same rhythm as every module: **Frame** → **Demo** → **Guided build** → **Check**. The lesson's "Now You" solo variation starts in class when there is time and **finishes at home**. Every lesson ends with a recipe card, a checklist and a journal prompt; use them as your closers.
- ◆ **One brand runs through the whole session.** The practice files follow "Northwind · Coastal Coffee Roasters," the example from Lesson 5.1. The type styles from L13 and the logo from L14 feed the L15 campaign, so learners should keep everything in **one project folder** from the first minute.
- ◆ **Name the mindset shift.** Modules 1–4 were about making one image excellent. Module 5 is about making decisions once and letting a system carry them: a style, a template, a linked file, a master. Say it out loud: "Today we stop editing things and start building things that edit themselves."
- ◆ **Teal** = hands-on, **amber** = watch-outs. In this module the watch-outs are mostly *structure* problems: a linked file that moved, replacement art at the wrong size, text that "won't snap," a master built small and scaled up.
- ◆ Shortcuts are given as **Windows / Mac**: Ctr⌘/⌘, Alt/⌘. Tool letters used today: Type T, Move / Artboard V, Rectangle U, Free Transform Ctr⌘/⌘+T, show grid Ctr⌘/⌘+ '.

THE #1 RISK OF SESSION 5: FILES THAT WANDER

Linked Smart Objects are the backbone of L14 and L15, and they break the moment a linked file is **moved or renamed**. Learners who save to the desktop, then "tidy up" into a folder, lose every link and lose the lesson's payoff. Start the session by having everyone create one project folder (suggested name `northwind-campaign/`) and copy `l14-logo-v1.png` into it as `logo.png`. **Say:** "Everything you make today lives in this folder, and nothing in it moves. The power of links comes with the discipline of not moving them." The second trap is Replace Contents with art of a different size: keep the practice designs at the placeholder's exact pixel size (the filenames say it: `...-1200x1800.png`).

BEFORE SESSION 5

- ◆ ☐ Copy `labs/module-5/practice/` to each machine (or a shared drive). Keep `labs/module-5/instructor/` (the two model results) for your screen only. Never hand out the course site's licensed `images/` folder.
- ◆ ☐ Remind learners (in the Session 4 wrap-up or by email) to bring: their **Module 4 composite** (the layered PSD, or a full-resolution export) as campaign key art; optionally their **own logo** as a transparent PNG; optionally a **photo of a surface to brand** (a mug, a poster on a wall, a laptop lid); and the **LUT** they saved in Module 4 if they want a shared grade. Details in `labs/module-5/README.md`.
- ◆ ☐ Check every machine is **signed in to Creative Cloud** so **Adobe Fonts** can activate fonts (L13 pairs a display face and a text face). Activate two families on the demo machine in advance in case the network is slow.
- ◆ ☐ Check the Photoshop version so menus match the lesson: in current versions the columns command is **View ► Guides ► New Guide Layout** (older versions: View ► New Guide Layout). Artboards, Place Linked and File ► Package are in every recent release.
- ◆ ☐ Set your demo machine's **Preferences ► Units & Rulers ► Type to Pixels** (L13 Step 1), and know where **Preferences ► Guides, Grid & Slices** is on both platforms (Edit ► Preferences on Windows, Photoshop ► Settings/Preferences on Mac).
- ◆ ☐ Open `lesson_13.html`, `lesson_14.html` and `lesson_15.html` in browser tabs for the before/after sliders, and have `l13-ransom-note-before.png` ready to project.

SESSION 5 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	NOTES
0:00–0:10	Warm-up: Module 4 recap + project folder	10	Two or three Module 4 composites on screen: "Next session this becomes a campaign." Everyone creates <code>northwind-campaign/</code> and copies the logo in.

CLOCK	SEGMENT	MIN	NOTES
0:10–0:57	L13 (Lesson 5.1) Type Systems, Styles & Grids	47	Build: a one-page layout driven by a scale, a grid and named styles.
0:57–1:42	L14 (Lesson 5.2) Smart-Object Templates & Linked Assets	45	Build: a reusable mockup template plus a linked-logo update.
1:42–1:55	Break	13	Leave the L14 checklist on the projector for stragglers.
1:55–2:55	L15 (Lesson 5.3) Campaign from One Master	60	Build: poster, square, story and banner artboards, exported in one pass (the Module 5 deliverable). Ends with a 4-minute "one edit" showcase.
2:55–3:00	Wrap-up & exit ticket	5	Recap the arc, assign homework, preview Session 6.
Total		180	No separate gallery: the "one edit updates everything" showcase inside L15 plays that role.

The lesson pages estimate $55 + 55 + 60 = 170$ minutes for a solo learner; the room has about 152 once warm-up, break and wrap are taken out. The room runs each lesson's frame, demo and Guided Build; the "Now You" solo variations start in class if there is time and finish at home. If you run long, **protect the L15 guided build and the one-edit proof** (it is the Module 5 deliverable): trim the L13 build to three roles (Display, H1, Body) and let learners add the rest at home, and in L14 skip the tote bag and build the poster-wall template only.

MODULE AT A GLANCE

LESSON	TOPIC	TIME	LEARNERS LEAVE ABLE TO...
L13 (5.1)	Type Systems, Styles & Grids	47m	Build a modular scale, create Paragraph and Character Styles, use real OpenType, and lay text on columns plus a document grid set to the body leading
L14 (5.2)	Smart-Object Templates & Linked Assets	45m	Build a warped Smart-Object mockup, swap art with Replace Contents, and update many placements from one linked file
L15 (5.3)	Campaign from One Master	60m	Hold four sizes on artboards, share art by Place Linked, recompose per aspect ratio, and export every artboard in one pass

One story: **decide the type once (styles) → decide the treatment once (templates) → decide the campaign once (a master with linked art)**. The through-line is the lesson mantra: make the decision once, name it, and let the system carry it everywhere.

L13

Type Systems, Styles & Grids (Lesson 5.1)

47 min

◆ OBJECTIVES

- ◆ Build a modular type scale and a clear hierarchy from it
- ◆ Create and apply Paragraph and Character Styles so type is consistent and editable
- ◆ Turn on real OpenType features: ligatures, true small caps, oldstyle figures
- ◆ Set up columns and a baseline grid (Photoshop's document grid, spaced at your leading) so text sits on a shared rhythm
- ◆ Pair typefaces from Adobe Fonts with intent, not by accident

◆ TIMING

3m	Frame: one voice, not twenty (ransom note vs system)
4m	Mental model: the modular scale and the four levers
4m	Demo: styles and real OpenType
4m	Demo: columns plus a document grid at the body leading
2m	Demo: edit one style, watch the page re-flow
26m	Guided build: a type system (Steps 1–4)
4m	Self-check, recipe card, journal prompt

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 Frame: one voice, not twenty (3m).** Project `L13-ransom-note-before.png`, then the lesson's slider. Ask: "Count the fonts. Count the alignments." **Say:** "Same words, completely different credibility. The ransom note isn't ugly because of any one choice. It's ugly because nothing agrees with anything. A type system replaces 'eyeball each label' with decisions you make once and reuse everywhere."
- 2 The modular scale (4m).** Write on the board: base 16 × ratio 1.25. Multiply up and down: **13 / 16 / 20 / 25 / 31 / 39**. Assign roles: Caption 13, Body 16, Subhead 20, H2 25, H1 31, Display 39. **Say:** "You never freehand a size again. The math guarantees any two sizes already agree." Then the four cheap levers: **size, weight, color, space**. **Say:** "Often a weight change or more space says 'this matters' more elegantly than making it bigger."

- 3 **Units first (part of Step 1).** Preferences ► Units & Rulers ► Type ► **Pixels**. **Say:** "Photoshop measures type in points by default. We'll work in pixels like the lesson; if you keep points, use the same numbers as pt."
- 4 **Styles and OpenType (4m).** Set one Body line: text face, 16 px, leading 24 px, tracking 0. Open the Character panel; show the **OpenType buttons** in the row at the bottom (also panel menu ► OpenType): Standard Ligatures, Oldstyle Figures, Small Caps. Type "fi fl" to show real ligatures. Then Window ► **Paragraph Styles:** with the line selected, create a new style from it and name it **Body**. Show Window ► **Character Styles** is a separate panel, for a word-level Emphasis or Link. **Say:** "A style bottles the whole spec: font, size, leading, tracking, color, OpenType. And never press Faux Bold or Faux Italic: they smear a regular weight. No real bold in the family? Pick another face."
- 5 **Columns plus a grid (4m).** View ► Guides ► **New Guide Layout** (older versions: View ► New Guide Layout): 3 or 6 columns, a gutter, margins. **Say:** "That's columns and rows, not a baseline grid. Photoshop has no baseline-grid feature like InDesign's, and text never snaps its baselines to a grid." Then Preferences ► Guides, Grid & Slices: **Gridline Every 24 px, Subdivisions 1**. Show it with View ► Show ► Grid (Ct r l / ⌘ + '). Zoom in and nudge a text layer with the arrow keys until its first baseline sits on a gridline. **Say:** "Leading equals the grid, so once the first baseline is on a line, every line after it lands on the grid too. Snap To ► Grid snaps a layer's edges, not its baselines, so check by eye."
- 6 **Prove the system (2m).** With a few styled blocks on the page, edit the **Body** style's size by one step. Everything re-flows. **Say:** "That propagation is the win. When the client says 'body a point bigger everywhere,' this is one edit, not forty."

◆ LAB: A TYPE SYSTEM (26 MIN)

Learners follow Steps 1–4 using the copy in `labs/module-5/practice/l13-layout-copy.txt` (Display, H1, Subhead, H2, Body, CTA, Caption). New document at a target size, two Adobe Fonts (one display, one text). The copy deck offers the lesson's numbers and, for an 1800 × 2400 px practice poster, the same scale ×3 (Body 48 on a 72 px grid). Save as a PSD (suggested: `northwind-campaign/type-system.psd`); its styles feed L15.

Definition of done (the lesson's checklist): a written modular scale with roles assigned; columns via New Guide Layout plus a document grid set to the body leading; Paragraph and Character Styles created and named; real OpenType on, no faux bold or italic; editing one style updates every instance.

Circulate for: text set by hand after the styles exist (the Paragraph Styles panel shows a "+" override next to the style name); a leading that isn't the grid increment or a multiple of it; Faux Bold toggled on; three or more families. Put `labs/module-5/instructor/l13-system-layout-INSTRUCTOR.png` on your screen as the model.

◆ MISTAKES TO PRE-EMPT

- ◆ **"My text won't snap to the grid."** It never will: Photoshop has no true baseline grid. Keep leading equal to the grid increment, align each block's first baseline by eye, zoomed in.

- ◆ **Local overrides.** Changing the size of styled text by hand breaks the link to the style. Clear the override (Paragraph Styles panel) and edit the style instead.
- ◆ **Points vs pixels confusion.** Units & Rulers ► Type still on Points while the plan says px. Change the preference or use the same numbers as pt, consistently.
- ◆ **Faux styling.** Faux Bold and Faux Italic distort letterforms. Real weight or a different face.
- ◆ **Typesetting a brochure.** A paragraph or two on a poster, yes. Long, text-heavy documents belong in InDesign.
- ◆ **The ransom note again.** Usually two families: one display, one text. Three is advanced; four is almost always the ransom note.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** C) 13 / 16 / 20 / 25 / 31 / 39; **Q2** A) edit the style once and every instance updates; **Q3** D) Preferences ► Guides, Grid & Slices, Gridline Every = body leading, Subdivisions 1; **Q4** B) Snap To Grid snaps layer edges, not baselines, so align the first baseline by eye; **Q5** C) use a face that has a real Bold, not Faux Bold; **Q6** A) columns and rows with gutters and margins, not a baseline grid. Full explanations in the Answer Key.

Journal (finish at home if needed): *Find two typefaces you think pair well and articulate why — contrast in structure, shared era, matching x-height? Naming the reason turns taste into a repeatable skill.*

◆ DIFFERENTIATION

Fast finishers (if time / homework): the lesson's solo variations: build the same layout with ratios 1.2 and 1.618 and compare the mood; restrict to one typeface and build hierarchy with weight and space only; rebuild an ugly flyer (or `l13-ransom-note-before.png`) on a scale and grid, keeping the content identical. Going further: audition a display + text pair in Adobe Fonts using the classification and property filters (x-height, contrast), then lock them into the styles.

Struggling learners: three roles only (Display, H1, Body), one grid, one column guide layout. The must-have is one named style edited once and seen updating everywhere. Hand them the recipe card and the copy deck's numbers so the math isn't a barrier.

◆ OBJECTIVES

- ◆ Explain what a Smart Object protects and why it is non-destructive
- ◆ Choose between embedded and linked Smart Objects for a job
- ◆ Build a mockup template with a warped Smart-Object placeholder
- ◆ Use Replace Contents and Place Linked to swap art in one move
- ◆ Update one linked file and have every design using it refresh

◆ TIMING

3m	Frame: change once, update everywhere (lesson slider)
3m	Mental model: a container you can refill; embedded vs linked
7m	Demo: placeholder, warp, Multiply/Screen, Replace Contents
3m	Demo: a linked logo updating in two places; Package
25m	Guided build: a reusable template (Steps 1–4)
4m	Self-check, recipe card, journal prompt

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 Frame (3m).** Lesson slider: one linked source feeding three deliverables vs three baked-in copies. **Say:** "A design you can only use once is a craft. A design that updates itself across ten files is a system. The linked model costs a little setup and repays it every time the art changes, which on real jobs is constantly."
- 2 A container you can refill (3m).** **Say:** "A Smart Object is a container. Scale it down and back up, it stays sharp. Filter it, the filter becomes a Smart Filter you can retune. Transforms, warps, filters and styles live on the container; Replace Contents swaps what's inside and keeps all of that." Then the catch: **Say:** "The container's transform is applied to whatever you swap in. Different pixel dimensions, different result. Build placeholders at the exact size of the art they'll hold." Embedded vs linked: embedded is stored inside the PSD, each copy independent; linked (File ► Place Linked) points at a file on disk that many documents share.
- 3 Placeholder and warp (Steps 1, 3m).** Open `l14-mockup-poster-wall.jpg`. Rectangle tool (U), click once on the canvas and enter **1200 × 1800 px**, the exact size of the practice designs. **Say:** "Exact size of the art, not the size it looks on the wall. You size it to the surface next." Right-click the layer ► **Convert**

to **Smart Object**. **Ctrl/⌘+T**, scale it down over the blank poster, then right-click ► **Warp** and drag the four corners onto the poster's corners (Perspective Warp also works).

- 4 **Light it (Step 2, 2m)**. Above the placeholder, a new layer set to **Multiply**; paint the poster's fold shadows and the window-bar shadow back over it (sample the dark from the photo). A new layer set to **Screen** for the lit fold ridges. **Alt/⌘-click** between each layer and the placeholder to **clip** it. **Say**: "Effects on the container and the layers around it, never baked into the art. The art stays a clean, swappable rectangle."
- 5 **Swap (Steps 3–4, 2m)**. Double-click the placeholder thumbnail: it opens in its own tab. Place **l14-art-a-sunrise-1200x1800.png**, save, close: it lands on the wall, warped and lit. Then **Layer ► Smart Objects ► Replace Contents ► l14-art-b-geometric-1200x1800.png**. Finally Replace Contents with **l14-art-d-landscape-1800x1200.png** and let the room see it come in wrong. **Say**: "Different shape, different result. For odd-sized art, double-click the placeholder and paste into its existing canvas instead." Undo.
- 6 **Linked logo (3m)**. New document (a quick "brand sheet"). **File ► Place Linked ► northwind-campaign/logo.png**; place it twice at different sizes. Point out the link badge in the Layers panel. Now update the source: open **logo.png**, change it (for example **Image ► Adjustments ► Hue/Saturation** on its single layer) and **File ► Save**; or, in the file manager, copy **l14-logo-v2.png** over **logo.png**. Back in the brand sheet, both placements refresh (if not: **Layer ► Smart Objects ► Update All Modified Content**). Then mention **File ► Package**: **Say**: "Package collects every linked file so nothing breaks on another machine. Move or rename a linked file and Photoshop loses it; the layer shows a warning, and Relink to File fixes it."

◆ LAB: A REUSABLE TEMPLATE (25 MIN)

Learners follow Steps 1–4 on **l14-mockup-poster-wall.jpg** (or their own surface photo; **l14-mockup-tote-bag.jpg** is the harder, curved alternative that genuinely needs Warp). Swap in at least two of the **...-1200x1800.png** designs, then prove the size trap with the landscape design. Save as a reusable template (suggested: **northwind-campaign/mockup-template.psd**) and make sure **logo.png** sits in the project folder for L15.

Definition of done (the lesson's checklist): the placeholder is a Smart Object, warped to the surface; shadows (Multiply) and highlights (Screen) on the container, clipped; art swaps via Replace Contents or double-click edit; shared art uses Place Linked where appropriate; saved as a reusable template, and Packaged if handing off.

Circulate for: a placeholder rasterized by accident (no Smart Object badge); shadow layers not clipped (they darken the wall too); a placeholder drawn at poster-on-screen size (about 600 × 900) so the art comes in at half size; the logo placed from Downloads instead of the project folder.

◆ MISTAKES TO PRE-EMPT

- ◆ **Treatment baked into the art**. Warping or shadowing the design itself means the next design needs it all again. Treat the container.

- ◆ **Replacement art at the wrong size.** The container's transform applies to new contents. Export art at the placeholder's exact pixel size, or paste into the existing canvas.
- ◆ **Moved or renamed linked files.** The link breaks. Keep one project folder, don't reshuffle, Package before handoff.
- ◆ **Embedded when linked was meant.** Place Embedded copies the art into the PSD; editing the original file changes nothing. Check for the link badge.
- ◆ **Expecting a PNG to keep layers.** If they edit `Logo.png` by adding layers, saving asks for another format. Change the single layer directly, or swap the file in the file manager.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** they stay on the container and apply to the new art; **Q2 D)** a linked Smart Object points to one file on disk that many documents share; **Q3 A)** the container's transform is applied to art with different pixel dimensions; **Q4 C)** on Multiply and Screen layers above the placeholder, clipped to it; **Q5 B)** File ► Package; **Q6 D)** the link breaks and the layer shows a warning (Relink to File). Full explanations in the Answer Key.

Journal: *Think of a repetitive design task you have done by hand. How would a Smart-Object template and a linked asset have collapsed it into a single edit?*

◆ DIFFERENTIATION

Fast finishers (if time / homework): the lesson's solo variations: a three-surface mockup (front, side, angled; try the wall and the tote together) fed by one shared linked file; a "brand sheet" PSD with a linked logo, updated from the source and seen refreshing on reopen; a favorite filter turned into a Smart Filter and retuned later. Going further: nest Smart Objects so a whole sub-design, with its L13 type system, becomes one swappable unit.

Struggling learners: poster wall only, and Free Transform corners rather than Warp if Warp is fighting them (the poster is flat). One Multiply shadow layer, clipped. The must-haves are a Smart-Object placeholder and one successful Replace Contents.

◆ OBJECTIVES

- ◆ Use artboards to hold multiple sizes in one document
- ◆ Share art across artboards with linked Smart Objects
- ◆ Adapt one design to portrait, square and landscape without redrawing it
- ◆ Export every artboard at once with Export As and presets
- ◆ Deliver a coherent, on-brand campaign set from one file

◆ TIMING

3m	Frame: one idea, every size (lesson slider)
4m	Mental model: master and instances; biggest first; one color mode
8m	Demo: artboards, Place Linked, styles, recompose, Export As
38m	Guided build: the campaign (Steps 1–4)
4m	Showcase: one source edit, every artboard updates, re-export
3m	Self-check, recipe card, journal prompt

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 Frame (3m).** Lesson slider: four drifting files (*poster_v3*, *insta_FINAL*, *banner_old*, *story_v2*) vs one master with four artboards. **Say:** "A real brief is never one image. It's a poster, a square, a story and a banner, all on-brand and due together. Consistency stops being willpower and becomes structure."
- 2 Master and instances (4m).** **Say:** "The master owns the shared truth: key art, logo, tagline, grade. Each artboard is a view onto it at a specific size and crop. Same idea as 5.1's styles and 5.2's templates, scaled up." Two rules on the board: **design at the biggest size, export down** (usually the print poster), and **every artboard shares one color mode and one resolution**. **Say:** "Keep the master RGB. If the printer wants CMYK, that's a separate, final step on an exported copy."
- 3 Artboards (Step 1, 2m).** File ► New, RGB, **Artboards** ticked. Build the poster at **3300 × 5100 px** (11 × 17 in at 300 ppi), then add IG square **1080 × 1080**, story **1080 × 1920** and banner **1500 × 500** with the Artboard tool (under the Move tool) using its options-bar presets where they exist and custom sizes otherwise. Save as `northwind-campaign/campaign-master.psd`.
- 4 Share the essentials (Step 2, 2m).** Select the poster artboard, File ► **Place Linked** the logo and `L15-key-art.png` (or a Module 4 composite). Repeat for each artboard (or duplicate the linked layer with

Ctrl/⌘+J and drag it onto the next artboard: a duplicate still points at the same file). Copy a styled text layer in from `type-system.psd`: its Paragraph Styles come with it. **Say:** "Placed by reference, not copied. It exists once and appears everywhere."

- 5 **Recompose, don't squash (Step 3, 2m).** On the banner: anchor the focal point first (key art left), then headline and CTA spread to the right. On the story: stack vertically and keep text inside the safe zone (drop `l15-safe-zones-story.png` on top as a temporary guide, then delete it). **Say:** "Every size clearly the same idea, not literally the same crop."
- 6 **Export the set (Step 4, 2m).** **File ▶ Export ▶ Export As:** every artboard is listed; pick format (PNG or JPG) and scale (1×; 2× if the platform wants it), Export. Mention **File ▶ Export ▶ Artboards to Files** as the batch alternative. Show the four files in the folder.

◆ LAB: THE CAMPAIGN (38 MIN), MODULE 5 DELIVERABLE

Learners follow Steps 1–4 using `labs/module-5/practice/l15-campaign-copy.txt` (sizes, copy, recomposition notes), the linked `logo.png` from L14, `l15-key-art.png` or their Module 4 composite, `l15-background-light.jpg`, and their L13 styles. Optional shared grade: the same adjustment layer at the top of each artboard, or the Module 4 LUT loaded into a Color Lookup layer on each (adjustment layers can't be linked, so each copy is updated by hand). Save the master PSD with its linked files in the same project folder.

Definition of done (the lesson's checklist): one master PSD with poster, square, story and banner artboards; logo and key art placed as linked Smart Objects; type driven by the 5.1 Paragraph/Character Styles; each size recomposed for its aspect ratio, safe zones respected; all artboards exported in one pass via Export As; a single source edit updates the whole set.

Circulate for: the poster artboard built small and scaled up; the key art upscaled past 100% on the poster (the practice PNG is 2400 px wide, so keep it at or below 100%); a banner that's the poster squashed; logo placed embedded on some artboards; text in the story's top or bottom band. Model result on your screen:

`labs/module-5/instructor/l15-campaign-reference-INSTRUCTOR.png`.

◆ SHOWCASE: ONE EDIT, EVERY ARTBOARD (4 MIN)

Two or three volunteers (or you) run the lesson's proof live: open the linked logo (or key art), make a visible change, save. Back in the master, every artboard updates (if one doesn't: Layer ▶ Smart Objects ▶ Update All Modified Content). Re-export with Export As. The room names what stayed identical across sizes and what was recomposed.

◆ MISTAKES TO PRE-EMPT

- ◆ **Small master, scaled up.** Scaling up for print softens everything. Biggest first, always.
- ◆ **Squashing instead of recomposing.** A 3:1 banner is not a shrunken poster. Anchor the focal point, rearrange the rest.
- ◆ **Expecting per-artboard color modes.** All artboards share one mode and one ppi. RGB master; CMYK for print is a separate final step.

- ◆ **Trying to link an adjustment layer.** It can't be linked. Duplicate it into each artboard or reload the same LUT; update each copy if the look changes.
- ◆ **Safe zones ignored.** Story UI overlays cover the top and bottom; banners crop at the edges. Keep text clear.
- ◆ **Moving the project folder mid-build.** Links break (see the module risk note).

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** A) the print poster, the largest; **Q2** C) Place Linked it into each artboard; **Q3** B) File ► Export ► Export As with all artboards; **Q4** D) anchor the focal point and spread the elements horizontally; **Q5** A) keep the master RGB and convert an exported copy to CMYK as a final step; **Q6** C) Layer ► Smart Objects ► Update All Modified Content. Full explanations in the Answer Key.

Journal: *Find a brand you admire and list every place its campaign appears (feed, story, print, web). What stays identical across them, and what is recomposed? That is the master-and-instances split in the wild.*

◆ DIFFERENTIATION

Fast finishers (if time / homework): the lesson's solo variations: two more sizes (a billboard and an email header, sized to the destination's spec) without recreating any shared element; a light and a dark variant of the whole set by swapping one linked background (`l15-background-light.jpg` → `l15-background-dark.jpg` with Relink to File; a Color Overlay layer style on the linked logo keeps it readable on dark); Package the project and have a "client" change the linked logo on their machine. Going further: pair it with data-driven graphics (Module 6) for localized versions.

Struggling learners: two artboards first (poster and square), both fed by the same linked logo and key art, exported with Export As. Add the story and banner once that works. The must-have is the one-edit proof: change the source, see both update.

CLOSING THE SESSION

MODULE 5 WRAP-UP (5 MIN, 2:55–3:00)

Recap the arc: **a named type voice (5.1) → self-updating templates (5.2) → a full campaign from a single master (5.3)**. Exit ticket (on a sticky note): "One thing I decided once today and will never set by hand again is ____." Homework from the lessons' "Before the Next..." lists: ship one campaign set (4+ sizes) from a single master file; prove a one-edit update propagates across every artboard; finish any solo variation started in class; write the three journal entries. Remind learners to keep `northwind-campaign/` intact (and to use File ► Package if they move it to another machine). Preview Session 6 (Module 6, Generative + Automation, Advanced): "You have systems for consistency; next you get systems for speed and volume: generative compositing, conditional Actions, and data-driven graphics that produce dozens of variations from one template."

Module 6: Generative + Automation, Advanced

How to teach the three lessons that add speed and scale to everything the course has built so far: generative tools as a fast, fallible compositing assistant (including Harmonize), conditional Actions batched over a mixed folder, and data-driven graphics from a template plus a CSV, with a first readable script. The theme of the day: **design the work once, then let the machine repeat it, and keep your judgment for the parts only you can do.**

3 lessons

Session 6 • 3 hrs incl. break

Lessons L16–L18

Level Intermediate

Needs internet + credits (L16)

HOW TO USE THIS GUIDE

- Same rhythm as every module: **Frame** → **Demo** → **Guided build** → **Solo variation** → **Check**. Each lesson plan gives a timing table, a demo script with **Say:** lines, the lab, the mistakes to pre-empt, a check for understanding, and differentiation notes.
- **Demo live, one step at a time**, and let the room do each step before you move on. Name every menu path and both shortcut forms: **Ct r l** on Windows is **⌘** on Mac, **Alt** is **⌥**.
- **Teal** = hands-on, **amber** = watch-outs. Today's watch-outs come in three flavors: **environmental** (sign-in, credits, network) for L16, **"recorded something absolute"** for L17, and **"the names don't match"** for L18.
- Lesson codes: the kit numbers lessons by file (**L16–L18**); the site calls them **Lessons 6.1–6.3**. Quiz answers below match the workbook's Self-checks; the Answer Key explains each one.

THE #1 RISK OF MODULE 6: THE GENERATIVE FEATURES ARE NOT AVAILABLE IN THE ROOM

Generative Fill, Generative Expand, and Harmonize run in Adobe's cloud. Each learner needs **Photoshop 2025 or later** (Harmonize is a recent addition, and its exact place in the interface can shift between releases), **an internet connection, to be signed in to an Adobe account**, and **generative credits** (every Generate spends them). Classroom networks and lab licenses are the usual failure points. **At least a day before**, on one room machine: sign in, generate once with Generative Fill, once with Generative Expand, and run Harmonize once on a placed layer. On the day, everyone signs in during the warm-up, not when L16 starts.

FALLBACK PLAN IF CREDITS, SIGN-IN, OR NETWORK FAIL

1. Reorder. L17 (Actions, conditionals, Batch, droplets) and L18 (Variables, Data Sets, scripting) are 100% offline. If the network is down at the start, teach L17 and L18 first and return to L16 when it recovers. **2. Share.** If only some accounts have credits, pair learners: one generates, both judge the variations, then swap for the Harmonize step. **3. Offline equivalents.** Run the generative demos from your machine (or a screen recording made during the readiness check) and have learners do the same jobs with tools they already own: **room** with the Crop tool dragged outward and **Fill: Content-Aware Fill; removal** with the Remove tool or Content-Aware Fill on a separate layer; **connective tissue** by hand (Clone Stamp for gravel, a painted contact shadow as in L11 Step 2); **Harmonize by hand** with the L11 method (a clipped 50% gray Dodge & Burn layer plus a clipped Color Balance or Photo Filter to push the subject's light toward the scene's), then the L12 unifying grade. The compositing judgment survives intact; only the "describe it in words" part is lost. **4. Homework.** Learners with credits at home repeat the L16 build there.

BEFORE SESSION 6

- ◆ ☐ Copy `labs/module-6/practice/` to each machine (or a shared drive). It holds the L16 composite parts, the L17 delivery folder `l17-input/` plus `l17-watermark.png`, the L18 data folder `l18-data/` (card parts, six photos, two CSVs), and `l18-resize-folder.jsx`. Keep `labs/module-6/instructor/` (the broken CSV and the reference card) on your machine only. As always, never hand out the course site's licensed `images/` folder.
- ◆ ☐ Remind learners (in the Session 5 wrap-up or by email) to bring: their **layered Module 4 composite** (or any busy, cramped photo); a folder of **8 to 12 photos, mixed landscape and portrait, as copies**; optionally their own watermark or logo PNG; and for L18 a **short real list** (team roster, event lineup) with photos they have permission to use. Details in `labs/module-6/README.md`.
- ◆ ☐ Check every machine is on **Photoshop 2025 or later** (Help ► About Photoshop on Windows, Photoshop ► About Photoshop on Mac), and run the generative readiness check above.
- ◆ ☐ On one room machine, confirm you can run a script (File ► Scripts ► Browse) and create and run a droplet (File ► Automate ► Create Droplet). Managed lab images sometimes block one or the other; better to find out now.
- ◆ ☐ Open `lesson_16.html`, `lesson_17.html`, `lesson_18.html`, and `recipes.html` (the "Generative, Automation & Multi-Shot" cards) in browser tabs for the sliders and the recipe cards.
- ◆ ☐ Prepare your own demo folder: a copy of `l17-input/`, an empty `l17-output/` beside it, and a second copy of `l17-input/` named `script-test/` for the L18 script (the script writes its `web_` files into the folder it reads from).

SESSION 6 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	NOTES
0:00–0:10	Warm-up: sign-in check + Module 5 recap	10	Everyone opens Photoshop and confirms they are signed in (fix problems now). Two volunteers show their Module 5 campaign set and prove a one-edit update. Frame the day: "assist, repeat, scale."
0:10–1:00	L16 Generative for Compositing	50	Build: give a cramped composite room, remove the sign, grow connective tissue, Harmonize, finish by hand. Now You starts in class, finishes at home.
1:00–1:45	L17 Advanced Actions & Conditionals	45	Build: a Web Delivery Action with a landscape/portrait conditional, batched over <code>L17-input/</code> , then a droplet.
1:45–1:58	Break	13	Leave the L17 checklist on the projector. Early finishers open <code>L18-data/</code> and look at <code>L18-cards.csv</code> in a text editor.
1:58–2:52	L18 Data-Driven Graphics & Scripting	54	Build: a card template bound to a CSV, exported as one file per row (the Module 6 deliverable); read and run the sample script.
2:52–3:00	Wrap-up & exit ticket	8	Recap the arc, assign homework, preview Session 7 (Multi-Shot + capstone kickoff).
Total		180	Generations take seconds each; fill that time by narrating what you are judging, not by starting the next step.

The lesson pages estimate $55 + 50 + 60 = 165$ minutes for a solo learner; the room has 149 minutes of lesson time after the warm-up, break, and wrap. The room runs each lesson's frame, demo, and Guided Build together, and each **"Now You" solo variation starts in class and finishes at home**. If you run long, protect the **L18 guided build and export** (it is the Module 6 deliverable): shorten L16 Step 3 to one connective-tissue pass, demo the L17 droplet rather than having everyone build one, and move the L18 script run to homework (it is also in the lesson's "Before the Next Module" list).

MODULE AT A GLANCE

LESSON	TOPIC	TIME	LEARNERS LEAVE ABLE TO...
L16 (6.1)	Generative for Compositing	50m	Give a composite room with Generative Expand, remove and connect with Generative Fill, Harmonize a placed subject, and finish the generated pixels by hand
L17 (6.2)	Advanced Actions & Conditionals	45m	Record a relative delivery Action, branch it by orientation with Insert Conditional, batch a mixed folder hands-off, and make a droplet

LESSON	TOPIC	TIME	LEARNERS LEAVE ABLE TO...
L18 (6.3)	Data-Driven Graphics & Scripting	54m	Bind Text, Pixel, and Visibility variables to CSV columns, proof and export one graphic per row, and read and adapt a simple ExtendScript

The module is one story: **let a model do the tedious pixels (6.1) → let an Action do the repeated steps (6.2) → let data do the repeated designs (6.3)**. Each lesson ends where the previous tool hits its wall: generation cannot repeat a workflow, Actions cannot read a spreadsheet, and Variables cannot loop or decide. Say that out loud at each transition.

L16

Generative for Compositing (Lesson 6.1)

50 min

◆ OBJECTIVES

- ◆ Frame generative fill as a compositing tool, not a replacement for craft
- ◆ Extend a background with **Generative Expand** to fit a layout
- ◆ Remove distractions and fill gaps with **Generative Fill** + a prompt
- ◆ Generate **connective elements** (shadow ground, foliage, debris) to sell a composite
- ◆ Use **Harmonize** to match a placed element's light and color to its background
- ◆ Judge, mask, and hand-finish generative output, and respect its limits and licensing

◆ TIMING

3m	The goal (lesson slider) + mental model: a fast, fallible assistant
3m	The generative workflow + Harmonize: where each command lives
3m	Limits, ethics & licensing; "it won't fix a weak composite"
34m	Guided build: Steps 1–5 (6 + 8 + 8 + 6 + 6)
3m	Now You: start "generate the same fill five times" (finish at home)
4m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (1m).** Show the L16 slider: cramped and signed on one side, extended and clean on the other.
Say: "Generative fill did the grunt work, new sky and ground and the sign gone, so the time goes into composition instead of cloning. Fill the gaps, not the vision."
- 2 Mental model (2m).** Write the loop on the board: **Select → prompt (short or empty) → 3 variations, pick the best → mask + finish by hand.** **Say:** "Treat it like a talented but literal intern. Superb at plausible texture, sky, grass, pavement, walls, and at removals. Unreliable at specifics: hands, text, exact brands, precise counts. It never ships raw."
- 3 Where things live (3m).** **Say:** "Make a selection and the contextual task bar offers Generative Fill. For Expand, it's the Crop tool with Fill set to Generative Expand in the options bar. Harmonize appears when a placed layer sits above a background: in the task bar or the layer's Properties panel. All three write to a new layer and cost credits." Show the Properties panel's variations list once.

- 4 **Limits and licensing (3m).** Faces, hands, readable text, logos, product details, and repeat patterns drift; generated regions can be softer than the capture; artifacts hide until 100%. **Say:** "Adobe designs its Firefly models to be commercially safe, but current versions can also offer third-party partner models with different terms. For client work, know which model you used and check the terms. And for journalism or documentary work, generative changes may be off-limits." Then the warning box: **Say:** "It fills texture; it doesn't reason about your scene. Wrong perspective, light, or scale from Module 4 is still yours to fix."
- 5 **Set up (part of Step 1).** Open `l16-cramped-scene.jpg`, **File ▶ Save As** `l16-composite-working.psd`. **File ▶ Place Embedded** `l16-pasted-subject.png` and drag it hard against the right edge with the feet near the bottom, as in `l16-cramped-composite-preview.jpg`. Name the layer `subject`. **Say:** "This is the awkward composite: framed too tight, a SALE sign fighting for attention, and a figure lit cold from the left in a warm scene lit from the right, with no shadow."
- 6 **Step 1: Give it room (6m).** Hide the `subject` layer (**Say:** "so the model continues the plaza, not the person"). Crop tool (C), options bar **Fill: Generative Expand**, drag the left and right edges outward (a modest amount), commit, leave the prompt empty, **Generate**. Cycle the variations in the Properties panel and keep the cleanest. Show the subject again. Zoom to 100% (Ctrl/⌘ + 1) and pan along both seams looking for repeats: the buildings and the wall's joints are the tell.
- 7 **Step 2: Remove distractions (8m).** Click the Background layer. Lasso (L) around the SALE sign and its post *and its cast shadow*, with a little margin. **Say:** "Leave the shadow behind and you've removed the sign but kept the evidence." Task bar ▶ Generative Fill, empty prompt, **Generate**. If a variation leaves a smudge, regenerate, or clean it with the Healing Brush on a new layer.
- 8 **Step 3: Grow connective tissue (8m).** Draw a flat, wide selection on the ground around and just left of the subject's feet (sun from the right means shadows fall left). Prompt `gravel with small shadows`, **Generate**. Paint the generative layer's mask so it only contributes near the feet. **Say:** "Short prompt, small selection. Ask for a whole scene and it flails." If no variation gives a believable contact shadow, paint one by hand on a layer below the subject, exactly as in L11.
- 9 **Step 4: Harmonize the subject (6m).** Select the `subject` layer, then **Harmonize** in the contextual task bar or the Properties panel. Compare the variations with the original by toggling the eye. Keep the best, then mask back to the original wherever Harmonize changed something it shouldn't: an edge, the face, the jacket's zip line. **Say:** "Same rule as every generative result: compare, keep, mask, finish."
- 10 **Step 5: Finish by hand (6m).** Add matching grain and a touch of sharpening to the generated regions; dodge & burn to marry their light to the scene. Then one unifying grade over everything (the Module 4 finish, L12). **Say:** "The single step that makes a generative fill invisible is matching the grain." Save the layered file with every generative layer and mask intact.

Guided build on the learner's own layered Module 4 composite, or on the practice parts (`l16-cramped-scene.jpg` + `l16-pasted-subject.png` , placed as in `l16-cramped-composite-preview.jpg`), saved as `l16-composite-working.psd` . Solo variation: generate the same fill five times and study the differences; remove three increasingly hard distractions from `l16-three-distractions.jpg` (a sign, a person, a hard shadow across the bicycle) and note where the AI succeeds and where they take over; turn a phone snapshot into a wide banner with Generative Expand. Start in class, finish at home. Details in `labs/module-6/README.md` .

Circulate for: every result on its own generative layer (nothing flattened); lasso selections with margin that include the sign's shadow; learners actually cycling variations; masks painted on the generative layers; someone zoomed to 100% on each seam.

Definition of done: the lesson's six-item checklist: canvas extended and edges audited; a distraction removed cleanly; a connective element generated and masked in; the subject harmonized (or light matched by hand) with unwanted changes masked out; generated regions grain- and sharpen-matched and hand-blended; one unifying grade over the whole result.

◆ MISTAKES TO PRE-EMPTY

- ◆ **"Generative Fill is grayed out" or "Harmonize isn't there."** Check sign-in, network, and version first. For Harmonize, the target must be a single pixel or Smart Object layer sitting above a background; a group, an adjustment layer, or the Background itself won't offer it. The lesson's FAQ: its location can shift between releases, so search Help ► Photoshop Help or the Discover panel. Two minutes of debugging, then the fallback plan.
- ◆ **Expanding with the subject visible.** The model tries to continue the person at the edge. Hide the subject while expanding and removing.
- ◆ **Removing the sign but not its shadow.** A shadow with nothing casting it is a bigger tell than the sign was.
- ◆ **One giant selection with a long prompt.** Mushy, low-detail results. Several small, well-selected passes with short prompts.
- ◆ **Accepting Harmonize wholesale.** It can alter edges, faces, and product details. Compare with the original and mask back what it shouldn't touch.
- ◆ **Flattening before blending.** Once flattened, the variations and masks are gone. Keep every generative layer until the seams are finished.
- ◆ **Judging at fit-to-screen.** Soft patches, repeats, and seams show only at 100%.
- ◆ **Expecting it to fix structure.** A subject with the wrong perspective or scale stays wrong after any amount of generation. Fix it by hand first (L10).

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 C)** Select, prompt short or empty, pick the best variation, then mask and finish by hand; **Q2 A)** Crop tool with Fill: Generative Expand, drag the edges outward, commit, Generate; **Q3 D)** Lasso the sign, its post, and its shadow with a little margin, empty prompt, Generate; **Q4 B)** Matching a placed element's light, color, and

shadows to its background; **Q5 D)** Add matching grain and a touch of sharpening; **Q6 A)** Which model produced it (Firefly or a partner model) and its terms for your account and deliverable. Cold-call: "The subject's perspective is wrong. Which generative tool fixes that?" (None: fix it by hand first.) Full set in the Answer Key. Journal (3 min, finish at home if needed): *Where does generative fill genuinely save you time, and where does trusting it cost you more time cleaning up than doing it by hand? Draw your own line between the two.*

◆ DIFFERENTIATION

Fast finishers: the lesson's "Going further": generate a busy background, then composite a cleanly cut subject (Module 1 masking) on top, so the subject stays pixel-perfect and only the surroundings are AI. Or the banner challenge: a snapshot to a wide banner by expanding in bites, with every seam fixed by hand until it is undetectable.

Struggling learners: Steps 1, 2, and 4 only: room, one removal, Harmonize, each checked at 100%. Skip connective tissue and hand-paint the contact shadow instead (they know how from L11). The must-haves are "separate layers, compare variations, mask back."

◆ OBJECTIVES

- ◆ Record a clean, reusable Action and re-order or re-record steps
- ◆ Insert stops, menu items, and modal pauses for input
- ◆ Add a **Conditional** so one Action handles landscape and portrait differently
- ◆ Batch an Action across a folder and save out automatically
- ◆ Build a **droplet**, a drag-and-drop executable for your workflow

◆ TIMING

3m	The goal (lesson slider) + mental model: record, branch, batch
3m	The Actions panel tour + "record relative, not absolute"
3m	What a Conditional can and cannot test; Batch settings for a hands-off run
30m	Guided build: Steps 1–4 (9 + 10 + 6 + 5)
2m	Now You: pick one challenge (finish at home)
4m	Self-check, recipe card, journal

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (1m).** Show the L17 slider: 200 images by hand versus one Action batched. **Say:** "If you've done a task twice, Photoshop can do it a thousand times. The setup cost is paid back the second time you run it."
- 2 Mental model (2m).** Board: **Record → if landscape? → Crop Wide / Crop Tall → Batch.** **Say:** "An Action is a recorded macro. A Conditional is an *if* inside it. A Batch points it at a folder. A droplet is that batch as an icon." Then the two reliability rules: record relative (percentages and Fit Image, not fixed pixels), and remember that Actions capture what you did, not what you meant, so record on a representative file and test on odd ones.
- 3 The panel (3m).** Window ► Actions. Point out the set folder, record, play, stop, the checkmark column (the step runs), and the dialog toggle (the step pauses for input). Double-click a step to re-record its values; drag steps to re-order. **Say:** "Image Size records fixed dimensions. Fit Image scales by the long edge, so it survives a portrait file."
- 4 Conditionals and Batch (3m).** Read the list from the lesson: Document Is Landscape, Portrait, or Square; Document Mode Is RGB or CMYK; Document Is 16 Bits Per Channel; Document Has Layers;

Layer Is Type or Smart Object. **Say:** "There is no condition for pixel size or for a layer's name. When you need that, you need a script, which is next lesson." Then the hands-off Batch settings: Override Action "Save As" Commands, Suppress File Open Options Dialogs, Suppress Color Profile Warnings, Errors: Log Errors To File.

- 5 **Step 1: Record the core (9m).** Open `l17-input/l17-delivery-01.jpg` (landscape). Actions panel: new set `Delivery`, new action `Web Delivery`, **Record**. File ► Automate ► **Fit Image** 2000 × 2000 px; Filter ► Sharpen ► **Unsharp Mask** (80% / 1.0 / 0, as in the lesson's figure); File ► **Place Embedded** `l17-watermark.png`, position it, commit; **Layer** ► **Flatten Image**; File ► **Save As JPEG**, quality 10, into `l17-output/`. **Stop.** **Say:** "Flatten is what lets Save As offer JPEG: a JPG can't hold the watermark layer. Skip it and the batch saves the wrong format or stops." (See the note in the Answer Key.)
- 6 **Step 2: Add the branch (10m).** Preferences ► Units & Rulers ► **Percent**. Record two tiny helper actions in the same set: `Crop Wide` (on the landscape file, a Crop tool crop, for example to 16:9) and `Crop Tall` (on `l17-delivery-02.jpg`, for example to 4:5). The lesson leaves the exact crops to you; pick shapes the room can recognize. Undo the test crops. In Web Delivery, select the **first** step, then panel menu ► **Insert Conditional...**: If Current **Document Is Landscape**, Then Play Action **Crop Wide**, Else Play Action **Crop Tall**. **Say:** "It lands *below* the selected step, so drag it to the top: crop first, then Fit Image, so the long edge is still 2000."
- 7 **Step 3: Batch it (6m).** File ► Automate ► **Batch**: Set Delivery, Action Web Delivery, Source Folder `l17-input/`, Destination Folder `l17-output/`, tick **Override Action "Save As" Commands**, both Suppress boxes, Errors: Log Errors To File, and a naming pattern (document name + extension). Turn off the Unsharp Mask dialog toggle if it's on. OK, and let the room watch the white keylines: landscape files lose their top and bottom edges to Crop Wide, portrait files to Crop Tall. **Say:** "The square file, 03, isn't landscape, so it takes the Else branch. That's what 'test on your weirdest file' is for."
- 8 **Step 4: Make a droplet (5m).** File ► Automate ► **Create Droplet**, point it at Web Delivery, save it to the desktop with the same Batch-style options. Drag a folder onto it. **Say:** "It runs when files are dropped on it; it doesn't watch a folder. Hand it to a colleague who never opens a menu." Finally, set Units & Rulers back to your usual unit (Pixels) before moving on.

◆ LAB: A DELIVERY ACTION, THEN NOW YOU

Guided build on a copy of `l17-input/` (eight JPEGs: four landscape, three portrait, one square, long edges 2000 to 3000 px, each labeled with its number, orientation, and size and framed by a white keyline) with `l17-watermark.png` kept **outside** the input folder and an empty `l17-output/` beside it; or the learner's own mixed folder of copies. Solo variation: three sizes into three subfolders in one run (Batch Destination **None**), a Document Has Layers conditional, or a colleague-proof droplet with a Stop step. Details in `labs/module-6/README.md`.

Circulate for: the red record dot actually on before the first step; Fit Image, not Image Size; Units & Rulers set to Percent *before* recording the crops; the conditional dragged to the top; Override ticked in Batch; eight files in `l17-output/` and the input folder unchanged.

Definition of done: the lesson's five-item checklist: a recorded, relative delivery Action (Fit Image, percent-based crops); a conditional branching landscape versus portrait; a batch that runs the folder hands-off with Override Save As; errors logged and no mid-run dialogs; a working droplet on the desktop.

◆ MISTAKES TO PRE-EMPT

- ◆ **Recording Image Size or a pixel-positioned crop.** Works on the recording file, breaks on the rest. Fit Image for size; Percent units before recording crops.
- ◆ **Expecting a percent crop to give the same ratio everywhere.** It trims the same *share* of each image. On files of different ratios the result differs a little; that's why the branch exists, and why you test on odd files.
- ◆ **Conditional left below the first step.** Fit Image runs first, then the crop shrinks the long edge below 2000. Drag the conditional to the top.
- ◆ **Watermark inside the input folder.** Batch processes it too, and the Place step may point at a file that moves. Keep it in a fixed place outside the input folder.
- ◆ **Batch stops halfway on a dialog.** A dialog toggle is still on (often Unsharp Mask), or a profile/open-options warning appeared. Toggles off, both Suppress boxes on, errors logged to a file.
- ◆ **Files land in the wrong folder.** Override Action "Save As" Commands not ticked; Batch obeyed the recorded Save As instead.
- ◆ **Forgetting to press Stop,** so browsing and fixes are recorded too. Delete the stray steps from the panel.
- ◆ **Rulers left in Percent for the rest of the course.** Every later dialog reads oddly (and scripts that read document size can misbehave). Reset to Pixels at the end of L17.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** Fit Image scales the long edge, so it works on landscape and portrait files alike; **Q2 C)** Set Preferences ► Units & Rulers to Percent; **Q3 D)** Document Is Portrait; **Q4 A)** Drag it to the top of the list so the crop runs before Fit Image; **Q5 C)** Override Action "Save As" Commands, both Suppress options, and Log Errors To File; **Q6 B)** None, and let the Action's own Save As steps decide. Full set in the Answer Key.

Journal (3 min): *List three repetitive tasks in your own workflow. For each, could an Action do it, or do you need a conditional or a script? Sorting them is the start of automating your practice.*

◆ DIFFERENTIATION

Fast finishers: the three-sizes Action (thumb, web, print into three subfolders; Batch Destination None, because Override sends every save to one folder), or a Document Has Layers conditional that plays the full delivery on layered masters and a lighter "already flat" version otherwise. Going further: list the tasks Actions couldn't handle; they are L18's opening.

Struggling learners: record Web Delivery and batch it *without* the conditional first (Fit Image handles both orientations on its own). Once that batch works, add the two crop helpers and the conditional. If recording goes wrong, delete the bad step and re-record only that step rather than starting over.

◆ OBJECTIVES

- ◆ Define **Variables** that bind layers to data: text, visibility, and pixel replacement
- ◆ Import a **Data Set** from a CSV and preview each row on the canvas
- ◆ **Export Data Sets as Files** to render one graphic per row
- ◆ Understand where scripting begins and read a simple **ExtendScript** / JS example
- ◆ Choose between Actions, Variables, and scripts for a given job

◆ TIMING

3m	The goal (lesson slider) + mental model: template + data, three variable types
4m	Variables & Data Sets tour + the naming rules
5m	Where scripting begins: read <code>resize-folder.jsx</code> aloud, line by line
34m	Guided build: build the template, then Steps 1–4 (6 + 10 + 9 + 9)
3m	Now You: run the sample script on a test folder (finish at home)
5m	Self-check, recipe card, journal, Module 6 summary

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 **The goal (1m).** Show the L18 slider. **Say:** "When the brief is 500 name badges, you stop designing files and start designing a template plus data. Ten or ten thousand costs the same effort."
- 2 **Mental model (2m).** Board: **data.csv columns → variables → layers**. Write the three types: **Text Replacement** (swap the words in a type layer), **Pixel Replacement** (swap the image in a layer, with fit rules), **Visibility** (show or hide a layer). **Say:** "There are exactly three kinds. Almost any personalized-graphics job breaks down into those three."
- 3 **Tour and naming rules (4m).** Image ► Variables ► Define; Image ► Variables ► Data Sets... ► Import; File ► Export ► Data Sets as Files. Open `l18-cards.csv` in a text editor on the projector. **Say:** "Each variable binds to the CSV column whose header matches its variable name. A stray space or a capital letter breaks it. Variable names can't contain spaces and must start with a letter, underscore, or colon. Visibility values are `true` or `false`; Photoshop rejects yes/no. Pixel values are full paths, or just file names with the images in the same folder as the CSV." Point out that the photo file names in the CSV match the files in `l18-data/` exactly.

- 4 **Where scripting begins (5m).** Show the lesson's `resize-folder.jsx` and read it as English: pick a folder; list its JPGs and PNGs; for each one open it, compute a scale for a 2000 px long edge, resize, save a `web_` copy at JPEG quality 10, close without saving. **Say:** "You can already follow this. That's the point: reading and adapting a snippet is a real skill before you ever write one. Scripts do loops, math, renaming by pattern, and decisions that Actions and Variables can't." Mention the two flavors: classic ExtendScript `.jsx` and newer UXP `.psjs`, both via File ► Scripts ► Browse.
- 5 **Build the template (part of Step 1).** Open `l18-data/l18-card-background.jpg`, **Save As** `l18-card-template.psd` inside `l18-data/`. Open `l18-ana.jpg` and drag it in as a normal pixel layer, sized to fill the white-framed window (it is 800 × 800 px, the same as the window). Type layer `Ana` centered in the name band. **File ► Place Embedded** `l18-sale-badge.png` at the top-right corner of the photo. **Say:** "Pixel Replacement needs a pixel layer. If you placed the photo as a Smart Object, Layer ► Rasterize ► Smart Object first."
- 6 **Step 1: Name everything (6m).** Rename the layers exactly `name`, `photo`, `sale`. **Say:** "The variables get the same names, and the CSV headers must match the variable names. Matching the layer names too is a helpful habit, not a requirement."
- 7 **Step 2: Define variables (10m).** Image ► Variables ► Define. Layer `name`: tick **Text Replacement**, variable name `name`. Layer `photo`: tick **Pixel Replacement**, variable name `photo`, method **Fit** (the options are Fit, Fill, As Is, and Conform). Layer `sale`: tick **Visibility**, variable name `sale`.
- 8 **Step 3: Import and proof (9m).** Switch to the Data Sets page (Image ► Variables ► Data Sets...), **Import** `l18-cards.csv`, tick **Use First Column For Data Set Names**. With **Preview** on, step through the rows: Ana has the badge, Ben doesn't, Cy does. **Say:** "The name tag inside each photo is there for proofing: the tag must match the name on the card." Then the troubleshooting demo with `instructor/l18-cards-broken.csv` (import it into a copy of the template, or talk through it on the projector): header `Name` with a capital N, `yes` / `no` in the sale column, and a photo name that doesn't exist (`l18-benn.jpg`). **Say:** "Those three faults are 90% of 'why didn't it bind?'"
- 9 **Step 4: Export the batch (9m).** File ► Export ► **Data Sets as Files**: choose an output folder (for example `l18-cards-out/`), Data Set: All Data Sets, File Naming: **Data Set Name** plus any fixed text. Export. Photoshop writes one PSD per row. **Say:** "Six cards from one template. The deliverable is this folder." Optional finish: run the PSDs through the L17 delivery Action to flatten to JPG; see the note in Mistakes about the Action's crop.

◆ **LAB: A DATA-DRIVEN BATCH (MODULE 6 DELIVERABLE), THEN NOW YOU**

Guided build on the practice parts in `l18-data/` (card background, six 800 × 800 photos named `l18-ana.jpg` ... `l18-fay.jpg`, the sale badge, and `l18-cards.csv` with the columns `name, photo, sale`), or on the learner's own template and a small CSV of their own. Solo variation: a second visibility variable (the VIP ribbon, `l18-vip-ribbon.png`) and a color swap via a visible/hidden tint layer, both driven by `l18-cards-vip.csv` (columns `name, photo, sale, vip, tint`); 20 social cards for a real list; and the script challenge on a **copy** of a test folder. Details in `labs/module-6/README.md`.

Circulate for: the photo layer is a pixel layer, not a Smart Object; variable names exactly `name`, `photo`, `sale`; the template saved in the same folder as the CSV and photos; Preview on and several rows proofed before exporting; File Naming set to Data Set Name.

Definition of done: the lesson's Module 6 checklist: variable names (and layers) matching the CSV headers exactly; Text, Pixel, and Visibility variables defined; the CSV imported as a data set with several rows proofed on canvas; Data Sets exported as files with a clear naming pattern; (optional) finished via a delivery Action or droplet.

◆ MISTAKES TO PRE-EMPT

- ◆ **Pixel Replacement isn't offered for the photo layer.** It's a Smart Object (Place makes one) or a type/shape layer. Rasterize it, or drag the photo in from an open document.
- ◆ **Header and variable name don't match.** `Name` vs `name`, a trailing space, or a space inside the name. The variable name is what binds, not the layer name.
- ◆ **Visibility written as yes/no, Y/N, or 1/0.** Only `true` and `false`.
- ◆ **"Can't find the image."** The photo file names in the CSV don't match the files exactly, or the images aren't in the CSV's folder. Bare file names only work beside the CSV; otherwise use full paths.
- ◆ **Spreadsheet app saves the CSV oddly.** Keep it UTF-8 with a header row. If Excel or Numbers changed it, open it in a text editor and check.
- ◆ **Running the cards through the L17 Web Delivery Action as-is.** Its conditional crop trims a share of every card (they are portrait, so they get Crop Tall), and Fit Image enlarges a 1080 × 1350 card to a 2000 px long edge. For cards, duplicate the Action and delete the conditional and Fit Image steps (keep a Flatten before the Save As), or use File ► Scripts ► Image Processor.
- ◆ **Running the sample script on the only copy of a folder.** It writes its `web_` files into the same folder, and a second run processes those too. Always a test copy. And if rulers are still in Percent from L17, set them back to Pixels first: the script reads the document's width and height, and those follow the ruler unit.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** D) Text Replacement, Pixel Replacement, and Visibility; **Q2** B) The variable name matching the CSV column header exactly; **Q3** A) `true` or `false`; **Q4** C) The images sit in the CSV's folder and the names match exactly (otherwise use full paths); **Q5** D) File ► Export ► Data Sets as Files, naming by Data Set Name; **Q6** B) Resize every file in a folder, skipping any whose name already starts with `web_`. Full set in the Answer Key.

Journal (3 min): *What in your work is really "one design, many values"? Sketch the template and the columns its data set would need. Seeing jobs as template-plus-data is the mindset shift of this module.*

◆ DIFFERENTIATION

Fast finishers: the VIP ribbon and tint layer from `l18-cards-vip.csv` (Ben, Eli, and Fay are VIP; Cy and Eli are tinted); then the script challenge: change the long edge and output format, and add a line that skips files already starting with `web_`. Going further: a data set feeding a template whose art lives in linked Smart Objects (Module 5), exported by an Action (L17), a genuine production pipeline.

Struggling learners: one variable first. Define only the Text Replacement on `name`, import, and step through the rows. When that works, add `sale` (Visibility), then `photo` (Pixel). Re-import the CSV after each change to the definitions. The script reading is "follow along," not "type along."

CLOSING THE SESSION

MODULE 6 WRAP-UP (8 MIN, 2:52–3:00)

Recap the arc: **a model for the tedious pixels → an Action for the repeated steps → data for the repeated designs → a script when you need loops, math, or logic.** Ask the room for one job that belongs to each tool.

Exit ticket (on a sticky note): "One job in my work that is really one design, many values, and the column headers its CSV would need." Collect them; they make a good opener for the capstone brief next week.

Homework, from the three lessons' "Before the Next..." lists: save the finished, layered L16 composite with generative layers and masks intact, and note one task where generation saved time and one where you had to take over; batch Web Delivery over a mixed folder with no mid-run dialogs and test the droplet on a fresh set; **ship one data-driven batch of 10 or more outputs from a single template**; adapt the sample script and run it on a test folder; finish the three solo variations as far as time allows; write the Learning Journal entries.

Preview Session 7 (Module 7, Multi-Shot Techniques): "You can now output at volume. Next we combine many frames into one: panoramas, HDR, focus stacking, and a short looping animation. The session also kicks off your capstone: we'll run Lesson 8.1, the brief and the plan, in class." Ask learners to bring any overlapping, bracketed, or focus-stack frames they have (practice sets are provided) and to start thinking about which capstone brief they want.

Module 7: Multi-Shot Techniques

How to teach the two lessons of the Going Further tier (panoramas, HDR and focus stacking, then Timeline animation and export) and how to run the capstone kickoff that closes Session 7. The theme of the day is "many frames, one result": three merges that beat the limits of a single frame, then layers turned into motion, then a brief that turns "make something cool" into a project learners can finish.

2 lessons + capstone kickoff

Session 7 • 3 hrs incl. break

Lessons L19–L20 (+ L21 kickoff)

Level Intermediate

Tier Going Further

HOW TO USE THIS GUIDE

- Same rhythm as every module: **Frame** → **Demo** → **Guided build** → **Solo variation** → **Check**. Demo one step, let the room do it on the practice files, then move on. Each lesson plan gives a timing table, a demo script with **Say**: lines, the lab, the mistakes to pre-empt, a check for understanding, and differentiation notes.
- Name every menu path out loud and give both shortcut forms: **Ctr l** on Windows is **⌘** on Mac, **Alt** is **⌥**.
- Teal** = hands-on, **amber** = watch-outs. In L19 the watch-outs are about **source frames** (overlap, a moved subject, a skipped focus distance); in L20 they are about **which track a layer has** and **GIF weight**.
- The last 42 minutes run **Lesson 8.1 (L21, The Brief & the Plan)** as the capstone kickoff. Its full lesson plan (objectives, demo script, mistakes, self-check answers) lives in the **Module 8 facilitator guide** (<source/module-8/facilitator-guide.html>); this guide gives only the kickoff segment and how it hands off to Session 8.
- Quiz answers below match the workbook's Self-checks; the Answer Key explains each one.

THE #1 RISK OF MODULE 7: THE MERGES EAT THE CLOCK

Photomerge, Merge to HDR Pro and Auto-Blend Layers are processor-heavy, and a learner who feeds them a folder of 45-megapixel raw files can lose ten minutes to one progress bar. **In class, everyone merges the practice sets** (1800 × 1200 px JPEGs, a few seconds each on a normal laptop); their own full-resolution sets are the take-home "Now You." The second risk is the session's shape: two lessons plus the kickoff leave no slack, so run the demos crisply and **protect the kickoff**. Session 8 is built on learners arriving with a written brief and a plan.

MERGE TO HDR PRO AND THE PRACTICE BRACKET

HDR Pro reads each frame's exposure from its metadata. The three practice frames carry generated exposure data (f/8, ISO 100, 1/250 s, 1/60 s, 1/15 s, a 2-stop bracket), so the merge normally runs without questions. If a learner's copy has lost its metadata (some cloud-sync tools and messaging apps strip it), Photoshop opens a **Manually Set EV** dialog. The answer is in `labs/module-7/README.md`: choose the shutter-speed option and enter f/8, ISO 100 and the three shutter speeds, darkest frame fastest. Rehearse this once on your own machine so you can talk a learner through it in thirty seconds.

BEFORE SESSION 7

- ◆ ☐ Copy `labs/module-7/practice/` to each machine (or a shared drive), keeping its five subfolders (`l19-pano/`, `l19-hdr/`, `l19-stack/`, `l20-anim/`, `l20-steam/`). Keep `labs/module-7/instructor/` for yourself: the pano and focus-stack references and a reference GIF of the target loop. As always, never hand out the course site's licensed `images/` folder.
- ◆ ☐ Remind learners (in the Session 6 wrap-up or by email) to bring: **optionally**, one multi-shot set of their own (3 or more overlapping frames of a wide view; a phone bracket of a bright window; or a close object shot 3 or 4 times with focus tapped near, middle and far); a **layered design** of their own worth animating (the Module 5 campaign master is ideal); and for the kickoff, **one or two capstone ideas** plus a handful of reference images. Details in `labs/module-7/README.md`.
- ◆ ☐ Print (or share as a PDF) the **L21 pages of the Module 8 participant workbook** for the kickoff: learners write their brief and plan on them tonight and bring them back to Session 8.
- ◆ ☐ On one room machine, rehearse the three merges on the practice sets and run a **Render Video** once (File ► Export ► Render Video). The first render on a machine can be slow to start; better you find that out than the room.
- ◆ ☐ Check free disk space: 16-bit HDR results and focus-stack files with four full layers add up quickly. A few gigabytes per learner is plenty.
- ◆ ☐ Open `lesson_19.html`, `lesson_20.html` and `lesson_21.html` in browser tabs for the before/after sliders and figures.

SESSION 7 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	NOTES
0:00-0:10	Warm-up: recap Module 6	10	Two volunteers show their data-driven output folder from L18. Ask: "Actions, Variables, or a script: which would you reach for to make 50 badges, and why?" Confirm everyone has the practice folders.

CLOCK	SEGMENT	MIN	NOTES
0:10–1:05	L19 (Lesson 7.1) Panoramas, HDR & Focus Stacking	55	Build: merge three ways on the practice sets, then clean the joins. Solo variation starts here, finishes at home.
1:05–1:18	Break	13	Leave the L19 checklist on the projector. Early finishers open L20-anim/ and read the L20 "Mental Model" section.
1:18–2:08	L20 (Lesson 7.2) Timeline Animation & Export	50	Build: a keyframed, seamless 3-second loop exported as a GIF and an MP4 (completes the Module 7 deliverable).
2:08–2:50	Capstone kickoff: L21 (Lesson 8.1) The Brief & the Plan	42	Brief written, moodboard started, plan drafted. Full lesson plan in the Module 8 guide.
2:50–3:00	Wrap-up & exit ticket	10	Recap the module, exit ticket, homework (finish the plan, build the capstone), preview Session 8.
Total		180	Journals for L19 and L20 are three minutes each inside the lessons; the L21 journal prompt goes home.

The lesson pages estimate 55 + 50 minutes for a solo learner, which the room matches once the demos run alongside the builds on small practice files. The pressure comes from the kickoff: L21 is a 45-minute lesson on the site, run here in 42 by starting the moodboard in class and finishing it at home. If you run long, protect two things: the **Module 7 deliverable** (one clean merge of each kind and a looping GIF and MP4) and the **kickoff's written brief**. Move to homework, in this order: the "Now You" variations, the MP4 render (it can run at home from the saved .psd), the L19 unifying grade, and the second half of the moodboard.

MODULE AT A GLANCE

LESSON	TOPIC	TIME	LEARNERS LEAVE ABLE TO...
L19 (7.1)	Panoramas, HDR & Focus Stacking	55m	Stitch a panorama, merge an exposure bracket, focus-stack a series, and clean every join at 100%
L20 (7.2)	Timeline Animation & Export	50m	Keyframe a seamless loop on the Video Timeline and deliver it as a GIF and an MP4
L21 (8.1)	Capstone kickoff: The Brief & the Plan	42m	Leave with a one-page brief, a started moodboard, and a staged plan (taught from the Module 8 guide)

The module is one story: **many frames into one still → layers into motion → everything you know into a plan**. The through-line of L19 is one sentence you will say many times: **align first, blend second**.

◆ OBJECTIVES

- ◆ Shoot overlapping frames and stitch a panorama with **Photomerge**
- ◆ Bracket exposures and combine them with **Merge to HDR Pro** for wider range
- ◆ Shoot a focus series and focus-stack it with **Auto-Align + Auto-Blend**
- ◆ See the shared engine (auto-align, then blend) behind all three
- ◆ Clean the seams, ghosts, and edges these merges leave behind

◆ TIMING

3m	The goal: beat a single frame (lesson slider)
5m	Mental model: align, then blend · three jobs, one engine
4m	The three front doors: Photomerge, HDR Pro, Load Files into Stack + Auto-Blend
10m	Guided build Step 1: panorama
12m	Guided build Step 2: HDR
10m	Guided build Step 3: focus stack
5m	Step 4: clean the joins (the unifying grade can finish at home)
6m	Self-check, recipe card, journal (Now You goes home)

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (3m).** Show the L19 slider: one frame catches a slice, three stitched frames hold the whole vista. **Say:** "A single frame has three limits: it is only so wide, it only holds so much range from shadow to highlight, and only so much of it is in focus. Today we beat all three with the same trick: shoot several frames and let Photoshop combine them."
- 2 Align, then blend (5m).** Draw the pipeline on the board: **source frames → Auto-Align → Blend → one image.** **Say:** "Auto-Align finds common detail and warps the frames into registration. Blend then picks, pixel by pixel, the best contribution: the overlap for a pano, the well-exposed tone for HDR, the sharpest region for a stack." Then the three-column table: **vary angle → wider field; vary exposure → wider range; vary focus → deeper focus.** **Say:** "Change only the one variable the technique needs."

Hold everything else still. That is why a tripod and a 2-second timer pay off, especially for HDR and stacking, where frames must register to the pixel."

- 3 **Three front doors (4m).** Point at each menu without running it yet: **File ► Automate ► Photomerge**; **File ► Automate ► Merge to HDR Pro**; **File ► Scripts ► Load Files into Stack** followed by **Edit ► Auto-Align Layers** and **Edit ► Auto-Blend Layers**.... Then the warning from the lesson: **Say:** "Garbage frames in, garbage merge out. Too little overlap and the pano won't stitch; something moves mid-bracket and HDR shows ghosts; skip a focus distance and the stack has a soft band."
- 4 **Step 1: panorama (10m).** **File ► Automate ► Photomerge → Browse →** select the four files in `l19-pano/`. Layout **Auto**; tick **Blend Images Together** and **Content Aware Fill Transparent Areas**; OK. While it runs: **Say:** "These frames overlap by about 35%. The dialog doesn't measure overlap for you; aim for roughly 30% when you shoot." Show the result's layer masks (one per frame): the stitch is just masks, the Module 1 idea again. Crop or accept the filled edges, then zoom to 100% on a seam and heal anything that doubled. Save as `pano-working.psd` (suggested name). **Say:** "The frames were deliberately shot with slightly different brightness and some vignetting. Blend Images Together is what hides that."
- 5 **Step 2: HDR (12m).** **File ► Automate ► Merge to HDR Pro → Browse →** the three files in `l19-hdr/`; leave **Attempt to Automatically Align Source Images** on; OK. In the merge dialog, tick **Remove Ghosts**. **Say:** "A bird flew across the window between shots. Toggle Remove Ghosts and watch the three faint birds become one." Set **Mode: 16 Bit** and choose **Local Adaptation**; nudge the sliders toward believable: window detail back, room detail visible, no halos around the window frame. **Say:** "Keep it believable. Want every stop kept for later? Set 32 Bit instead and tone-map afterwards with Camera Raw." OK; save as `hdr-working.psd`. If a learner gets the **Manually Set EV** dialog, walk them through the README values.
- 6 **Step 3: focus stack (10m).** **File ► Scripts ► Load Files into Stack → Browse →** the four files in `l19-stack/`; tick **Attempt to Automatically Align Source Images**; OK. **Say:** "The frames breathe: the image grows a hair as focus moves, which is what the auto-align fixes." Select all four layers, **Edit ► Auto-Blend Layers... → Stack Images**, tick **Seamless Tones and Colors →** OK. Show the masks Photoshop painted: white where each frame was sharpest. Zoom to 100% on the ruler and the wall chart's fine print. **Say:** "If a patch is soft, paint the mask of the frame that had it sharp. Auto-Blend can only choose from the sharp regions you captured." Save as `stack-working.psd`.
- 7 **Step 4: clean the joins (5m).** On each result, 100% zoom: heal a stitch artifact, a ghost remnant, or a soft band on a separate layer. Then the lesson's finish: a **Module 4 unifying grade** so all three feel like your work. **Say:** "One careful shoot beats any repair, but every merge still gets a 100% inspection." If time is short, the grade moves to homework; the inspection does not.

◆ LAB: MERGE THREE WAYS, THEN NOW YOU

Guided build on the three practice sets in `labs/module-7/practice/`: `l19-pano/` (4 overlapping frames), `l19-hdr/` (-2 / 0 / +2 bracket), `l19-stack/` (4 frames, near to far). Learners with their own small

sets may use them for any one step. Solo variation (at home): a vertical panorama; a subtle HDR and a deliberately overcooked one; a focus stack of a small object compared with a single f/8 frame. Going further: a bracketed panorama (HDR each position, then stitch). Details in [labs/module-7/README.md](#).

Circulate for: Photomerge ticks set before OK (Blend Images Together, Content Aware Fill Transparent Areas); Remove Ghosts on for the bracket; Mode deliberately chosen (16 Bit to tone-map now, 32 Bit to keep range); all four stack layers selected before Auto-Blend, with Stack Images (not Panorama); learners actually zooming to 100%.

Definition of done: the lesson's five-item checklist: a stitched panorama with invisible seams; an HDR merge with recovered highlights and shadows and no ghosts; a focus stack sharp front to back; edges cropped or filled and joins healed at 100%; a consistent grade across all three.

◆ MISTAKES TO PRE-EMPT

- ◆ **Transparent, ragged edges on the panorama.** Content Aware Fill Transparent Areas was unticked. Crop them off, or rerun with the tick. Either is fine; checking the filled edges at 100% is not optional.
- ◆ **The pano bows or bends.** Wide pans on a flat projection curve straight lines. Try **Cylindrical** or **Spherical** in Photomerge's Layout list; for their own shoots, keep the horizon centered; **Filter ► Adaptive Wide Angle** straightens afterwards.
- ◆ **Three ghost birds in the window.** Remove Ghosts was off. Re-open the merge and tick it; for their own brackets, the lesson's fix is prevention (shoot when nothing moves).
- ◆ **HDR result looks radioactive.** Every slider pushed to the right. Ask: "Would a viewer believe this room?" Pull Detail and Vibrance back; the overcooked version is a Now You exercise, not the deliverable.
- ◆ **Auto-Blend grayed out or odd.** Only one layer selected, or the layers are still the locked Background. Select all four (click the top one, Shift-click the bottom) and run it again.
- ◆ **"Panorama" chosen in Auto-Blend.** That is the other Blend Method. Stack Images is the focus-stacking option.
- ◆ **A soft band across the table.** On the practice set this only happens if a frame was left out (try it deliberately: drop frame 2 and watch the band appear). On real shoots: reshoot with smaller focus steps, or mask in a sharper frame by hand.
- ◆ **Feeding full-resolution raw sets in class.** Long progress bars stall the room. Practice files now, own sets at home.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** B) Photoshop auto-aligns the frames, then blends the best of each; **Q2** C) The focus distance; **Q3** D) Content Aware Fill Transparent Areas; **Q4** A) Tick Remove Ghosts in Merge to HDR Pro; **Q5** B) 32 Bit, then tone-map later in Camera Raw; **Q6** D) A focus distance was skipped, so reshoot with smaller steps or mask in a sharper frame. Full set in the Answer Key.

Journal (3 min, finish at home if needed): *Which single-frame limit frustrates you most: width, range, or depth? Plan the shoot that would beat it with a multi-shot merge.*

◆ DIFFERENTIATION

Fast finishers: rerun Photomerge with **Cylindrical** and **Reposition** layouts and compare the horizon; rerun HDR at 32 Bit and tone-map in Camera Raw; drop frame 2 from the stack to see the soft band the lesson's FAQ describes. Or start the Going Further: HDR-then-stitch on their own bracketed pan.

Struggling learners: one merge done well beats three done badly. Prioritize Photomerge (the most forgiving), then the focus stack; let them watch the HDR demo and do it at home. Give them the recipe card, and the mantra: align first, blend second.

◆ OBJECTIVES

- ◆ Choose between **Frame Animation** and the **Video Timeline**
- ◆ Build a frame loop with tweening
- ◆ Animate layer position, opacity, and style with keyframes (and scale, via Smart Objects)
- ◆ Loop cleanly and shape timing so motion feels natural
- ◆ Export a GIF (Save for Web) or an MP4 (Render Video)

◆ TIMING

3m	The goal: make it move (lesson slider)
5m	Mental model: frames vs keyframes (flip-book vs smooth path)
4m	The Timeline panel tour + the two export doors
6m	Guided build Step 1: start the timeline
14m	Step 2: keyframe the moves
8m	Step 3: loop and shape the timing
6m	Step 4: export both (the MP4 render may finish at home)
4m	Self-check, recipe card, journal (Now You goes home)

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The goal (3m).** Show the L20 slider and, if you like, the reference loop in [Labs/module-7/instructor/L20-reference-loop.gif](#). **Say:** "Photoshop is an animator hiding in plain sight. Everything that moves in this loop is a layer you already know how to make."
- 2 Frames vs keyframes (5m).** Two columns on the board. **Frame Animation** = the flip-book: discrete frames, each a layer arrangement, each with its own hold time, Tween to fill in-betweens. **Video Timeline** = the smooth path: each layer gets a track, you set keyframes and Photoshop interpolates. **Say:** "Short, loopy GIF with exact frames: Frame Animation. Smooth glides and fades, or you plan to export video: the Video Timeline. You can convert between them, but picking the right one first saves rework."
- 3 Panel tour + export doors (4m).** **Window ► Timeline.** Point at the single button (**Create Video Timeline**) and its dropdown arrow (**Create Frame Animation**). Twirl a layer open: normal layers show

Position, Opacity, Style; a Smart Object shows **Transform** instead of Position. **Say:** "No scale track on a normal layer, at all. If it's going to grow or rotate, make it a Smart Object first. And there is no ease curve: keyframes are linear, or Hold for a jump." Then the two doors out: **File ► Export ► Save for Web (Legacy)** (Ctrl+Alt+Shift+S / ⌘+⌥+Shift+S) for GIF; **File ► Export ► Render Video** for MP4.

- 4 **Step 1 (6m).** **File ► Scripts ► Load Files into Stack** → the three files in `120-anim/` (the L19 command, reused). Put the background at the bottom, badge and headline above; save as `Loop-working.psd` (suggested name). Right-click the badge layer ► **Convert to Smart Object**. **Say:** "The badge will scale, so it becomes a Smart Object now, not after I find the missing track." Then **Create Video Timeline;** panel menu ► **Set Timeline Frame Rate ► 24 fps**; drag the right end of each layer's bar to the 3-second mark, then drag the work-area end handle above the ruler to match.
- 5 **Step 2 (14m).** Twirl open the badge. Playhead at frame 0; press Ctrl/⌘ + T and scale the badge small, commit, then click the **Transform** stopwatch. Move the playhead to about 0.6 seconds, scale it back to full size in its spot; **Say:** "Watch the diamond appear. After the stopwatch is on, any change at a new time drops a keyframe for you." Headline: at about 0.4 seconds click the **Opacity** stopwatch with Opacity 0% and the **Position** stopwatch with the headline nudged about 40 px down; at the 1-second mark set 100% and move it up into place. Scrub to show both moves.
- 6 **Step 3 (8m).** **Say:** "A loop is seamless when the last frame matches the first." Hold the finished pose until about the 2-second mark, then add keyframes that return each layer to its frame-0 state by the last frame (small badge, headline invisible and low). Then the fake ease: add one extra keyframe just before the end of each move with only a small change left. **Say:** "Photoshop moves at constant speed. Put most of the move early and a little at the end, and the finish slows down." Press the play button (or Space) and adjust until it feels natural, not robotic.
- 7 **Step 4 (6m).** **File ► Export ► Save for Web (Legacy)** → GIF, a modest color count, a little dither, **Looping Options: Forever**; show the file size readout in the preview's corner. **Say:** "Watch the size. If it balloons past a couple of megabytes, shrink the Image Size in this same dialog, or deliver an MP4." Save as `Loop.gif`. Then **File ► Export ► Render Video** → H.264, a high-quality preset, the frame rate → **Render** → `Loop.mp4`. **Say:** "The lesson says 1080p, which is the 16:9 case. Our practice design is square, so keep the Size at Document Size and the loop stays 1080 × 1080. One animation, two delivery formats."

◆ LAB: A LOOPING ANIMATION, THEN NOW YOU

Guided build on the three pieces in `120-anim/` (or a learner's own simple layered design: a logo, a badge, a headline). Solo variation (start in class if time allows, finish at home): a **cinemagraph** from the six frames in `120-steam/`: steam loops while the cup stays still, masked so only the steam animates; a pulsing layer-style glow; or the same 3-second loop rebuilt once in Frame Animation and once on the Video Timeline. Details in `labs/module-7/README.md`.

Circulate for: the badge converted to a Smart Object *before* keyframing; stopwatches clicked on the right property (Transform on the badge, Opacity and Position on the headline); the work area trimmed to 3 s so the GIF doesn't carry dead seconds; the last frame matching the first.

Definition of done: the lesson's checklist: a video timeline with keyframed Transform (or Position) and Opacity; transformed layers converted to Smart Objects (stay sharp); a seamless loop with natural-feeling timing; a GIF looping Forever and an MP4 exported; the GIF short and reasonably small. With L19, this completes the Module 7 deliverable.

◆ MISTAKES TO PRE-EMPTY

- ◆ **"There's no scale property."** The layer is a normal layer, which only has Position, Opacity and Style. Convert to Smart Object; the Transform track appears.
- ◆ **Scaled a normal layer anyway, now it's soft.** Pixels were resampled. Undo, convert to a Smart Object, redo: it re-renders crisply at every frame.
- ◆ **Changed a value but no keyframe appeared.** The stopwatch was never clicked, so the change applied to the whole clip. Click the stopwatch first, then change values at new times.
- ◆ **The loop jumps at the wrap-around.** The last frame doesn't match frame 0, or the work area runs past the last keyframes. Copy the frame-0 values to the end and trim the work-area handle.
- ◆ **Robotic, constant-speed motion.** Expected: Photoshop keyframes are linear. Add the extra near-the-end keyframe to fake the ease.
- ◆ **A 15 MB GIF.** Too many pixels, colors, or seconds. Shrink Image Size in Save for Web, lower the color count, shorten the loop, or ship the MP4.
- ◆ **Banded gradients in the GIF.** GIF is limited to 256 colors. More colors and a little dither help; for gradient-heavy work, the MP4 is the better deliverable.
- ◆ **Hunting for Save for Web in Export As.** It is File ► Export ► Save for Web (Legacy), a separate command.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** B) The Video Timeline; **Q2** C) Convert the badge to a Smart Object so it gets a Transform track; **Q3** A) Returning each layer to its frame-0 state by the last frame; **Q4** D) Add an extra keyframe near the end of each move with only a small change left; **Q5** B) File ► Export ► Save for Web (Legacy), GIF, Looping Options: Forever; **Q6** C) An MP4 made with Render Video (H.264). Full set in the Answer Key.

Journal (3 min, finish at home if needed): *What is one still design of yours that a few seconds of motion would improve, and which single element should move? Restraint matters in animation too.*

◆ DIFFERENTIATION

Fast finishers: the cinemagraph from [120-steam/](#): Load Files into Stack, **Create Frame Animation**, panel menu ► **Make Frames From Layers**, set each frame's delay, and mask the top frames so only the steam area changes. Or animate a **Style** track (an Outer Glow pulsing on the badge). Going further (from the lesson): assemble a data-driven sequence from Module 6 on the Timeline.

Struggling learners: animate one property on one layer (headline Opacity 0 → 100 → 0) until the stopwatch-then-move rhythm clicks, then add the badge scale. Export only the GIF in class; the MP4 is one menu command at home.

CAPSTONE KICKOFF (2:08–2:50)

WHAT THIS SEGMENT IS, AND WHERE ITS LESSON PLAN LIVES

This 42-minute block runs **Lesson 8.1 (L21), The Brief & the Plan**, in class, so learners go home with the thinking done and a clear finish line. The **full L21 lesson plan** (objectives, demo script, mistakes to pre-empt, self-check answers, differentiation) is in the **Module 8 facilitator guide**, labelled "taught in the Session 7 kickoff, reviewed in Session 8." Teach from that plan; the table below is how it fits into tonight's clock.

CLOCK	KICKOFF BLOCK	MIN	WHAT HAPPENS
2:08–2:12	A spec, not a wish	4	The L21 slider (vague sticky notes vs a structured brief). Say: "The capstone begins on paper. A brief is the contract with your future self at 2 a.m., wondering if you're done."
2:12–2:16	Anatomy of a brief	4	The six parts: objective · audience · deliverables + specs · constraints · success criteria · references. The four project types from "Now You": key art poster, product campaign, editorial portrait, your own.
2:16–2:30	Step 1: write the brief	14	Learners fill the six parts on the Module 8 workbook's L21 page. Circulate and pin adjectives: "What would 'bold' look like? Which reference shows it?"
2:30–2:40	Step 2: start the moodboard	10	Collect the first 3 or 4 of the 6–12 references and annotate each with the one thing it contributes. The rest is homework.
2:40–2:48	Step 3: draft the plan	8	Stages mapped to module skills, assets to source or shoot, two or three milestones, stretch goals marked optional.
2:48–2:50	Pair swap	2	Partners read each other's success criteria aloud. "Could you grade my project with this list?"

The site estimates 45 minutes for L21 (15 + 15 + 10 for the three steps). Tonight's 42 minutes trims the moodboard to a start; finishing it is the first homework item. **Circulate for** the commonest failure the lesson names: **over-scoping**. A campaign of twelve deliverables built in a week of evenings is a warning sign; ask "Which one piece would you still be proud of if everything else got cut?" and make the rest stretch goals. Learners who want to fold tonight's techniques into the capstone (a stitched backdrop, an HDR interior, a short motion version) should plan them as stretch goals, not requirements.

CLOSING THE SESSION

MODULE 7 WRAP-UP + CAPSTONE HOMEWORK (10 MIN, 2:50–3:00)

Recap the arc: **many frames into one still (align first, blend second) → layers into motion (keyframe the start and end, let Photoshop travel the middle) → everything you know into a brief and a plan.** Exit ticket (on a sticky note): "Panorama, HDR, focus stack: which variable changes in each?" (Angle, exposure, focus distance.)

Homework, in two parts. 1. Finish the plan (first evening): complete the one-page brief and have someone else read the success criteria; finish the annotated moodboard (6–12 references); finalize the staged plan with milestones; gather source assets and confirm the license on anything they didn't shoot; write the L21 journal entry (the one-sentence objective). **2. Build it** (before Session 8): work through Lesson 8.2, **Studio Session: Build It**, in passes (block-in, refine, polish) with named groups and versioned saves. Say it plainly: **"Session 8 opens with a plan review in pairs and then a studio block for finishing, so arrive with the piece well underway, and with your brief, moodboard, and plan in hand."**

Also from the lessons' "Before the Next Module": produce one multi-shot still of their own (pano, HDR, or stack) and one short animation exported as a GIF and an MP4, and write the L19 and L20 journal entries. Preview Session 8 (Module 8): "No new tools. You review plans in pairs, build in the studio, then present your capstone with a case study and give each other real critique."

Module 8: Capstone

How to run the course finale: one self-directed professional project, taken from a one-page brief to a presented case study. There are no new tools. Your job shifts from demonstrator to studio coach and editor: hold learners to a checkable brief, keep scope finishable, protect the file discipline under deadline pressure, and run a critique that is honest, kind, and specific.

3 lessons

Session 8 • 3 hrs incl. break

Lessons L21–L23 (8.1–8.3)

Level Intermediate

L21 taught in the Session 7 kickoff

Most of the build is homework

HOW TO USE THIS GUIDE

- ◆ **The capstone spans two sessions.** L21 (Lesson 8.1, The Brief & the Plan) is **taught in class during the Session 7 capstone kickoff** (42 minutes near the end of Session 7; the Module 7 guide points here). Learners leave with a brief, a moodboard, and a plan started. Between sessions they finish the plan and build most of the piece at home (L22). Session 8 is a quick plan review, a studio block for finishing and getting help, case-study assembly, presentations with structured peer critique, and the course close.
- ◆ **Say this plainly, twice** (at the Session 7 kickoff and in a midweek message): "Session 8 is not enough time to build a capstone from scratch. Arrive with Pass 1 and Pass 2 done and Pass 3 under way, about 70 to 80% of the piece. The studio block is for finishing, fixing, and getting help. Capture a screenshot at the end of every pass: you need them for the case study."
- ◆ **Teal** = hands-on, **amber** = watch-outs. In this module the risks are a brief nobody can check, over-scoping, polishing one corner while the rest is rough, and file chaos under pressure.
- ◆ **One rubric for everything.** The kit's `final-assessment.html` scores the capstone on 12 rows (1–3 each, max 36). Rows 2–5 are the lesson's own fresh-eyes checks, rows 6–7 and 8–11 are the L22 and L23 checklists. Use the same row numbers and names in check-ins, on the reviewers' rubric cards, and in written feedback, so learners hear one language all the way through.
- ◆ Quiz answers below match the workbook's Self-checks and the Answer Key. The L21 quiz is homework after the kickoff; L22's is homework during the build; L23's is done after the session.

THE #1 RISK OF MODULE 8: A CAPSTONE THAT CAN NEVER BE "DONE"

Learners at this level can execute almost anything, so the failure is rarely a skill; it is a brief with vague adjectives ("make it pop") and no checkable success criteria, which turns the build into endless fiddling and the presentation into an apology. Hold the line at the kickoff: **"If you can't check it, it isn't in the brief."** The second risk is generative steps. Generative Fill and Expand need internet, an Adobe sign-in, and credits; tell learners to finish them at home before Session 8, never to plan a piece that depends on generating in class, and to disclose any generative assistance in the case study.

BEFORE SESSION 8

- ◆ ☐ **End of Session 7:** run the L21 kickoff (lesson plan below). Show the four project options, hand out `L21-brief-template.txt`, and state the "Pass 1 and Pass 2 done" expectation.
- ◆ ☐ **Copy** `labs/module-8/practice/` to each machine (or a shared drive): the brief template, the case-study outline, the backup briefs and their assets, and the three mockups. Keep `labs/module-8/instructor/` for yourself. As always, never hand out the course site's licensed `images/` folder.
- ◆ ☐ **Midweek (optional but powerful):** ask learners to email the brief's objective and success criteria. Reply with one sentence: a green light, a scope trim ("One hero and two sizes, not six"), or the one criterion that is not yet checkable.
- ◆ ☐ **What learners bring:** the layered master and all incremental versions, the brief and moodboard, the end-of-pass screenshots and a true "before," their own photos and any licensed stock, and fonts they are licensed to use. Details in `labs/module-8/README.md`.
- ◆ ☐ **Version and sign-in check:** anyone who used Generative Fill or Expand should have finished those steps at home. Confirm the projector machine opens JPG, PNG, and PDF, and that learners know where their Content Credentials option is (Export As or the Content Credentials panel) if they want it.
- ◆ ☐ **Room prep:** a shared folder (or USB drive) for presentation files, a visible timer, help-flag cards in three colors, and printed rubric cards (final-assessment rows 2–5 and 8–10) for reviewers. Draft the presentation order and put it on the board; put a confident learner first.
- ◆ ☐ Open `lesson_21.html`, `lesson_22.html`, and `lesson_23.html` in browser tabs for the before/after sliders and the checklists.

SESSION 8 AGENDA (3 HOURS)

CLOCK	SEGMENT	MIN	NOTES
0:00–0:15	Plan review in pairs (L21 · 8.1)	15	Welcome (2). Pairs swap briefs and try to grade the current file against the partner's success criteria (8). Learners write their studio goal (3). Timeline on the board (2). You triage red / yellow / green.

CLOCK	SEGMENT	MIN	NOTES
0:15–1:25	Studio block (L22 · 8.2)	70	Launch (5), finishing with two circulation rounds and timed cues (55), the finish gate: tick success criteria, save the master, export deliverables at spec (10).
1:25–1:37	Break	12	No more editing. You check the shared folder against the presentation order.
1:37–2:02	Case-study assembly (L23 · 8.3, Steps 1–2)	25	The five beats (4), hero into a mockup + before/after to the shared folder (8), write the beats and the 30-second version (10), critique norms and rubric cards (3).
2:02–2:52	Presentations + peer critique (L23)	50	About 4 minutes per learner for a class of about 12; pods for 13 or more.
2:52–3:00	Course close & exit ticket	8	What you have learned, keep growing, final journal, exit ticket, celebration.
Total		180	The L21 kickoff (42 min) runs at the end of Session 7 and is not counted here.

The lesson pages estimate 45 minutes for L21, "self-paced (a focused session or several)" for L22, and 50 minutes for L23, which is far more than the room can hold alongside 12 presentations. Resolve it the way the capstone is designed: L21 runs in the Session 7 kickoff and finishes at home; L22 is mostly homework, and the studio block finishes it; L23's Steps 1–2 (assets and the story) happen in class so learners can present, and Steps 3–4 (lay out, export and publish) are homework due within a week. The presentation block is fixed; protect it. If the plan review or studio runs long, take the time from the studio block, never from presentations. If many learners arrive under 50% done, see "If the Room Arrives Behind" below.

MODULE AT A GLANCE

LESSON	TOPIC	WHERE	LEARNERS LEAVE ABLE TO...
L21 (8.1)	The Brief & the Plan	S7 45m + home + S8 15m	Write a six-part, checkable brief, fix the visual direction with an annotated moodboard, and stage a finishable plan
L22 (8.2)	Studio Session: Build It	home + 70m	Build in passes in a clean, non-destructive, versioned file, judge it with fresh eyes, and grade it against the brief
L23 (8.3)	Present & Case Study	25m + 50m + close	Tell the work's story in five beats, present it in context, and publish a portfolio-ready case study

The module is one story: **a brief is the contract → raise the whole image a level at a time → show the thinking, not just the pixels.** The deliverable, from the course homepage: a self-directed professional brief with a written case study, presented.

L21

The Brief & the Plan (Lesson 8.1)

S7 kickoff 45 min • homework • S8 review 15 min

WHEN THIS RUNS

The full lesson below is **taught in the Session 7 capstone kickoff** (the Module 7 agenda gives it 42 minutes, 2:08–2:50 of Session 7). Learners finish the brief, moodboard, and plan at home, then start building. It is **reviewed at the start of Session 8** in the 15-minute pair review. The timing below is written for 45 minutes; to fit the 42-minute slot, cut the pair warm-up to 5 minutes and the Step 2–3 start to 4.

◆ OBJECTIVES

- ◆ Write a tight creative brief: objective, audience, deliverables, constraints, success criteria
- ◆ Translate a vague idea into specific, checkable requirements
- ◆ Build a moodboard that fixes the visual direction before you start
- ◆ Draft a staged plan that maps each phase to the skills you have learned
- ◆ Scope realistically so the project is ambitious but finishable

◆ TIMING: SESSION 7 KICKOFF (45 MIN)

4m	The goal: a spec, not a wish (lesson slider) + the brief is the contract
6m	Warm-up in pairs: fix the sticky-note brief (<code>l21-vague-brief.txt</code>)
5m	Anatomy of a brief (six parts) + the moodboard rule
4m	The plan: map skills to stages (Figure 3) + scope for finishing
3m	Now You: pick your project (four options)
15m	Guided build Step 1: write the brief (in class)
6m	Steps 2–3 started: moodboard file set up, stages listed; finish at home
2m	Homework, the "Pass 1 and Pass 2 done" rule, journal prompt

◆ TIMING: SESSION 8 PLAN REVIEW (15 MIN, 0:00–0:15)

2m	Welcome; the day's shape
8m	Pairs, 4 minutes each: grade the file against the partner's success criteria

3m Studio goal written in the workbook

2m Timeline on the board; your triage list complete

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT (SESSION 7 KICKOFF)

- 1 **A spec, not a wish (4m).** Drag the L21 slider from the vague sticky notes ("make it pop," "you'll know it when you see it") to the structured brief. **Say:** "A brief is the agreement between the vision and the work, with a client or just with your future self at 2 a.m. wondering if you are done. Its job is to make success checkable. 'Make it modern' cannot be checked. 'A 3000×2000 hero image for a coffee brand's homepage, warm and editorial, readable at a glance on mobile, delivered as web + print by Friday' can." Write the mantra on the board: **if you can't check it, it isn't in the brief.**
- 2 **Warm-up: fix the sticky notes (6m).** Put `l21-vague-brief.txt` on screen. In pairs, learners pin each phrase to a spec, a named constraint, or a reference image. Take three answers. **Say:** "'Soon-ish' became a date and a time. 'Our colors, I guess?' became two hex values. 'Young people should like it' became a named group, a device, and one second of attention." Your answer is in `Labs/module-8/instructor/l21-vague-brief-fixed.txt`; accept any version where every line is checkable.
- 3 **Anatomy of a brief (5m).** The six parts: **objective** (the one sentence on what this must achieve and why), **audience** (who, their context, what should make them stop), **deliverables + specs** (exact outputs, sizes, formats, where they will live), **constraints** (brand rules, colors, fonts, budget, and the deadline), **success criteria** (the checkable tests that mean "done and good"), **references / moodboard**. **Say:** "Keep it to one page. A page forces the clarity that three pages let you avoid." Then the moodboard: 6–12 references for tone, color, type, and composition, each with a note on why it is there. **Say:** "It aligns you, and a client, while change is still cheap. Later it is the yardstick you grade your drafts against."
- 4 **The plan (4m).** Show Figure 3: assets & cutouts (Module 1) → color base (Module 2) → subject retouch (Module 3) → composite (Module 4) → type & system (Module 5) → automate & export (Module 6, maybe 7) → present (8.3). **Say:** "Not every project needs every stage, but naming them turns a blank canvas into a checklist. The commonest capstone failure is over-scoping. A tightly executed simple idea beats an ambitious half-finished one every time. Stretch goals are fine if you can cut them without breaking the deliverable."
- 5 **Pick your project (3m).** The four options: **key art poster** (a composited hero with title treatment; Modules 1, 3, 4, 5), **product campaign** (a hero plus a sized social set from one master; Modules 2, 4, 5, 6), **editorial portrait** (a retouched, graded portrait with layout and type; Modules 2, 3, 5), or **your own**. **Say:** "No client? Invent one: a fictional brand, a favorite band's next album, a film that doesn't exist. Just hold yourself to the same specificity a client would demand." Point hesitant learners to `l21-backup-briefs.txt`.
- 6 **Step 1: write the brief (15m).** Everyone fills `l21-brief-template.txt` (or the workbook page). Circulate. At the 10-minute mark, **say:** "Now write the success criteria as a checklist you could hand to

someone else to grade." In the last 2 minutes, learners swap and read each other's success criteria aloud: any criterion the partner cannot check gets rewritten.

- 7 **Steps 2–3 started (6m).** Create the moodboard file (a contact-sheet PSD or a simple grid is fine) and drop in the first two references with one-line annotations; list the stages and the module skill each uses; note assets to source or shoot. **Say:** "Anything you did not shoot: check the license and the credit terms now, not the night before."
- 8 **Homework and journal (2m).** Finish the brief, the moodboard (6–12 annotated references), and the plan with two or three milestones and a finish date; have someone else read the success criteria; gather source assets. Then build: **Pass 1 and Pass 2 done, Pass 3 under way**, with a screenshot at the end of each pass and a true "before" saved. Journal prompt: the capstone's one-sentence objective.

◆ **LAB: THE BRIEF, MOODBOARD & PLAN (KICKOFF), THEN THE SESSION 8 PLAN REVIEW (15 MIN)**

Kickoff: `l21-vague-brief.txt` warm-up, then the learner's own brief on `l21-brief-template.txt` or the workbook. Details in `Labs/module-8/README.md`.

Session 8 review, 0:00–0:15. Welcome (2). In pairs (4 minutes each), the partner reads the brief's success criteria and tries to grade the current file against them, criterion by criterion: met, not yet, or "I can't check this one." Learners then write their **studio goal** in the workbook (the one or two things they will finish today, and what they need help with) (3). Put the day's timeline on the board (2).

Triage as you walk: **red** = no brief, or less than Pass 1 built; **yellow** = built but stuck on a specific skill; **green** = in Pass 3, finishing. Write initials in three columns; this list drives the first circulation round.

Definition of done: the lesson's checklist: a one-page brief with all six parts; every vague term pinned to a spec or reference; success criteria written as a checkable list; an annotated moodboard of 6–12 references; a staged plan mapping skills, assets, and milestones. This is final-assessment **row 1 • Brief & plan**.

◆ **MISTAKES TO PRE-EMPT**

- ◆ **Adjectives left floating.** "Bold, clean, modern" with no reference image. Ask: "Show me the picture that means 'bold' to you."
- ◆ **Success criteria that only the author can judge** ("it feels premium"). Rewrite as something a stranger could tick: "product name readable on a phone-sized thumbnail."
- ◆ **Scope for a series.** Six campaign sizes, a poster, and an animation. Scale to one deliverable plus cuttable stretch goals; the lesson's "Now You" and L22's stretch list are where extras belong.
- ◆ **A moodboard as a copy list**, or a pile of 40 images. 6–12, each annotated with the one thing being taken from it.
- ◆ **Unlicensed assets.** Stock grabbed from a search. The plan step asks for license and credit terms checked for anything they did not shoot.
- ◆ **A plan with no dates.** Two or three milestones and a realistic finish date make it possible to cut stretch goals early.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** C) objective, audience, deliverables + specs, constraints, success criteria, references/moodboard; **Q2** A) pin it to a reference image, a measurable spec, or a named constraint; **Q3** D) 6–12 references, each annotated with the one thing you take from it; **Q4** B) "The product name is readable on a phone-sized thumbnail"; **Q5** A) it can be cut without breaking the deliverable; **Q6** D) over-scoping. Full set in the Answer Key.

Journal (homework): *Write your capstone's one-sentence objective. If you can't say what success looks like in a sentence, the idea isn't ready; keep refining until you can.*

◆ DIFFERENTIATION

- ◆ **Confident learners:** the lesson's "Going further": write the brief as if a real client sent it, then a second version as their own artist statement, and compare the success criteria.
- ◆ **Hesitant learners:** hand them a backup brief (`121-backup-briefs.txt` : "The Salt Road" key art poster or the Halden Cold Brew campaign). The six parts are already written; their job is the moodboard, the plan, and one success criterion of their own.

◆ OBJECTIVES

- ◆ Execute a project in passes: block-in, refine, polish
- ◆ Keep ruthless layer hygiene: groups, names, color labels
- ◆ Stay non-destructive with Smart Objects and adjustment layers throughout
- ◆ Version and save so you can always step back safely
- ◆ Use fresh-eyes tricks to judge your own work honestly

◆ TIMING (0:15–1:25)

5m	Studio launch at the projector: passes, hygiene, fresh eyes, help flags, timed cues
15m	Round 1: reds first, then yellows
20m	Round 2 check-ins; 0:45 whole-room cue: fresh-eyes checks
20m	Round 2 continues; 1:05 whole-room cue: polish pass only, no new elements
10m	The finish gate: tick success criteria, save the master, export at spec, hero to the shared folder

◆ TEACHING SEQUENCE & STUDIO SCRIPT

- 1 Work in passes (launch, 2m).** Drag the L22 slider: forty layers named "Layer 12 copy" against tidy groups. **Say:** "Same artwork; only one is still editable next week when the client asks for 'one small change.' Professionals work the whole image in passes: block-in gets every element placed roughly right, refine fixes selections, light, and color across the frame, polish adds the small, high-value details last. The image is done when a pass changes almost nothing, not when you run out of things you could do."
- 2 Hygiene, shown once (launch, 1m).** Groups by role, top-to-bottom in render order (Grade · Type & layout · Subject · Atmosphere & light · Background); **Ctrl/⌘ + G** to group, double-click to name, right-click the eye icon to color-label. **Say:** "Every adjustment a live layer, every scaled element a Smart Object, every filter a Smart Filter. Save capstone_v01, v02 at each pass, and keep a flattened reference JPG of milestones."
- 3 Help flags and cues (launch, 2m).** A card on each monitor: **green** = working, **yellow** = a question when you are free, **red** = blocked for five minutes. Announce the cues now: 0:45 fresh eyes, 1:05 polish only, 1:15 the finish gate.
- 4 Round 1, 0:20–0:35: reds first.** Three minutes max each. Goal: a finishable target for today. **Say:** "Which success criteria can we meet by 1:15? Everything else becomes a stretch goal." Then yellows.

- 5 **Round 2, 0:35–1:15: everyone.** Two to three minutes per learner, the same four prompts: "Show me your brief's success criteria." · "Which pass are you in, and is it done across the whole frame?" · "What is your next move?" · **file health:** groups named and labeled, adjustments live, a version saved this session? Keep your hands off the mouse. If three or more learners are stuck on the same skill, run a 5-minute clinic at the projector for those learners only.
- 6 **0:45 whole-room cue: fresh eyes. Say:** "Everyone, four checks. View ► Flip Horizontal: it flips only the view; in older versions it is Image ► Image Rotation ► Flip Canvas Horizontal, so flip back after. Zoom to fit and squint: does the focal point survive at thumbnail size? Add a Black & White adjustment layer at the very top of the stack, look, then toggle it off. And tonight, walk away and look again tomorrow." Those are final-assessment rows 3, 4, and 5.
- 7 **1:05 whole-room cue: polish only. Say:** "No new elements now. Unifying grade on top, final details, tighten any mask at 100%, name anything unnamed. Save a version."
- 8 **1:15–1:25 the finish gate. Say:** "Open your brief's success criteria and tick each one on the artwork. Grade against the brief, not your mood." Save the layered master; export every deliverable at spec; export a hero JPG in sRGB named `firstname-hero.jpg` to the shared folder; make sure the end-of-pass screenshots and the true "before" are in one folder. Track it: tick each name when their file lands, and chase anyone without a tick at 1:22.

◆ LAB: FINISH THE CAPSTONE (STUDIO BLOCK)

The learner's own capstone file, built at home through Pass 1 and Pass 2. Learners who arrive behind work from a backup brief: the key art pair (`l21-backup-keyart-plate.jpg` + `l21-backup-keyart-figure.jpg`) or the campaign set (`l21-backup-product.jpg` + `l21-backup-product-surface.jpg` + `l21-backup-brand-logo.png`). Details in `labs/module-8/README.md` .

Circulate for: a pass finished across the frame before the next begins; groups named and color-labeled; adjustments and filters still live; more than one version on disk; a Black & White check layer left switched on by accident.

Definition of done: the lesson's checklist: built in passes, whole-image at each level; grouped, named, color-labeled, non-destructive layer stack; incremental versions saved along the way; fresh-eyes checks (flip, squint, desaturate, walk away) done; every brief success criterion met and ticked; layered master saved and deliverables exported at spec. These are final-assessment **rows 2–7**.

◆ MISTAKES TO PRE-EMPTY

- ◆ **One perfect corner.** Hours on an eye at 300% while the background is still a placeholder. Ask "Which pass is the whole frame in?" and zoom them out.
- ◆ **"Layer 12 copy 3."** Make naming and grouping a condition of your check-in: you help once the stack is named.
- ◆ **Baked effects under pressure.** A filter run on the pixel layer, or an adjustment applied from Image ► Adjustments. Ask them to rebuild the one key edit as a Smart Filter or an adjustment layer.

- ◆ **The check layer left on.** A Black & White layer still visible at export gives a gray deliverable. It is a temporary layer; toggle it off.
- ◆ **Huge, slow file.** The lesson's mid-project rescues: move heavy, finished elements into linked Smart Objects (Layer ► Smart Objects ► Convert to Linked), rasterize truly final elements, Edit ► Purge ► Histories, sensible working resolution. Save a full layered version first. An embedded Smart Object tidies the stack but still stores every pixel, so only linked ones shrink the file.
- ◆ **Fiddling past done.** Moving things a pixel back and forth. **Say:** "You're past done: save it and present it."
- ◆ **New generative experiments in class.** Credits and network are not guaranteed; steer to finishing with what is already in the file.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1 B)** raise the whole image a level at a time: block-in, refine, polish, each across the frame; **Q2 D)** save incrementally at each pass and keep a flattened reference JPG of milestones; **Q3 A)** live adjustment layers, Smart Objects, Smart Filters, grouped by role and color-labeled; **Q4 C)** it flips only the view, so composition and balance errors leap out; **Q5 B)** a temporary Black & White adjustment layer at the very top, toggled off when done; **Q6 D)** when a fresh-eyes pass changes almost nothing and every success criterion is met. Full set in the Answer Key.

Journal (homework): *Which pass was hardest to leave "good enough" and move on from? Learning when to stop refining a corner is a professional skill worth naming.*

◆ DIFFERENTIATION

- ◆ **Ahead of schedule:** the lesson's stretch list: an alternate grade or a day/night variant from the same master (Modules 4.3, 5.2); spin the hero into a sized campaign set (Module 5.3) or a short animated version (Module 7.2); record the repetitive export as an Action or data set (Module 6).
- ◆ **Behind:** agree a "minimum presentable" version: the focal element through Pass 2, one unifying grade, clean edges at 100%, and the success criteria it can meet today. Unfinished side elements get hidden, not rushed; the missed criteria go into the case study's reflection.

◆ OBJECTIVES

- ◆ Structure a case study: challenge → approach → process → result → reflection
- ◆ Present the final piece in context with mockups, not floating in space
- ◆ Use before/after and progress shots to show your thinking
- ◆ Write about your work clearly, for clients and employers
- ◆ Package and export a portfolio-ready presentation

◆ TIMING (1:37–3:00)

4m	The work plus its story (lesson slider) + the five beats + show the decisions
8m	Step 1: hero into a mockup, a before/after pair, both to the shared folder
10m	Step 2: write the five beats and the 30-second spoken version
3m	Critique norms and rubric cards
50m	Presentations + structured peer critique
8m	Course close & exit ticket

◆ TEACHING SEQUENCE & LIVE-DEMO SCRIPT

- 1 The work plus its story (4m).** Drag the L23 slider from "image only" to "case study." **Say:** "A portfolio full of pretty pictures says 'I can operate Photoshop.' A portfolio of case studies says 'I can solve problems.' Every real job is a problem, not a picture." Write the arc: **Challenge → Approach → Process → Result → Reflection.** **Say:** "A case study is a highlight reel, not a tutorial. Pick the three or four moments that show judgment: the tricky mask, the grade choice, the composition call."
- 2 Step 1: assemble (8m).** Demo on `l23-mockup-wall.jpg`, using the Module 5.2 recipe: draw a rectangle over the gray opening, Convert to Smart Object, double-click its thumbnail, place the hero in the tab that opens, save and close. **Say:** "On a real surface you would Warp or Perspective Warp the Smart Object to fit, and add the surface's shadows on Multiply and highlights on Screen above it, clipped. To swap art later, Layer ► Smart Objects ► Replace Contents; the treatment stays." The phone mockup is rotated, so its post needs a Free Transform rotate as well. Learners export `firstname-context.jpg` and `firstname-before-after.jpg` to the shared folder. **Say:** "Keep the before/after honest: the real starting point and the real steps."
- 3 Step 2: write the story (10m).** Workbook or `l23-case-study-outline.txt`: the five beats, a few sentences each, then the 30-second spoken version. **Say:** "Lead with the outcome or the idea, not the

tools. 'A warm, editorial key art that reads on mobile' beats 'I used Curves and Generative Fill.' If any part used Generative Fill or Expand, say so in the process beat." While they write, load the context and before/after files on the projector in presentation order.

- 4 **Critique norms (3m).** On the board: critique the work, not the person · point at the pixels · judge against the presenter's brief, not your taste · the presenter listens and says thank you, no defending · kind is not the same as vague. Hand out the rubric cards: final-assessment rows 2–5 (meets the brief, composition & balance, focal point & read, tonal structure) and 8–10 (hero & context, the story, honest process).

◆ PRESENTATION & STRUCTURED PEER CRITIQUE PROTOCOL (ABOUT 4 MIN PER LEARNER)

- 1 **Present (90 s).** The hero in context, the brief in a line, the before/after, and the 30-second story: challenge, approach, result.
- 2 **Squint test (15 s).** The whole room squints at the projector. You ask: "What do you see first?" Two or three people call it out. Does it match the brief's focal point?
- 3 **Two reviewers (90 s).** Assigned in advance: the next two people in the order. Each starts from the presenter's brief ("Which success criterion does it meet most clearly?"), then picks **one rubric row** and gives **one glow** (what works, specifically, pointing at the pixels) and **one grow** ("I wonder if..." or "What if you tried...").
- 4 **Presenter notes (15 s).** The presenter writes the grow in the workbook's critique box and says thank you. You add at most one sentence, only if the reviewers missed something important. About 30 seconds of changeover completes the 4 minutes.

Group size: 50 minutes fits 12 learners at about 4 minutes each. For **13 or more**, split into pods of 4 to 5 around a laptop with printed rubric cards and a pod timekeeper; run the same protocol in parallel for 35 minutes while you float, then each pod nominates one piece for a 15-minute whole-room showcase. For groups **under 7**, give each presenter 6 to 7 minutes and let everyone review.

Definition of done (homework, within one week): the lesson's checklist: a big, clean hero shot leading the piece; the five story beats written tight and outcome-first; honest before/after and 3–5 progress shots; deliverables shown in context via mockups; laid out with your type system, exported, and published; any generative-AI assistance disclosed honestly. Steps 3–4 (lay it out; export web images via File ► Export ► Export As and a PDF via File ► Save a Copy ► Photoshop PDF, or File ► Export ► Artboards to PDF) happen at home. These are final-assessment **rows 8–11**; the presentation is **row 12**.

◆ MISTAKES TO PRE-EMPT

- ◆ **Apologizing before showing** ("It's not finished, I ran out of time..."). Coach it out during Step 2: start with the challenge, not the excuse. Missed criteria belong in the reflection beat.
- ◆ **A tool list instead of a story.** "First I used Select Subject, then Curves..." Redirect to outcome-first and to three or four decisions.

- ◆ **Art pasted flat over the mockup.** No Smart Object, so no warp and no shadow; it looks stuck on. Rebuild inside the mockup's Smart Object.
- ◆ **A fabricated "before."** A deliberately worsened starting image. It costs all credibility; use the real one.
- ◆ **Reviewers who only say "I love it"** or give taste ("I'd have made it blue"). Ask: "Which rubric row, what exactly on screen, and what does the brief say?"
- ◆ **Undisclosed generative content.** Ask presenters to name it in the process beat; Content Credentials can be attached on export.

◆ COURSE CLOSE & EXIT TICKET (8 MIN, 2:52-3:00)

- 1 **What you have learned (2m).** Read the lesson's arc: advanced selection and masking → color and tone, deep → high-end retouching → commercial compositing → design systems and type → generative and automation → multi-shot techniques → the capstone. **Say:** "You arrived able to operate Photoshop. You leave able to work like a professional, and with the judgment to know which tool a problem needs and when to stop."
- 2 **Keep growing (2m).** Build a portfolio of case studies; practice one weak spot deliberately; study work you admire and name why it works; seek critique; specialize or broaden. **Say:** "Your next project: slightly beyond comfortable, briefed the way you learned in 8.1, finished and presented the way you learned in 8.3."
- 3 **Exit ticket (2m).** On a card: "One thing I can do now that I couldn't at the start of this course," and the date they will publish the case study.
- 4 **Final journal and celebration (2m).** Assign the final prompt to finish at home. Thank them; finishing an entire course of deliberate practice is rare. Applause for every piece.

◆ CHECK FOR UNDERSTANDING: SELF-CHECK + JOURNAL

Answers: **Q1** C) challenge → approach → process → result → reflection; **Q2** A) a big, clean hero shot of the finished piece; **Q3** D) "A warm, editorial key art that reads on mobile"; **Q4** B) context tells the reviewer what it is for and makes it look shipped rather than sketched; **Q5** A) File ► Save a Copy ► Photoshop PDF; **Q6** C) say so in the process section; Content Credentials can be attached on export. Full set in the Answer Key.

Final journal (homework): *Read back through your whole journal, from Module 1 to here. What can you do now that you couldn't when you started? Write that down; it is the real deliverable of this course.*

◆ DIFFERENTIATION

- ◆ **Confident presenters:** the lesson's "Now You": publish the case study publicly, turn two or three earlier projects into mini case studies, and (going further) ask someone in the field for honest feedback, then revise.

- ◆ **Anxious presenters:** let them present from their seat, or read the 30-second version from the workbook while you drive the screen. Seat them mid-order, never first or last.

IF THE ROOM ARRIVES BEHIND

PLAN B FOR LEARNERS WHO DID NOT BUILD AT HOME

Some learners will arrive with a brief but little built, or with nothing. Do not let them derail the session. **Scope to one deliverable and three success criteria.** Offer a ready-made brief from `l21-backup-briefs.txt` : "The Salt Road" key art (place `l21-backup-keyart-figure.jpg` on `l21-backup-keyart-plate.jpg` : cutout, scale to the horizon, light from the left, contact and cast shadows) or the Halden Cold Brew hero (`l21-backup-product.jpg` on `l21-backup-product-surface.jpg` with `l21-backup-brand-logo.png`). They work through Pass 1 and Pass 2 in the studio block, present honestly as a "studio sketch" with the missed criteria in the reflection, and receive the same critique. They can complete within the one-week case-study window. If more than half the room is behind, shorten each presentation to 3 minutes and give the saved time to the studio block.

CLOSING THE COURSE

AFTER SESSION 8

Homework: L23 Steps 3–4 (lay out the case study top to bottom with the Module 5 type system, export web images and a PDF, publish it and share the link with at least one person) within one week; the final journal entry; the L23 self-check. **Assessment:** score rows 1–7 and 12 of `final-assessment.html` from Session 8, and rows 8–11 when the published case study arrives. Pass = no row below 2, plus the deliverables exported at spec and a saved layered master. Below the bar, give feedback tied to rows and allow one resubmission. Collect case-study links (with permission) for a class gallery, and send each learner their critique notes and the Recipes II link. Recap the whole arc: **masks that select by tone → color you can trust → invisible retouching → believable composites → design systems → automation → many frames into one → a briefed, built, and presented portfolio piece.**